

Comparing a Full Structured Copula Model with Several Sample Splitting Based Copula Models for Multivariate Clustered Survival Data

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Contents

1	Introduction	5
2	Methodology	6
2.1	Data Description	7
2.2	Statistical Model	7
2.2.1	Model Description	7
2.2.2	Archimedean Copula	9
2.2.3	Parameter Estimation	10
2.2.4	Sunclarco Model	11
2.3	Full Data Analysis	11
Part 1:	Random Sample Splitting	13
2.3.1	Sample Splitting Strategy	13
2.4	Combination of Estimates	13
2.4.1	Information-Type Weighting Based on Estimated Precision	14
2.4.2	Inverse Variance Weighting	15
2.4.3	Sample Size Weighting	15
2.4.4	Variance of the Combined Estimator	15
2.5	Evaluation Metrics	16
2.5.1	Difference in Regression Estimates	16
2.5.2	Relative Difference	16
2.5.3	Standard Error Comparison	17
2.5.4	Dependence Parameter Comparison	17
2.6	Comparison Across Weighting Methods	17
Part 2:	Quartile-Based Splitting Strategy	18
2.7	Quartile Construction Based on Herd Size	18
2.7.1	Splitting of Largest Quartile (Q1)	18
2.7.2	Splitting of Second-Largest Quartile (Q2)	19
2.8	Analysis Framework	19
Part 2:	Quartile-Based Splitting Strategy	20
2.9	Computational Cost Evaluation	20

3	Results	21
3.1	Data Overview	22
3.2	Full Data Analysis	22
3.2.1	Marginal Cox Model	22
3.2.2	Cluster-Robust Cox Model	22
3.2.3	Sunclarco Copula Model	23
3.2.4	Comparison of Full Data Model Results	24
3.2.5	Computational Cost of the Full Data Copula Model	24
3.2.6	Interpretation of the Full Data Analysis	24
Part 1:	Random Sample Splitting	25
3.3	Three-Subset Analysis	25
3.3.1	Subset-Specific Model Results	25
3.3.2	Combined Estimates from the Three Subsets	25
3.3.3	Comparison with the Full-Data Analysis	26
3.3.4	Standard Error Comparison for the Three-Subset Analysis	26
3.3.5	Computational Cost Analysis	27
3.3.6	Interpretation of the Three-Subset Analysis	27
3.4	Five-Subset Analysis	29
3.4.1	Subset-Specific Model Results	29
3.4.2	Combined Estimates from the Five Subsets	29
3.4.3	Comparison with the Full-Data Analysis	30
3.4.4	Standard Error Comparison for the Five-Subset Analysis	31
3.4.5	Computational Cost Analysis	31
3.4.6	Interpretation of the Five-Subset Analysis	32
3.5	Eight-Subset Analysis	33
3.5.1	Subset-Specific Model Results	33
3.5.2	Combined Estimates from the Eight Subsets	33
3.5.3	Comparison with the Full-Data Analysis	34
3.5.4	Standard Error Comparison for the Eight-Subset Analysis	35
3.5.5	Computational Cost Analysis, Eight Subset	37
3.5.6	Interpretation of the Eight-Subset Analysis	37
Part 2:	Quartile-Based Splitting Strategy	38
3.6	Data Description After Quartile-Based Splitting	38
3.7	Random Sample Splitting After Q1 Splitting	38
3.7.1	Results of Quartile-Based Splitting (Q1)	39
3.7.2	Computational Cost, Q1-Based Splitting	39
3.7.3	Interpretation	39
3.8	Three-Subset Analysis After Q1 Splitting	40
3.8.1	Subset-Specific Model Results	40

3.8.2	Combined Estimates from the Three Subsets	40
3.8.3	Comparison with the Q1-Split Full-Data Analysis	40
3.8.4	Standard Error Comparison with the Q1-Split Full-Data Analysis	41
3.8.5	Computational Cost of the Three-Subset Analysis	41
3.9	Five-Subset Analysis After Q1 Splitting	41
3.9.1	Subset-Specific Model Results	41
3.9.2	Combined Estimates from the Five Subsets	42
3.9.3	Comparison with the Q1-Split Full-Data Analysis	42
3.9.4	Standard Error Comparison for the Five-Subset Analysis	42
3.9.5	Computational Cost of the Five-Subset Analysis	43
3.10	Eight-Subset Analysis After Q1 Splitting	43
3.10.1	Subset-Specific Model Results	43
3.10.2	Combined Estimates from the Eight Subsets	43
3.10.3	Comparison with the Q1-Split Full-Data Analysis	44
3.10.4	Standard Error Comparison for the Eight-Subset Analysis	44
3.10.5	Computational Cost of the Eight-Subset Analysis	44
3.11	Data Description After Combined Q1 and Q2 Splitting	44
3.11.1	Results of Combined Q1 and Q2 Splitting	45
3.11.2	Computational Cost of the Combined Q1 and Q2 Splitting Analysis	46
3.12	Three-Subset Analysis After Q1 and Q2 Splitting	46
3.12.1	Data Description for the Three-Subset Analysis	46
3.12.2	Subset-Specific Model Results	47
3.12.3	Combined Estimates from the Three Subsets	47
3.12.4	Comparison with the Q1+Q2 Full-Data Analysis	47
3.12.5	Standard Error Comparison for the Three-Subset Analysis	48
3.12.6	Computational Cost of the Three-Subset Analysis	48
3.13	Five-Subset Analysis After Q1 and Q2 Splitting	48
3.13.1	Data Description for the Five-Subset Analysis	48
3.13.2	Subset-Specific Model Results	48
3.13.3	Combined Estimates from the Five Subsets	49
3.13.4	Comparison with the Q1+Q2 Full-Data Analysis	49
3.13.5	Standard Error Comparison for the Five-Subset Analysis	49
3.13.6	Computational Cost of the Five-Subset Analysis	49
3.14	Eight-Subset Analysis After Q1 and Q2 Splitting	50
3.14.1	Data Description for the Eight-Subset Analysis	50
3.14.2	Subset-Specific Model Results	50
3.14.3	Combined Estimates from the Eight Subsets	50
3.14.4	Comparison with the Q1+Q2 Full-Data Analysis	50
3.14.5	Standard Error Comparison for the Eight-Subset Analysis	51



3.14.6	Computational Cost of the Eight-Subset Analysis	51
3.15	Overall Summary of Full Data and Sample-Splitting Results	52
3.15.1	Heifer Estimate Comparison	52
3.15.2	Standard Error (Heifer) Comparison	52
3.15.3	Kendall's τ Comparison	52
3.15.4	Standard Error τ Comparison	53
3.15.5	Computational Cost Comparision	53
4	Discussion	54
	Acknowledgement	55
	Appendix for Chapter 3	56
.1	PART 1 TABLES	56
.2	PART 2 TABLES	66

Chapter 1

Introduction



Chapter 2

Methodology



2.1 Data Description



The analysis is conducted in R using the **Sunclarco** package. The application dataset used in this thesis is the **insem** dataset, which is available in the **Sunclarco** package. This dataset contains clustered survival data, where individual cows are grouped within herds. Because cows belonging to the same herd may share similar management, environmental, or biological characteristics, the observations within a herd are likely to be correlated. This makes the dataset suitable for the study of multivariate clustered survival models.

The dataset contains the following variables:

- **Cowid**: identification number of each cow,
- **Time**: observed survival or follow-up time,
- **Status**: event indicator, where a value of 1 denotes that the event occurred and a value of 0 denotes censoring,
- **Herd**: cluster identifier indicating the herd to which each cow belongs,
- **Heifer**: covariate indicating whether the cow is a heifer.

In this study, the outcome variable is the survival time represented by **Time**, together with the event indicator **Status**. These two variables jointly define the survival outcome in the form **Surv(Time, Status)** in R. The main dependent variable is therefore the time until the event of interest, while accounting for possible right censoring.

The explanatory variable considered in the analysis is **Heifer**. This covariate is used to examine whether the hazard of the event differs according to heifer status. The clustering variable is **Herd**, which is essential because the dependence among cows within the same herd must be taken into account in the modelling process.

Thus, the data have a hierarchical structure in which cows are nested within herds, and the survival responses are potentially dependent within each cluster. For this reason, both the marginal effect of the covariate and the within-cluster association need to be considered in the data analysis.

2.2 Statistical Model

2.2.1 Model Description

The data considered in this study have a multivariate clustered survival structure. In clustered survival data, multiple individuals are grouped within the same cluster, and the event times of individuals from the same cluster may be correlated. Therefore, a

modelling approach is required that accounts for both survival outcomes and within-cluster dependence.

For subject j in cluster i , let T_{ij} denote the survival time and let $\mathbf{X}_{ij}$ denote the corresponding vector of covariates. The observed data are given by

$$Y_{ij} = \min(T_{ij}, C_{ij}), \quad (2.1)$$

with event indicator

$$\delta_{ij} = I(T_{ij} \leq C_{ij}), \quad (2.2)$$

where C_{ij} denotes the censoring time and $I(\cdot)$ is the indicator function. Thus, the survival outcome is represented by the pair (Y_{ij}, δ_{ij}) .

In survival analysis, interest lies in modelling the distribution of the time-to-event variable. The survival function is defined as

$$S(t) = P(T > t), \quad (2.3)$$

which gives the probability that the event time exceeds time t . The corresponding hazard function is

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}, \quad (2.4)$$

which represents the instantaneous risk of experiencing the event at time t , given survival up to that time.

A common marginal model for survival data is the Cox proportional hazards model, given by

$$h(t \mid \mathbf{X}_{ij}) = h_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{X}_{ij}), \quad (2.5)$$

where $h_0(t)$ is an unspecified baseline hazard function and $\boldsymbol{\beta}$ is the vector of regression coefficients. This model describes the marginal effect of the covariates on the hazard. However, by itself, the Cox model does not explicitly model the dependence between individuals within the same cluster.

For this reason, a copula model is used in this thesis. Copula models are especially useful for multivariate clustered survival data because they allow the marginal survival behaviour and the dependence structure to be modelled separately. This is an important advantage, since the effect of covariates on survival can be described through a marginal survival model, while the association between observations in the same cluster can be captured through a separate dependence parameter.

Let

$$S_{ij}(t_{ij}) = P(T_{ij} > t_{ij} \mid \mathbf{X}_{ij}) \quad (2.6)$$

denote the marginal survival function for subject j in cluster i . Then the joint survival

function for the n_i individuals in cluster i can be written using a copula as

$$S(t_{i1}, t_{i2}, \dots, t_{in_i}) = C(S_{i1}(t_{i1}), S_{i2}(t_{i2}), \dots, S_{in_i}(t_{in_i}); \theta), \quad (2.7)$$

where $C(\cdot)$ is the copula function and θ is the association parameter governing the within-cluster dependence.

Hence, the copula-based survival framework used in this thesis consists of two parts: a marginal survival model that describes the effect of covariates on the hazard, and a copula function that captures the dependence among survival times within the same cluster. This makes the copula approach particularly suitable for multivariate clustered survival data, where both regression effects and within-cluster association are of interest.

2.2.2 Archimedean Copula

In clustered survival data, the number of individuals within each cluster may vary. Let n_i denote the number of individuals in cluster i , where $i = 1, 2, \dots, m$. Because cluster sizes may differ across clusters, the dependence model must be able to accommodate a varying number of survival times within each cluster. Archimedean copulas are particularly suitable for this purpose because they allow a flexible multivariate dependence structure while remaining computationally tractable.

Let $S_{ij}(t_{ij})$ denote the marginal survival function of subject j in cluster i , for $j = 1, \dots, n_i$. Using an Archimedean copula, the joint survival function of the n_i individuals in cluster i can be written as

$$S(t_{i1}, t_{i2}, \dots, t_{in_i}) = C(S_{i1}(t_{i1}), S_{i2}(t_{i2}), \dots, S_{in_i}(t_{in_i})), \quad (2.8)$$

where $C(\cdot)$ denotes the copula function. For Archimedean copulas, the joint survival function can be expressed using a generator function $\phi(\cdot)$ as

$$C(u_1, u_2, \dots, u_{n_i}) = \phi^{-1}(\phi(u_1) + \phi(u_2) + \dots + \phi(u_{n_i})), \quad (2.9)$$

where $\phi(\cdot)$ is a decreasing convex generator function and $\phi^{-1}(\cdot)$ denotes its inverse.

This representation makes Archimedean copulas particularly convenient for clustered survival data with varying cluster sizes, since the joint survival distribution can be constructed for any number of individuals n_i within a cluster.

In this study, the Clayton copula, which belongs to the Archimedean copula family, is used to model the dependence structure between survival times within clusters. The Clayton copula generator function is given by

$$\phi(u) = u^{-\theta} - 1, \quad \theta > 0, \quad (2.10)$$

and the corresponding multivariate Clayton copula for a cluster of size n_i is

$$C(u_1, \dots, u_{n_i}; \theta) = \left(\sum_{j=1}^{n_i} u_j^{-\theta} - n_i + 1 \right)^{-1/\theta}. \quad (2.11)$$

The parameter θ controls the strength of dependence among survival times within the same cluster. When $\theta = 0$, the survival times are independent, while larger values of θ indicate stronger positive dependence. The dependence can also be expressed in terms of Kendall's tau,

$$\tau = \frac{\theta}{\theta + 2}. \quad (2.12)$$

2.2.3 Parameter Estimation

Parameter estimation in the copula-based survival model is carried out using a two-stage estimation procedure. This approach separates the estimation of the marginal survival model from the estimation of the dependence structure, which simplifies computation and improves interpretability.

In the first stage, the marginal survival model is estimated using a semi-parametric Cox proportional hazards model. The Cox model allows the effect of covariates on the hazard function to be estimated without specifying the baseline hazard function. Let T_{ij} denote the survival time for subject j in cluster i , and let $\mathbf{X}_{ij}$ denote the corresponding vector of covariates. The marginal hazard function is given by

$$h(t|\mathbf{X}_{ij}) = h_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{X}_{ij}), \quad (2.13)$$

where $h_0(t)$ represents the baseline hazard function and $\boldsymbol{\beta}$ is the vector of regression coefficients.

The regression parameters $\boldsymbol{\beta}$ are estimated using the Cox partial likelihood. The partial likelihood for the Cox model can be written as

$$L(\boldsymbol{\beta}) = \prod_{k=1}^D \frac{\exp(\boldsymbol{\beta}^\top \mathbf{X}_{(k)})}{\sum_{l \in R(t_k)} \exp(\boldsymbol{\beta}^\top \mathbf{X}_l)}, \quad (2.14)$$

where D denotes the total number of observed events, $R(t_k)$ represents the risk set at time t_k , and $\mathbf{X}_{(k)}$ corresponds to the covariates of the individual experiencing the event at time t_k .

In the second stage, the dependence between survival times within clusters is estimated using the copula model. After obtaining the marginal survival estimates, the copula parameter θ is estimated by maximizing the copula likelihood.

This two-stage estimation procedure allows the marginal covariate effects and the dependence structure to be estimated separately, making the copula-based framework

particularly suitable for clustered survival data with varying cluster sizes.

2.2.4 Sunclarco Model

To analyse the clustered survival data in this study, the **Sunclarco** model is used. This model is designed for multivariate survival outcomes grouped in clusters of variable size, and is based on the Archimedean copula framework. It combines a marginal survival model for the time-to-event outcome with a copula function that captures the dependence between observations within the same cluster.

The marginal survival component of the **Sunclarco** model can be specified in different ways. In this thesis, the Cox proportional hazards model is used as the marginal model.

The corresponding marginal survival function is

$$S_{ij}(t_{ij} \mid \mathbf{X}_{ij}) = \exp \left[-\Lambda_0(t_{ij}) \exp \left(\boldsymbol{\beta}^\top \mathbf{X}_{ij} \right) \right], \quad (2.15)$$

where $\Lambda_0(t)$ denotes the cumulative baseline hazard function.

To model the association between individuals within the same cluster, the **Sunclarco** model uses an Archimedean copula. This representation is particularly useful for clustered data with variable cluster size, because the same copula form can be applied for any number of individuals n_i within a cluster. In this study, the Clayton copula is used. The Clayton copula belongs to the Archimedean copula family and has generator function

$$\phi(u) = u^{-\theta} - 1, \quad \theta > 0. \quad (2.16)$$

The dependence parameter θ measures the strength of within-cluster association and can be interpreted through Kendall's tau. A larger value of θ corresponds to stronger positive dependence between survival times within the same cluster.

The **Sunclarco** model can be estimated using either a one-stage or a two-stage approach. In the one-stage approach, the marginal survival parameters and the copula dependence parameter are estimated jointly. In the two-stage approach, the marginal model is estimated first, after which the copula dependence parameter is estimated using the marginal survival estimates. In this thesis, the two-stage approach is used with a Cox marginal model and a Clayton copula, as it allows the marginal covariate effects and the within-cluster dependence structure to be handled separately.

2.3 Full Data Analysis

Before applying the sample splitting strategy, the copula-based survival model is first estimated using the complete dataset. The purpose of this analysis is to obtain reference

estimates of the regression parameters and the dependence parameter using the full information available in the data. These estimates serve as a benchmark for evaluating the performance of the sample-splitting approaches.

In the full data analysis, the copula-based survival model described in the previous section is fitted to the entire dataset without partitioning the clusters. The marginal survival component is estimated using a Cox proportional hazards model, which provides estimates of the regression coefficient β representing the effect of the covariate **Heifer** on the hazard of the event.

After estimating the marginal survival model, the dependence structure among survival times within the same herd is modeled using the Clayton copula. The copula parameter θ captures the strength of association between observations belonging to the same cluster. For interpretation purposes, the dependence is also expressed through Kendall's tau.

Thus, the full data analysis produces estimates of both the marginal regression effects and the within-cluster dependence using the entire dataset. These results provide a baseline against which the estimates obtained from the sample-splitting procedures can be compared in the subsequent sections.

Part 1: Random Sample Splitting

2.3.1 Sample Splitting Strategy

To evaluate whether sample splitting can provide results similar to the full data analysis, the clustered dataset was randomly divided at the herd level into several subsets. Since Herd is the clustering variable, the splitting was performed on whole herds rather than on individual cows, so that all observations from the same herd remained in the same subset. This preserves the within-herd dependence structure in each subset and avoids breaking the natural clustering of the data. The full dataset contains 181 unique herds, and these herds were randomly allocated into three different sample-splitting scenarios: 3 subsets, 5 subsets, and 8 subsets. Within each scenario, the subsets were made as balanced as possible with respect to the number of herds.

After splitting the data, the same modelling strategy that was used for the full data analysis was applied separately to each subset. More specifically, for each subset, the marginal Cox survival model was fitted first and then the Clayton copula model was estimated through the two-stage Sunclarco approach. In this way, each subset produced its own estimate of the regression effect and the dependence parameter.

Once the subset-specific analyses had been completed, the results from the different subsets were combined to obtain overall estimates for the corresponding sample-splitting scenario. These combined estimates were then compared with the estimates from the full data analysis. By repeating this procedure for 3, 5, and 8 subsets, it becomes possible to assess how the number of subsets affects the parameter estimates, standard errors, and the overall agreement with the benchmark full data analysis.

2.4 Combination of Estimates

After fitting the same survival model separately to each randomly created subset, the subset-specific estimates were combined to obtain one overall estimate for each sample-splitting scenario. Suppose that the data are divided into S subsets, where $S \in \{3, 5, 8\}$. Let $\hat{\beta}_s$ denote the regression estimate obtained from subset s , for $s = 1, \dots, S$. The general divide-and-combine estimator is written as a weighted average of the subset-specific estimates:

$$\hat{\beta}_{DC} = \left(\sum_{s=1}^S W_s \right)^{-1} \left(\sum_{s=1}^S W_s \hat{\beta}_s \right), \quad (2.17)$$

where W_s is the weight assigned to subset s . In this thesis, four weighting strategies were considered: information matrix weighting, inverse variance weighting, and sample size weighting (Herd, Cows).



2.4.1 Information-Type Weighting Based on Estimated Precision

The first approach combines subset-specific estimates using weights proportional to their estimated precision. In general, if the observed information matrix is available for subset s , then subsets with larger information receive greater weight because they contribute more statistical information to the final estimator. The divide-and-combine estimator can be written as

$$\hat{\beta}_{DC}^{(I)} = \left(\sum_{s=1}^S I_s(\hat{\beta}_s) \right)^{-1} \left(\sum_{s=1}^S I_s(\hat{\beta}_s) \hat{\beta}_s \right), \quad (2.18)$$

where $I_s(\hat{\beta}_s)$ denotes the observed information matrix for subset s . In the case of a single regression coefficient, this reduces to a scalar weighted average, so that subsets with larger information contribute more to the final estimate.

In practice, however, the exact observed information matrix is not always directly available for every fitted model. For the standard Cox model, the model-based variance estimate is available, and the corresponding precision weight can be written as the inverse of the estimated variance. For the clustered Cox model, one may use either the naive model-based variance or, more appropriately in clustered settings, the robust sandwich variance under a working independence assumption.

For the Sunclarco copula model, the package provides *jackknife-based* standard errors rather than an explicitly extracted Hessian-based observed information matrix. Therefore, for the copula model, the information-type weighting used in this thesis is based on the inverse of the jackknife variance,

$$w_s^{(I)} = \frac{1}{\widehat{\text{Var}}_{JK}(\hat{\beta}_s)},$$



where $\widehat{\text{Var}}_{JK}(\hat{\beta}_s) = \widehat{SE}_{JK}(\hat{\beta}_s)^2$. Thus, for the copula analysis, the so-called information-type weighting should be interpreted as a precision-based weighting constructed from jackknife standard errors, rather than from a directly computed Hessian-based information matrix.

For a scalar parameter, the combined estimator therefore becomes

$$\hat{\beta}_{DC}^{(I)} = \frac{\sum_{s=1}^S w_s^{(I)} \hat{\beta}_s}{\sum_{s=1}^S w_s^{(I)}}. \quad (2.19)$$

This approach ensures that subset estimates with smaller uncertainty, as reflected by smaller estimated variances, receive larger weights in the final combined estimate.

2.4.2 Inverse Variance Weighting

The second approach combines the subset estimates by weighting them according to the inverse of their estimated variances. This is similar to the usual fixed-effect meta-analysis approach and gives more weight to more precise subset estimates. The combined estimator is

$$\hat{\beta}_{DC}^{(V)} = \left(\sum_{s=1}^S \widehat{\text{Var}}(\hat{\beta}_s)^{-1} \right)^{-1} \left(\sum_{s=1}^S \widehat{\text{Var}}(\hat{\beta}_s)^{-1} \hat{\beta}_s \right). \quad (2.20)$$

For a single parameter, if $\widehat{\text{Var}}(\hat{\beta}_s) = SE(\hat{\beta}_s)^2$, then this can be written as

$$\hat{\beta}_{DC}^{(V)} = \frac{\sum_{s=1}^S w_s \hat{\beta}_s}{\sum_{s=1}^S w_s}, \quad w_s = \frac{1}{SE(\hat{\beta}_s)^2}. \quad (2.21)$$

Thus, a subset with a smaller standard error receives a larger weight in the combined estimate.

2.4.3 Sample Size Weighting

The third approach uses the subset sample sizes as weights. Let n_s denote the number of observations or, in the clustered setting, the number of clusters assigned to subset s . Then the combined estimator is

$$\hat{\beta}_{DC}^{(n)} = \frac{\sum_{s=1}^S n_s \hat{\beta}_s}{\sum_{s=1}^S n_s}. \quad (2.22)$$

This method is simple and intuitive because larger subsets contribute more to the final estimate. When the subsets are approximately equal in size, this estimator is close to the simple average of the subset-specific estimates.

2.4.4 Variance of the Combined Estimator

For inverse variance weighting, the variance of the combined estimator is estimated by

$$\widehat{\text{Var}}(\hat{\beta}_{DC}^{(V)}) = \left(\sum_{s=1}^S \widehat{\text{Var}}(\hat{\beta}_s)^{-1} \right)^{-1}, \quad \text{Figure 2.4.4: Variance of the combined estimator for inverse variance weighting.}$$

and therefore the combined standard error is

$$SE(\hat{\beta}_{DC}^{(V)}) = \sqrt{\widehat{\text{Var}}(\hat{\beta}_{DC}^{(V)})}. \quad (2.24)$$

Similarly, combined standard errors were obtained for the other weighting approaches and used to compare the performance of the different sample-splitting strategies against the full data analysis.

The purpose of considering these three weighting schemes was to evaluate whether different combination rules lead to estimates close to those obtained from the full dataset. Information matrix weighting reflects the amount of information contributed by each subset, inverse variance weighting reflects the precision of each subset estimate, and sample size weighting reflects the relative size of each subset. Comparing these methods helps assess which approach provides the best agreement with the benchmark full data analysis.

2.5 Evaluation Metrics

To assess the performance of the sample-splitting approach, the combined estimates obtained from the subsets were compared with the corresponding estimates from the full data analysis. Since the full data analysis serves as the benchmark, the main objective was to evaluate how closely the divide-and-combine results reproduced the full-data results under different splitting scenarios, namely 3, 5, and 8 subsets.

Several evaluation metrics were considered.

2.5.1 Difference in Regression Estimates

The first criterion was the difference between the combined regression estimate and the estimate obtained from the full dataset. Let $\hat{\beta}_{full}$ denote the regression estimate from the full data analysis and let $\hat{\beta}_{DC}$ denote the combined estimate from the subset analysis. Then the absolute difference is

$$\text{Absolute Difference} = \left| \hat{\beta}_{DC} - \hat{\beta}_{full} \right|. \quad (2.25)$$

A smaller value indicates that the sample-splitting approach produces results closer to the full data analysis.



2.5.2 Relative Difference

To express the discrepancy on a relative scale, the relative difference was also considered:

$$\text{Relative Difference} = \frac{\left| \hat{\beta}_{DC} - \hat{\beta}_{full} \right|}{\left| \hat{\beta}_{full} \right|}. \quad (2.26)$$

This metric is useful for comparing the magnitude of the error across different models or parameters.

2.5.3 Standard Error Comparison

The precision of the combined estimator was assessed by comparing its standard error with that from the full data analysis. Let $SE(\hat{\beta}_{DC})$ be the standard error of the combined estimate and $SE(\hat{\beta}_{full})$ the standard error from the full data model. Then the difference in standard errors is

$$\Delta SE = |SE(\hat{\beta}_{DC}) - SE(\hat{\beta}_{full})|. \quad (2.27)$$

A smaller difference indicates that the sample-splitting approach preserves the precision of the full-data estimator.

2.5.4 Dependence Parameter Comparison

For the copula model, the dependence structure was evaluated through Kendall's tau. Let $\hat{\tau}_{DC}$ denote the combined Kendall's tau estimate from the subset analyses and $\hat{\tau}_{full}$ the estimate from the full data analysis. Their difference was measured by

$$\text{Difference in Kendall's tau} = |\hat{\tau}_{DC} - \hat{\tau}_{full}|. \quad (2.28)$$

This measure shows whether the sample-splitting approach is able to recover the within-cluster dependence observed in the full dataset.

2.6 Comparison Across Weighting Methods

The above metrics were calculated for each of the three weighting methods: information matrix weighting, inverse variance weighting, and sample size weighting. The best performing method was identified as the one producing estimates and standard errors closest to those from the full data analysis. In this way, the evaluation framework made it possible to assess both the effect of the number of subsets and the effect of the weighting strategy on the final combined results.

Part 2: Quartile-Based Splitting Strategy

2.7 Quartile Construction Based on Herd Size

In order to investigate the impact of cluster size on model estimation, the dataset was partitioned into quartiles based on the number of cows within each herd. Since the observations are clustered at the herd level, it is essential to perform this partitioning using the cluster sizes rather than individual-level observations.

Let n_i denote the number of cows in herd i , for $i = 1, 2, \dots, m$, where m is the total number of herds. The herds were first ordered in decreasing order according to their sizes n_i . Based on this ordering, the herds were divided into four approximately equal groups, referred to as quartiles.

The quartiles are defined as follows:



- **Q1 (Largest Quartile):** Herds with the largest number of cows
- **Q2 (Second-Largest Quartile):** Herds with moderately large sizes
- **Q3 (Third Quartile):** Herds with moderately small sizes
- **Q4 (Smallest Quartile):** Herds with the smallest number of cows

This quartile-based grouping allows for a structured investigation of how cluster size influences both the marginal regression estimates and the dependence structure in multivariate survival models.

By constructing quartiles at the herd level, the natural clustering of the data is preserved, ensuring that all observations belonging to the same herd remain within the same quartile. This is crucial for maintaining the validity of subsequent analyses, particularly when applying cluster-based models such as the Cox model with robust variance and the copula-based Sunclarco model.

The quartile classification serves as the foundation for the subsequent splitting strategies, where selected quartiles are further partitioned into subclusters in order to assess the sensitivity of the model results to changes in cluster structure.

2.7.1 Splitting of Largest Quartile (Q1)

In the first scenario, herds belonging to the largest quartile (Q1) were further partitioned into three approximately equal subclusters. This approach specifically targets large clusters, which are expected to have a strong influence on the estimation of the dependence parameter.

For each herd in Q1, observations were divided such that the resulting subclusters were of approximately equal size. A new cluster identifier was constructed to represent these subclusters, while herds in the remaining quartiles were left unchanged.

2.7.2 Splitting of Second-Largest Quartile (Q2)

In the second scenario, herds belonging to the second-largest quartile (Q2) were divided into two approximately equal subclusters. This represents a less aggressive splitting strategy compared to Q1 and allows the assessment of how moderate cluster sizes affect the results.

As before, a new cluster identifier was defined for the resulting subclusters, ensuring that the clustering structure remains valid for subsequent analysis.

2.8 Analysis Framework

The same modelling framework introduced in Part 1 was applied to both splitting scenarios. Specifically, the standard Cox model, the cluster-robust Cox model, and the copula-based Sunclarco model were fitted using the modified cluster structures.

By keeping the analytical framework unchanged, any differences observed in the results can be attributed to the changes in cluster structure induced by the quartile-based splitting.

In addition, similar to Part 1, the modified Q1 and Q1+Q2 cluster structures were further divided into 3, 5, and 8 random subsets. Models were fitted separately within each subset, after which the subset-specific estimates were combined using information matrix, inverse variance, cluster-number, and cow-number weighting approaches. The combined estimates were then compared with both the corresponding quartile-split full-data analysis and the original full-data benchmark analysis.

Part 3: Computational Cost and Time Evaluation

2.9 Computational Cost Evaluation

Alongside the statistical evaluation of the models, a computational cost assessment was conducted to investigate the efficiency of the proposed sample-splitting strategies. Since copula-based clustered survival models can become computationally demanding for large clustered datasets, evaluating computational performance is important for understanding the practical advantages of subset-based estimation approaches.

Computational cost was assessed for the original full-data analysis, the quartile-based splitting approaches, and the subsequent 3-, 5-, and 8-subset analyses. The same modelling framework was maintained across all analyses so that differences in computational performance could be directly attributed to the modified cluster structures and sample-splitting strategies.

For each analysis, the following computational measures were recorded:

- **User Time:** The amount of processor time spent performing computations directly associated with model estimation in R.
- **System Time:** The amount of time used by the operating system for internal processes related to the analysis, including memory allocation and system-level operations.
- **Elapsed Time:** The elapsed computation time specifically associated with fitting the Sunclarco Clayton copula model. This measure was considered separately because copula estimation is substantially more computationally intensive than standard Cox model estimation.
- **Approximate Memory Usage:** An approximate measure of the memory consumed during model fitting, providing an indication of the computational resources required by each analysis strategy.

The recorded computational measures were compared across the full-data analyses and the subset-based approaches in order to evaluate the trade-off between computational efficiency and statistical performance. Particular attention was given to determining whether increasing the number of subsets reduced computational burden while maintaining results comparable to the full-data benchmark analyses.

Chapter 3

Results



3.1 Data Overview

The full data analysis was conducted using the `insem` dataset from the `Suncclarco` package in R. The dataset contains 10,513 observations and 5 variables, namely `Cowid`, `Time`, `Status`, `Herd`, and `Heifer`. The data are clustered at the herd level, with a total of 181 unique herds, Table 3.1.

The event indicator `Status` showed that 9,939 observations experienced the event, whereas 574 observations were censored. This corresponds to a censoring proportion of 5.46%. The covariate `Heifer` was distributed as follows: 6,313 observations belonged to the reference group (`Heifer` = 0) and 4,200 observations belonged to the heifer group (`Heifer` = 1).

The observed survival times ranged from 0.5 to 329 time units, with a median of 91 and a mean of 111.6. The herd sizes were unbalanced, confirming the presence of clusters of varying size, which supports the use of the Archimedean copula framework.

3.2 Full Data Analysis

3.2.1 Marginal Cox Model

A marginal Cox proportional hazards model was first fitted to evaluate the effect of `Heifer` on the hazard of the event. The estimated regression coefficient for `Heifer` was

$$\hat{\beta} = -0.0603,$$

with standard error

$$SE(\hat{\beta}) = 0.0205.$$

The corresponding hazard ratio was

$$\exp(\hat{\beta}) = 0.9414.$$

This result indicates that the hazard for cows with `Heifer` = 1 is approximately 5.9% lower than that for cows with `Heifer` = 0. The effect was statistically significant ($p = 0.0033$), suggesting evidence of an association between heifer status and the time-to-event outcome.

3.2.2 Cluster-Robust Cox Model

To account for within-herd correlation, a Cox proportional hazards model with cluster-robust standard errors was then fitted. The estimated regression coefficient remained

unchanged,

$$\hat{\beta} = -0.0603,$$

with a robust standard error of

$$SE(\hat{\beta}) = 0.0210.$$

The estimated hazard ratio was again

$$\exp(\hat{\beta}) = 0.9414.$$



Thus, after adjusting for herd-level clustering, the estimated effect of **Heifer** remained very similar to that obtained from the standard marginal Cox model. The corresponding p-value was 0.0040, indicating that the covariate effect remained statistically significant after accounting for clustering.

3.2.3 Sunclarco Copula Model

The full copula-based survival model was fitted using the `SunclarcoModel()` function with a two-stage approach, a Cox marginal survival model, and a Clayton copula. The model converged successfully, with a log-likelihood value of

$$\log L = -40.3241.$$

The fitted model produced the following parameter estimates:

$$\hat{\theta} = 0.4475, \quad SE(\hat{\theta}) = 0.0630,$$

and

$$\hat{\beta}_{\text{Heifer}} = -0.0603, \quad SE(\hat{\beta}_{\text{Heifer}}) = 0.0210.$$

The estimate of Kendall's tau was

$$\hat{\tau} = 0.1828, \quad SE(\hat{\tau}) = 0.0210.$$

The regression estimate for **Heifer** in the copula model was identical to that obtained from the marginal Cox model, which is expected under the two-stage approach with a Cox marginal specification. The negative coefficient again indicates a lower hazard for cows with **Heifer** = 1. In addition, the copula model quantified the within-herd dependence through the association parameter and Kendall's tau. The estimated Kendall's tau of 0.1828 suggests a moderate positive dependence between survival times of cows within

the same herd.

3.2.4 Comparison of Full Data Model Results

Table 3.2 summarizes the results obtained from the marginal Cox model, the cluster-robust Cox model, and the full copula model.

The regression results were highly consistent across the three fitted models. The coefficient estimate for **Heifer** was negative in all cases, and the corresponding hazard ratio remained close to 0.94. This suggests that the effect of **Heifer** on the hazard is stable across modeling approaches. The main additional information provided by the copula model is the estimate of within-cluster dependence, which cannot be obtained from the standard Cox model alone.

3.2.5 Computational Cost of the Full Data Copula Model



Table 3.3 presents the computational summary for the full data copula analysis. Fitting the Clayton copula model to the complete data required 220.95 seconds of user CPU time, 5.86 seconds of system CPU time, and 260.12 seconds of elapsed clock time. This shows that the full copula analysis is computationally intensive. The estimated memory use was approximately 0.1 MB.

3.2.6 Interpretation of the Full Data Analysis

The full data analysis indicates that heifer status is associated with a slightly reduced hazard of the event, and that survival times within the same herd are positively correlated. The standard Cox model and the cluster-robust Cox model gave nearly identical regression estimates, while the Sunclarco copula model additionally captured the herd-level dependence through the Clayton copula parameter and Kendall's tau.

These full-data estimates serve as the benchmark results for the subsequent sample splitting analyses. In the following sections, the same modeling framework has been applied to subsets of the clustered data, and the resulting estimates has been compared with the full-data estimates.

Part 1: Random Sample Splitting

3.3 Three-Subset Analysis

After completing the full data analysis, the clustered dataset was randomly divided at the herd level into three approximately equal subsets. This resulted in 61 herds in subset 1, 60 herds in subset 2, and 60 herds in subset 3. The corresponding numbers of observations were 3664, 3212, and 3637, respectively. The numbers of events were 3442, 3051, and 3446, while the numbers of censored observations were 222, 161, and 191. This confirms that the random splitting produced three reasonably balanced subsets while preserving the clustered herd structure. Table 3.4

3.3.1 Subset-Specific Model Results



The subset-specific regression estimates for the Heifer effect were negative in all three subsets. In subset 1, the estimated coefficient was -0.0192 , corresponding to a hazard ratio of 0.9809. In subset 2, the estimate was -0.0787 , with hazard ratio 0.9244. In subset 3, the estimate was -0.0889 , with hazard ratio 0.9149. The same pattern was observed for the cluster-robust Cox model and the copula model, as expected under the two-stage estimation approach.

Table 3.5, For the dependence structure, the Sunclarco copula model produced Kendall's tau estimates of 0.2048, 0.1748, and 0.1613 for subsets 1, 2, and 3, respectively.

The subset-specific results show some variability across subsets, which is expected after random partitioning. Nevertheless, all three subsets produced negative estimates for the Heifer effect, consistent with the direction of the effect observed in the full data analysis. The dependence estimates were also positive in all subsets, indicating that the within-herd association remained present after splitting the data.

3.3.2 Combined Estimates from the Three Subsets

The subset-specific estimates were then combined For the standard Cox model, the combined estimates were -0.0615 under both information matrix weighting and inverse variance weighting, -0.0620 under cluster-number weighting, and -0.0615 under cow-number weighting. These estimates were all very close to the full-data benchmark estimate of -0.0603 . (Table 3.6).

For the cluster-robust Cox model, the combined estimate was -0.0652 under both information matrix weighting and inverse variance weighting, -0.0620 under cluster-number weighting, and -0.0615 under cow-number weighting. For the copula model, the combined estimate was -0.0652 under information matrix weighting and inverse

variance weighting, -0.0620 under cluster-number weighting, and -0.0615 under cow-number weighting. For Kendall's tau, the combined estimates were 0.1747 , 0.1747 , 0.1804 , and 0.1806 under information matrix, inverse variance, cluster-number, and cow-number weighting, respectively.

3.3.3 Comparison with the Full-Data Analysis

The full-data analysis produced the benchmark estimates $\hat{\beta} = -0.0603$ for the Heifer effect and $\hat{\tau} = 0.1828$ for Kendall's tau. The combined estimates from the three-subset analysis were generally close to these benchmark values.

For the standard Cox model, the absolute difference from the full-data estimate was 0.0012 under information matrix weighting, 0.0012 under inverse variance weighting, 0.0017 under cluster-number weighting, and 0.0012 under cow-number weighting. For the cluster-robust Cox model, the smallest absolute difference was obtained under cow-number weighting (0.0012), followed by cluster-number weighting (0.0017), whereas both information matrix weighting and inverse variance weighting produced a larger deviation of 0.0048 . The copula model showed the same pattern, with the smallest absolute difference under cow-number weighting (0.0012), followed by cluster-number weighting (0.0017), and a larger deviation of 0.0048 under both information matrix weighting and inverse variance weighting.

For Kendall's tau, cow-number weighting again gave the closest agreement with the full-data benchmark, with an absolute difference of 0.0022 , followed by cluster-number weighting with 0.0024 Table 3.7. In contrast, both information matrix weighting and inverse variance weighting produced a larger absolute difference of 0.0081 . Overall, the three-subset analysis showed that cow-number weighting provided the closest approximation to the full-data benchmark, while cluster-number weighting also performed reasonably well.

3.3.4 Standard Error Comparison for the Three-Subset Analysis

Table 3.8 compares the combined standard errors from the three-subset analysis with the corresponding full-data standard errors. Overall, the combined standard errors were very close to the full-data benchmark, indicating that the three-subset sample-splitting procedure preserved the precision of the full-data analysis reasonably well.

For the standard Cox model, all weighting methods produced standard errors almost identical to the full-data standard error. This shows that, for the marginal Cox model, splitting the data into three subsets did not lead to a noticeable loss of precision.

For the cluster-robust Cox model and the copula model, the combined standard er-

rors were slightly smaller than the full-data standard error. The differences were still small, ranging approximately from 0.0007 to 0.0009. Among these methods, cow-number weighting gave the closest agreement with the full-data standard error for the regression coefficient.

For Kendall’s tau, the cluster-number weighting method reproduced the full-data standard error most closely. The combined standard error was approximately 0.0210, which was almost identical to the full-data standard error of 0.0210. In contrast, information matrix and inverse variance weighting gave slightly smaller standard errors.

Overall, the standard error comparison suggests that the three-subset analysis retained most of the statistical precision of the full-data analysis. The results also indicate that cow-number weighting performed best for the regression coefficient, whereas cluster-number weighting performed best for the dependence measure Kendall’s tau.

3.3.5 Computational Cost Analysis

The computational results in Table 3.9 show that the three-subset copula analysis required substantially less time for each separate subset compared with the full data copula analysis. Each subset contained approximately 60 herds and between 3212 and 3664 observations. The elapsed time ranged from 25.64 to 35.00 seconds, with the second subset taking the longest time. This indicates that after splitting the data at the herd level, the copula model becomes easier to fit because each likelihood calculation is performed on a smaller number of clusters and observations. The memory use was approximately 0.3 MB for each subset, suggesting that the main computational difference between the full data and subset analyses is in computation time rather than memory.

3.3.6 Interpretation of the Three-Subset Analysis

The three-subset analysis reproduced the main findings of the full-data analysis very well. Across all combined analyses, the estimated Heifer effect remained negative, indicating that cows with Heifer = 1 had a slightly lower hazard of the event compared to cows in the reference group. The corresponding hazard ratios remained close to those obtained from the full-data analysis, demonstrating that both the direction and magnitude of the covariate effect were well preserved after sample splitting.

The dependence estimates also remained positive, indicating that the herd-level association structure was still successfully captured after dividing the data into subsets. This confirms that the clustering information was not lost under the sample-splitting framework.

In addition to the point estimates, the comparison of standard errors shows that the statistical precision of the full-data analysis was largely retained. For the standard Cox model, all weighting methods produced standard errors that were almost identical to the

full-data benchmark, indicating virtually no loss of efficiency. For the cluster-robust Cox model and the copula model, the combined standard errors were slightly smaller than the full-data standard error, but the differences remained very small (on the order of 10^{-3}), suggesting only minor deviations in precision.

For Kendall's tau, the cluster-number weighting method provided the closest agreement with the full-data standard error, while information matrix and inverse variance weighting slightly underestimated the variability. Nevertheless, these differences were modest and did not materially affect the overall inference.

Among the weighting methods, sample size (cow-number) weighting showed the closest agreement with the full-data benchmark for the copula regression effect and Kendall's tau, while information matrix weighting performed best for the standard Cox and cluster-robust Cox regression estimates. This pattern is consistent for both point estimates and standard errors, reinforcing the robustness of these weighting strategies.

Taken together, these results indicate that random splitting into three subsets can provide combined estimates that closely approximate both the magnitude and precision of the full-data analysis, especially when appropriate weighting methods are applied. Therefore, the three-subset sample-splitting strategy appears to be a reliable and efficient approximation to the full-data analysis in this dataset, while still preserving the key regression and dependence structures observed in the complete clustered survival data.

3.4 Five-Subset Analysis

Following the three-subset analysis, the clustered dataset was repartitioned at the herd level into five approximately equal subsets. This allocation produced 37 herds in Subset 1 and 36 herds in each of Subsets 2 to 5. The corresponding numbers of observations across the five subsets were 2239, 2165, 1928, 2064, and 2117, respectively.

The event counts were 2090, 2084, 1798, 1993, and 1974, while the numbers of censored observations were 149, 81, 130, 71, and 143. Overall, the repartitioning maintained the clustered herd structure and achieved a reasonably balanced distribution of observations and events across the five subsets, as shown in Table 3.10.

3.4.1 Subset-Specific Model Results

For each of the five subsets, the same modelling approach was implemented. The estimated Heifer coefficients varied across the subsets, indicating greater heterogeneity compared with the three-subset analysis. In Subset 1, the estimated coefficient was -0.0678 , corresponding to a hazard ratio of 0.9345. Subset 2 produced a positive estimate of 0.0303 with a hazard ratio of 1.0307, whereas the remaining subsets produced negative estimates of -0.0839 , -0.1114 , and -0.0947 , with hazard ratios of 0.9196, 0.8946, and 0.9097, respectively.

The cluster-robust Cox model and the copula model produced identical regression estimates, while again small differences were observed in the standard errors. The Sunclarco model produced Kendall's τ estimates of 0.2248, 0.1438, 0.2189, 0.1210, and 0.1783 for Subsets 1 to 5, respectively, indicating that positive within-herd dependence remained present across all subsets, Table 3.11.

Compared with the three-subset analysis, the five-subset partitioning resulted in noticeably larger variability between subsets. In particular, the positive Heifer coefficient observed in Subset 2 suggests that smaller subset sizes can introduce greater instability in the regression estimates after repeated random partitioning.

3.4.2 Combined Estimates from the Five Subsets

The subset-specific estimates from the five random subsets were subsequently combined using the four weighting approaches. For the standard Cox model, the combined regression estimates were -0.0636 under both information matrix and inverse variance weighting, -0.0655 under cluster-number weighting, and -0.0645 under cow-number weighting. Although these estimates remained close to the benchmark full-data estimate of -0.0603 , the deviations were slightly larger than those observed in the three-subset analysis.

A similar pattern was observed for the cluster-robust Cox model. The combined estimates were -0.0692 under information matrix and inverse variance weighting, while

cluster-number and cow-number weighting produced estimates of -0.0655 and -0.0645 , respectively.

For the copula model, the combined regression estimates followed the same trend across the weighting schemes, indicating that the different combination methods produced highly comparable results.

The combined dependence estimates also remained relatively stable. Kendall's τ estimates were 0.1446, 0.1446, 0.1776, and 0.1773 under information matrix, inverse variance, cluster-number, and cow-number weighting, respectively, Table 3.12. Overall, the combined analyses preserved both the marginal covariate effect and the within-herd dependence structure reasonably well, despite the slightly increased variability introduced by the five-subset partitioning strategy.

3.4.3 Comparison with the Full-Data Analysis

Compared with full-data benchmark values, the five-subset combined estimates remained reasonably close, although the agreement was weaker than in the three-subset analysis.

For the standard Cox model, the absolute difference from the full-data estimate was 0.0033 under both information matrix and inverse variance weighting, 0.0052 under cluster-number weighting, and 0.0042 under cow-number weighting.

For the cluster-robust Cox model, the smallest absolute difference was obtained under cow-number weighting 0.0042 and cluster-number weighting 0.0052, whereas inverse variance weighting and information matrix produced the largest deviation 0.0088.

For the copula model, a similar pattern was observed. Information matrix and inverse variance weighting both resulted in the largest deviation 0.0088, while cluster-number 0.0052 and cow-number weighting 0.0042 provided estimates closer to the full-data benchmark.

For Kendall's tau, the largest deviations were observed under information matrix and inverse variance weighting 0.0382, indicating substantial loss in accuracy for dependence estimation. In contrast, cluster-number 0.0052 and cow-number weighting 0.0055 showed considerably better agreement with the full-data value.

These results suggest that while the five-subset sample-splitting approach provides reasonable approximations for regression parameters, the choice of weighting method plays a crucial role. In particular, cluster-number and cow-number weighting consistently yielded estimates closer to the full-data benchmark, especially for the dependence parameter, whereas inverse variance weighting was more prone to larger deviations Table 3.13.

3.4.4 Standard Error Comparison for the Five-Subset Analysis

The five-subset analysis shows that the combined standard errors for the Standard Cox model remained extremely close to the full-data standard error of 0.02050. Under information matrix weighting and inverse variance weighting, the combined standard error was 0.02053, giving an absolute difference of only 0.00002 and a relative difference of 0.0012. Cluster-number weighting produced a slightly larger combined standard error of 0.02056, with an absolute difference of 0.00006, whereas cow-number weighting gave 0.02053, with an absolute difference of 0.00003. These results indicate that sample splitting into five subsets had almost no effect on the precision of the Standard Cox estimator.

For the Cluster-Robust Cox and copula models, the differences were more noticeable under information matrix and inverse variance weighting. In both cases, the combined standard error was 0.01954 compared with the full-data standard error of 0.02096, corresponding to an absolute difference of 0.00142 and a relative difference of about 0.0677–0.0678. In contrast, cluster-number weighting improved the agreement substantially, giving a combined standard error of 0.02012 with an absolute difference of 0.00084, while cow-number weighting produced 0.02027 with an absolute difference of 0.00069. Thus, for these two models, cluster-number and cow-number weighting preserved the full-data precision better than the other two weighting methods.

The largest discrepancies were observed for Kendall’s tau. Information matrix and inverse variance weighting both produced a combined standard error of 0.01414, which was considerably smaller than the full-data standard error of 0.02103, resulting in an absolute difference of 0.00689 and a relative difference of 0.3276. By comparison, cluster-number weighting gave a combined standard error of 0.02041, with an absolute difference of 0.00062, and cow-number weighting gave 0.02066, with the smallest absolute difference of 0.00037. Therefore, in the five-subset analysis, the dependence parameter was much more sensitive to the weighting method, and the best preservation of full-data precision was achieved by cow-number weighting, followed closely by cluster-number weighting Table 3.14.

3.4.5 Computational Cost Analysis

The computational results in Table 3.15 show that the five-subset copula analysis further reduced the computational burden compared with both the full data analysis and the three-subset analysis. The elapsed computation time ranged from 8.22 to 13.44 seconds across the subsets, which is substantially smaller than the computation time observed in the three-subset scenario. This reduction occurs because the copula likelihood is evaluated on smaller subsets of clustered observations, making the estimation procedure computationally simpler. The memory usage remained very small, ranging from 0.1 MB to 0.2 MB, again indicating that computation time rather than memory is the main limitation

in fitting the copula model.

3.4.6 Interpretation of the Five-Subset Analysis

The five-subset analysis still reproduced the main direction of the full-data findings, since the combined Heifer effect remained negative under all weighting methods. However, compared with the three-subset analysis, the five-subset results showed greater instability at the subset level and somewhat larger deviations from the full-data benchmark after combination.

The subset-specific variation was more pronounced in the five-subset setting, as shown by the positive regression coefficient in subset 2 and the wider spread of Kendall's tau values across subsets. This suggests that increasing the number of subsets reduces the amount of information available within each subset and may therefore lead to less stable subset-specific estimates.

Among the three weighting methods, information matrix weighting performed best for the standard Cox and cluster-robust Cox regression estimates, while sample size weighting performed best for the copula regression effect and Kendall's tau. In particular, sample size weighting gave the closest agreement with the full-data benchmark for the dependence structure, whereas information matrix and inverse variance weighting substantially underestimated Kendall's tau in the five-subset analysis.

Taken together, these results suggest that random splitting into five subsets still provides a reasonable approximation to the full-data analysis, but with some loss of accuracy compared with the three-subset approach. The regression effect of Heifer remained broadly consistent with the benchmark full-data result, but the dependence structure was recovered less accurately unless sample size weighting was used. Therefore, increasing the number of subsets appears to introduce greater variability and somewhat weaker agreement with the full-data analysis.

3.5 Eight-Subset Analysis

After the five-subset analysis, the clustered dataset was further randomly divided at the herd level into eight approximately equal subsets. This resulted in subsets containing either 22 or 23 herds. The corresponding numbers of observations were 1497, 1261, 1308, 1277, 1124, 1213, 1487, and 1346, respectively. The numbers of events were similarly distributed across subsets, indicating that the random splitting preserved the clustered herd structure while creating relatively small partitions of the original data, Table 3.16.

3.5.1 Subset-Specific Model Results

For each of the eight subsets, the same modelling framework as in the previous analyses was applied.

The subset-specific Heifer effects showed noticeably greater variability than in the three-subset and five-subset analyses. In subset 1, the estimated coefficient was -0.1144 , corresponding to a hazard ratio of 0.8919. In subset 2, the estimate was positive, equal to 0.0254, with hazard ratio 1.0257. In subset 3, the estimate was also positive, equal to 0.0502, with hazard ratio 1.0514. In subsets 4 to 8, the estimated coefficients were -0.0688 , -0.1336 , -0.0823 , -0.0652 , and -0.0998 , respectively, with hazard ratios below one.

As in the previous analyses, the same Heifer estimates were obtained in the cluster-robust Cox model and the copula model, whereas the standard errors differed slightly because of robust variance adjustment and the dependence modelling in the copula framework. For the dependence structure, the Sunclarco model produced Kendall's tau estimates of 0.2268, 0.1840, 0.1549, 0.1777, 0.2228, 0.1245, 0.1103, and 0.2051 for subsets 1 to 8, respectively. These values again indicate positive within-herd dependence in all subsets, although the magnitude of dependence varied substantially across the smaller partitions Table 3.17.

The subset-specific results show the greatest variability among the all splitting strategies considered. In particular, subsets 2 and 3 produced positive Heifer coefficients, whereas the remaining six subsets produced negative estimates. This suggests that when the data are divided into many smaller subsets, the subset-specific estimates become less stable and more sensitive to the random allocation of herds.

3.5.2 Combined Estimates from the Eight Subsets

The subset-specific estimates from the eight-subset analysis were then combined. For the Standard Cox model, the combined estimate was -0.0603 under both information matrix weighting and inverse variance weighting, -0.0613 under cluster-number weighting, and -0.0611 under cow-number weighting. The corresponding hazard ratios were 0.9414,

0.9414, 0.9406, and 0.9407, respectively. These results show that all four weighting methods produced very similar regression estimates, with information matrix and inverse variance weighting being almost identical to the full-data benchmark, while cluster-number and cow-number weighting gave only slightly more negative estimates.

For the Cluster-Robust Cox model, the combined estimate was -0.0695 under both information matrix weighting and inverse variance weighting, -0.0613 under cluster-number weighting, and -0.0611 under cow-number weighting. The corresponding hazard ratios were 0.9329, 0.9329, 0.9406, and 0.9407. This indicates that information matrix and inverse variance weighting produced a somewhat stronger estimated heifer effect, whereas cluster-number and cow-number weighting yielded estimates much closer to the full-data result.

A similar pattern was observed for the copula model. The combined estimate for β_{Heifer} was -0.0695 under information matrix and inverse variance weighting, -0.0613 under cluster-number weighting, and -0.0611 under cow-number weighting. The corresponding hazard ratios were 0.9329, 0.9329, 0.9406, and 0.9407. Thus, for the copula model, the cluster-number and cow-number weighting approaches again produced estimates closer to the results of full-data analysis than the information-based weighting approaches.

For Kendall's tau, the combined estimate was 0.1358 under both information matrix weighting and inverse variance weighting, 0.1760 under cluster-number weighting, and 0.1753 under cow-number weighting. These results suggest that the dependence parameter was more sensitive to the choice of weighting method than the regression parameter. In particular, information matrix and inverse variance weighting substantially underestimated the within-cluster dependence, whereas cluster-number and cow-number weighting gave values much closer to the full-data estimate.

The eight-subset analysis shows that the combined regression estimates remained relatively stable across weighting methods, but the dependence estimate was more strongly affected by the way in which subset results were combined. Among the four weighting methods, cluster-number and cow-number weighting provided the closest agreement with the full-data analysis, particularly for the Cluster-Robust Cox model, the copula model, and Kendall's tau Table 3.18.

3.5.3 Comparison with the Full-Data Analysis

Compared with full-data benchmark values, the eight-subset combined results showed different levels of agreement across the weighting methods.

For the Standard Cox model, both information matrix weighting and inverse variance weighting reproduced the full-data estimate almost exactly, with a combined estimate of -0.0603 and an absolute difference of only 0.0000. Cluster-number weighting gave a combined estimate of -0.0613 , with an absolute difference of 0.0009, while cow-number

weighting gave -0.0611 , with an absolute difference of 0.0008 . Hence, all weighting methods performed well for the Standard Cox model, although information matrix and inverse variance weighting gave the closest agreement.

For the Cluster-Robust Cox model, information matrix weighting and inverse variance weighting both produced a combined estimate of -0.0695 , which differed from the full-data estimate by 0.0091 . In contrast, cluster-number weighting gave -0.0613 and cow-number weighting gave -0.0611 , with absolute differences of 0.0009 and 0.0008 , respectively. Thus, for the Cluster-Robust Cox model, cluster-number and cow-number weighting were clearly closer to the benchmark than the information-based weighting approaches.

For the copula model. Information matrix weighting and inverse variance weighting both produced a combined estimate of -0.0695 , again differing from the full-data estimate by 0.0091 . Cluster-number weighting yielded -0.0613 , with an absolute difference of 0.0009 , while cow-number weighting yielded -0.0611 , with an absolute difference of 0.0008 . Therefore, for the copula regression parameter, the best agreement with the full-data analysis was achieved by cow-number weighting, followed very closely by cluster-number weighting.

For Kendall's tau, the differences between weighting methods were more pronounced. Information matrix weighting and inverse variance weighting both produced a combined estimate of 0.1358 , which was substantially lower than the full-data estimate of 0.1828 and corresponded to an absolute difference of 0.0470 . Cluster-number weighting improved the agreement considerably, with a combined estimate of 0.1760 and an absolute difference of 0.0068 . Cow-number weighting gave 0.1753 , with an absolute difference of 0.0076 . Hence, for the dependence parameter, cluster-number weighting performed best, although cow-number weighting was also much closer to the benchmark than the information-based methods.

The eight-subset analysis indicates that information matrix weighting and inverse variance weighting were most accurate for the Standard Cox model, whereas cluster-number and cow-number weighting performed better for the Cluster-Robust Cox model, the copula model, and Kendall's tau. In particular, cow-number weighting gave the closest agreement for the clustered Cox and copula regression estimates, while cluster-number weighting gave the closest agreement for Kendall's tau. Table 3.19

3.5.4 Standard Error Comparison for the Eight-Subset Analysis

The eight-subset analysis showed that the precision of the combined estimators depended strongly on the weighting method. For the Standard Cox model, all four weighting approaches remained very close to the full-data standard error of 0.02050 . Information matrix weighting and inverse variance weighting both produced a combined standard

error of 0.02052, with an absolute difference of only 0.00002 and a relative difference of 0.0010. Cluster-number weighting gave 0.02062, corresponding to an absolute difference of 0.00012 and a relative difference of 0.0059, while cow-number weighting gave 0.02053, with an absolute difference of 0.00003 and a relative difference of 0.0012. Thus, for the Standard Cox model, the loss of precision from sample splitting into eight subsets was negligible.

For the Cluster-Robust Cox model, the full-data standard error was 0.02096. Information matrix weighting and inverse variance weighting both produced a much smaller combined standard error of 0.01809, giving an absolute difference of 0.00287 and a relative difference of 0.1369. In contrast, cluster-number weighting gave a combined standard error of 0.01989, with an absolute difference of 0.00107 and a relative difference of 0.0511, while cow-number weighting gave 0.01958, with an absolute difference of 0.00138 and a relative difference of 0.0659. Hence, cluster-number weighting preserved the full-data precision best for the Cluster-Robust Cox model, followed by cow-number weighting.

The same numerical pattern was observed for the copula model, since the full-data standard error was again 0.02096. Information matrix weighting and inverse variance weighting both gave a combined standard error of 0.01809, differing from the full-data value by 0.00287, with a relative difference of 0.1369. Cluster-number weighting improved the agreement substantially, yielding a standard error of 0.01989 and an absolute difference of 0.00107, whereas cow-number weighting produced 0.01958 with an absolute difference of 0.00138. Therefore, for the copula regression parameter, cluster-number weighting again gave the closest agreement with the benchmark precision.

The largest discrepancies were observed for Kendall's tau. The full-data standard error was 0.02103, whereas information matrix weighting and inverse variance weighting both produced a much smaller combined standard error of 0.01282. This corresponds to an absolute difference of 0.00821 and a large relative difference of 0.3904. By contrast, cluster-number weighting gave a combined standard error of 0.02117, differing from the full-data standard error by only 0.00014, while cow-number weighting gave 0.02143, with an absolute difference of 0.00040. These results show that the dependence parameter was much more sensitive to aggressive sample splitting than the regression coefficient, but that cluster-number and cow-number weighting still preserved the benchmark precision very well, with cluster-number weighting performing best overall.

Overall, the numerical results indicate that the Standard Cox model remained highly stable across all weighting methods even under eight subsets, whereas the Cluster-Robust Cox, copula, and especially Kendall's tau estimates were more sensitive to the weighting rule. In particular, information matrix weighting and inverse variance weighting tended to underestimate the standard errors for the clustered and dependence parameters, while cluster-number and cow-number weighting produced values much closer to the full-data benchmark Table 3.20.

3.5.5 Computational Cost Analysis, Eight Subset

The computational results in Table 3.21 show that the eight-subset copula analysis achieved the lowest computational burden among all sample-splitting scenarios. Each subset contained approximately 22 to 23 herds and between 1124 and 1497 observations. The elapsed computation times ranged from 2.87 to 8.23 seconds, which is substantially smaller than both the three-subset and five-subset analyses, as well as the full data analysis. This demonstrates that increasing the number of subsets considerably reduces the computational complexity of fitting the copula model because the likelihood calculations are performed on much smaller clustered datasets. The approximate memory usage remained constant at 0.1 MB across all subsets, indicating that the primary computational improvement obtained through sample splitting is reduced execution time rather than reduced memory consumption.

3.5.6 Interpretation of the Eight-Subset Analysis

The eight-subset analysis still preserved the general direction of the full-data Heifer effect after combination, since all combined estimates remained negative. However, the subset-specific variability was clearly larger than in the three-subset and five-subset analyses, reflecting the smaller amount of information available in each partition.

For the standard Cox and cluster-robust Cox models, information matrix weighting produced combined estimates that were almost identical to the full-data benchmark. In contrast, for the copula regression effect and Kendall's tau, sample size weighting gave the best agreement with the benchmark full-data results. Information matrix weighting and inverse variance weighting performed less well for the dependence-related quantities, especially for Kendall's tau, where the combined estimate was substantially lower than the full-data value.

These findings suggest that dividing the data into eight subsets  increases instability at the subset level but does not necessarily worsen the combined regression estimate when an appropriate weighting method is used. In particular, for the standard Cox and cluster-robust Cox models, the combined regression estimates remained extremely close to the full-data benchmark. For the copula model and the dependence measure, sample size weighting again appeared to be the most suitable combination strategy.

Taken together, the eight-subset results indicate that sample splitting into many smaller subsets is still feasible, but it leads to greater variability in the subset-specific analyses and less reliable recovery of the dependence structure unless a suitable weighting rule is chosen. Compared with the three-subset and five-subset analyses, the eight-subset approach appears to maintain the regression effect reasonably well but to recover the within-herd dependence less accurately under information matrix or inverse variance weighting.

Part 2: Quartile-Based Splitting Strategy

3.6 Data Description After Quartile-Based Splitting

In the part 2 of Data analysis, after constructing quartiles based on herd size, the largest quartile (Q1) was further divided into three approximately equal subclusters for each herd. This modification resulted in a change in the clustering structure.

The original dataset consists of 10,513 observations distributed across 181 herds. After splitting the largest quartile, the number of clusters increased to 273, reflecting the division of 46 herds in Q1 into three subclusters each.

Table 3.22 summarises the distribution of observations across quartiles after the splitting procedure.

The results show that the splitting procedure only affects the largest quartile, where each herd is divided into three subclusters, resulting in a total of 138 clusters in Q1. The remaining quartiles retain their original clustering structure.

Importantly, the overall characteristics of the dataset remain unchanged. The total number of observations is preserved at 10,513, with 9,939 events and 574 censored observations, corresponding to a censoring proportion of 5.46%. The distribution of the covariate Heifer also remains unchanged.

3.7 Random Sample Splitting After Q1 Splitting

After dividing the largest quartile (Q1) into three subclusters per herd, the modified dataset containing 273 clusters was further randomly divided into 3, 5, and 8 subsets. The splitting was performed at the cluster level so that all observations belonging to the same cluster remained within the same subset. This approach preserved the within-cluster dependence structure required for the copula analysis.

For each splitting scenario, the subsets were constructed to be as balanced as possible with respect to the number of clusters. The corresponding distribution of observations and clusters across the subsets is summarized in Table 3.23.

The results show that the random splitting procedure produced relatively balanced subsets within each scenario. In the three-subset analysis, each subset contained exactly 91 clusters and approximately one-third of the observations. Similarly, the five-subset and eight-subset analyses produced nearly equal distributions of clusters and observations across subsets. As the number of subsets increased, the number of observations and clusters within each subset decreased, which was expected to reduce the computational burden of fitting the copula model.

3.7.1 Results of Quartile-Based Splitting (Q1)

The models were re-estimated after splitting the largest quartile (Q1) into three subclusters per herd. The results are summarised in Table 3.24 and compared with the full data analysis.

The estimated regression coefficient for the covariate *Heifer* remains unchanged across all models, with $\hat{\beta} = -0.0603$, indicating that the marginal effect is robust to modifications in the cluster structure. Similarly, the standard error from the standard Cox model remains virtually identical, while a slight reduction is observed in the cluster-robust standard error after splitting.

For the copula-based model, the dependence parameter is estimated as $\hat{\theta} = 0.4433$ with a standard error of 0.0470, compared to a slightly higher value in the full data analysis. A similar pattern is observed for Kendall's tau, where $\hat{\tau} = 0.1814$ with a reduced standard error.

In the end, the results suggest that while the regression effect remains stable, the dependence parameter is sensitive to the cluster structure. The reduction in both θ and its standard error indicates that part of the observed dependence in the full data model may be driven by the presence of large clusters.

3.7.2 Computational Cost, Q1-Based Splitting

The computational results in Table 3.25 correspond to the initial Q1 full analysis performed without random sample splitting. In this analysis, the largest herd quartile (Q1) was divided into smaller subclusters before fitting the copula model, resulting in a total of 273 clusters while retaining all 10,513 observations. Compared with the original full data analysis, the computational burden increased substantially, with an elapsed computation time of 371.72 seconds and memory usage of 4.9 MB. This increase occurs because splitting the largest herds into additional subclusters increases the number of clusters that must be incorporated into the copula likelihood calculation. Consequently, although the total number of observations remains unchanged, the larger number of clusters makes the dependence structure more computationally intensive to estimate.

3.7.3 Interpretation

The results show that the regression coefficient remains virtually identical across all models, indicating that the marginal effect of the covariate is not influenced by the splitting of large clusters.

In contrast, the dependence parameter reflects the within-cluster association structure and is sensitive to changes in cluster definition. By splitting the largest clusters into smaller subclusters, the estimated dependence is slightly reduced compared to the full

data model, suggesting that part of the observed dependence may be driven by large cluster sizes. Overall, these findings indicate that while marginal effects remain stable, the dependence structure in copula-based models is influenced by the size and composition of clusters.

3.8 Three-Subset Analysis After Q1 Splitting

3.8.1 Subset-Specific Model Results

The subset-specific results show that the estimated regression coefficients remained negative across all three subsets, indicating a consistent marginal covariate effect after Q1 splitting. The estimated coefficients ranged from -0.0851 in Subset 1 to -0.0451 in Subset 2, while Subset 3 produced an intermediate estimate of -0.0514 . The corresponding hazard ratios varied between 0.9184 and 0.9559 , showing that the estimated hazard for the heifer group remained lower than that of the reference group in all subsets. Standard errors were relatively similar across subsets, ranging from 0.0336 to 0.0367 , indicating a stable level of estimation precision after splitting the data Table 3.26 .

The estimated within-cluster dependence also remained consistent across subsets. Kendall's tau estimates ranged from 0.1725 in Subset 2 to 0.1914 in Subset 1, with Subset 3 producing a value of 0.1838 . The corresponding standard errors ranged between 0.0242 and 0.0290 , suggesting only moderate variability in the estimated dependence structure across the three subsets.

3.8.2 Combined Estimates from the Three Subsets

The combined estimates remained very close across all weighting methods, with regression estimates ranging from -0.0065 to -0.0069 and hazard ratios between 0.9409 and 0.9413 . The corresponding standard errors were also highly consistent, varying only between 0.0209 and 0.0212 .

Similarly, the combined Kendall's tau estimates ranged from 0.1821 to 0.1826 , with standard errors between 0.0210 and 0.0214 (Table 3.27). Overall, the results indicate that the three-subset Q1 splitting approach preserved both the marginal covariate effect and the within-cluster dependence structure very well, with only minimal loss of statistical precision compared with the benchmark analysis.

3.8.3 Comparison with the Q1-Split Full-Data Analysis

Table 3.28 compares the combined three-subset estimates after Q1 splitting with the corresponding Q1-split full-data benchmark.

Overall, the combined estimates remained very close to the Q1-split full-data results. For the regression coefficient, the absolute differences ranged from 0.0002 to 0.0007, with relative differences below 1.1% across all weighting methods. For Kendall's τ , the combined estimates ranged from 0.1821 to 0.1826, compared with the Q1-split full-data estimate of 0.1814. This indicates that both the marginal Heifer effect and the within-herd dependence structure were well preserved after the three-subset Q1 splitting procedure.

3.8.4 Standard Error Comparison with the Q1-Split Full-Data Analysis

Table 3.29 compares the combined standard errors from the three-subset Q1-splitting analysis with the corresponding Q1-split full-data standard errors.

The standard errors remained close to the Q1-split full-data benchmark. For the Standard Cox model, the combined standard errors ranged from 0.0209 to 0.0210, compared with the Q1-split full-data value of 0.0205. For the Cluster-Robust Cox and Copula models, the combined standard errors ranged from 0.0211 to 0.0212, while the Q1-split full-data benchmark was 0.0200. For Kendall's τ , the combined standard errors ranged from 0.0210 to 0.0214, compared with the Q1-split full-data value of 0.0157. This shows that the three-subset analysis preserved the general precision pattern, although the dependence standard errors were somewhat larger than in the Q1-split full-data analysis.

3.8.5 Computational Cost of the Three-Subset Analysis

The computational results Table 3.30 show that the three-subset analysis substantially reduced the computational burden compared with the Q1-split full-data analysis. Although the total number of observations remained unchanged, fitting the copula model separately on smaller subsets reduced the complexity of the likelihood estimation. The elapsed computation times ranged from approximately 40 to 43 seconds per subset, while memory usage remained low between 0.3 MB and 0.4 MB. This indicates that random sample splitting can considerably improve computational efficiency while maintaining stable model estimates.

3.9 Five-Subset Analysis After Q1 Splitting

3.9.1 Subset-Specific Model Results

The Q1-split dataset was further divided into five random subsets at the cluster level. Each subset contained approximately 54 to 55 clusters. Table 3.31 presents the subset-specific model results.

The subset-specific estimates remained relatively consistent across all five subsets after Q1 splitting. The regression coefficients ranged from -0.0888 to -0.0384 , with hazard ratios between 0.9150 and 0.9624 , while standard errors varied from 0.0412 to 0.0478 .

Similarly, Kendall's τ estimates ranged from 0.1454 to 0.2155 , with standard errors between 0.0288 and 0.0387 , indicating that the within-cluster dependence structure remained present across all subsets. .

3.9.2 Combined Estimates from the Five Subsets

The subset-specific estimates were combined using information matrix weighting, inverse variance weighting, cluster-number weighting, and cow-number weighting. The combined results are shown in Table 3.32. The regression estimates ranged from -0.0604 to -0.0612 , with hazard ratios between 0.9406 and 0.9414 , while the standard errors varied only slightly from 0.0197 to 0.0205 .

Similarly, the combined Kendall's τ estimates ranged from 0.1773 to 0.1807 , with standard errors between 0.0156 and 0.0159 . Overall, the results indicate that the five-subset Q1 splitting approach preserved both the marginal covariate effect and the within-cluster dependence structure reasonably well.

3.9.3 Comparison with the Q1-Split Full-Data Analysis

Overall, Table 3.33 the five-subset combined estimates remained close to the Q1-split full-data benchmark. The absolute differences for the regression estimates ranged from 0.0001 to 0.0009 , while the relative differences remained below 1.5% across all weighting methods. For Kendall's τ , the absolute differences ranged from 0.0007 to 0.0041 , with the smallest deviation observed under cluster-number weighting. These results indicate that the five-subset splitting strategy maintained strong agreement with the benchmark full-data analysis.

3.9.4 Standard Error Comparison for the Five-Subset Analysis

Table 3.34 compares the combined standard errors with the corresponding Q1-split full-data standard errors.

For the Standard Cox model, the combined standard errors ranged from 0.0205 to 0.0205 , compared with the benchmark value of 0.0210 , resulting in absolute differences of 0.0005 . For the Cluster-Robust Cox and Copula models, the combined standard errors ranged from 0.0197 to 0.0199 , with relative differences between 6.1% and 7.1% .

For Kendall's τ , the combined standard errors ranged from 0.0156 to 0.0159 , compared with the benchmark value of 0.0157 .

3.9.5 Computational Cost of the Five-Subset Analysis

The computational results, Table 3.35, show that the five-subset analysis further reduced the computation time compared with the three-subset analysis and the Q1-split full-data analysis. Each subset required approximately 15 to 18 seconds of elapsed time, with low memory usage. This confirms that increasing the number of random subsets reduces the computational cost of fitting the copula model while still producing stable combined estimates.

3.10 Eight-Subset Analysis After Q1 Splitting

3.10.1 Subset-Specific Model Results

The Q1-split dataset was further divided into eight random subsets at the cluster level. Each subset contained 34 to 35 clusters. Table 3.36 presents the subset-specific model results.

The subset-specific estimates showed greater variability than in the three- and five-subset analyses. The regression coefficients ranged from -0.1316 to -0.0050 , with hazard ratios between 0.8767 and 0.9950, while standard errors varied from 0.0507 to 0.0676.

Similarly, Kendall's τ estimates ranged from 0.1394 to 0.2551, with standard errors between 0.0366 and 0.0527. Despite the increased variability, the regression estimates remained negative and Kendall's τ remained positive across all subsets, indicating preservation of both the marginal covariate effect and the within-cluster dependence structure after eight-subset Q1 splitting.



3.10.2 Combined Estimates from the Eight Subsets

Table 3.37 presents the combined estimates from the eight-subset analysis after Q1 splitting. The combined estimates from the eight-subset analysis remained reasonably close to the Q1-split full-data results across all weighting methods. The regression estimates ranged from -0.0567 to -0.0587 , with hazard ratios between 0.9433 and 0.9449, while standard errors varied from 0.0193 to 0.0206.

Similarly, the combined Kendall's τ estimates ranged from 0.1767 to 0.1778, with standard errors between 0.0156 and 0.0159. Although the regression estimates remained relatively stable, the dependence estimates were slightly lower than the benchmark full-data Kendall's τ , indicating some loss of stability when the data were divided into a larger number of smaller subsets.

3.10.3 Comparison with the Q1-Split Full-Data Analysis

Overall, Table 3.38, the eight-subset analysis preserved the main direction of the regression effect, but the deviations from the Q1-split full-data benchmark were larger than those observed in the three- and five-subset analyses. The absolute differences for the regression estimates ranged from 0.0019 to 0.0036, with relative differences between 3.1% and 5.7%.

For Kendall's τ , the absolute differences ranged from 0.0037 to 0.0047, while the relative differences varied between 2.0% and 2.6%. Among the weighting methods, cluster-number weighting produced the closest agreement with the benchmark analysis for both the regression coefficient and Kendall's τ .

3.10.4 Standard Error Comparison for the Eight-Subset Analysis

The combined standard errors, Table 3.39, from the eight-subset analysis remained close to the Q1-split full-data benchmark. For the Standard Cox model, the combined standard errors ranged from 0.0205 to 0.0206, compared with the benchmark value of 0.0210, resulting in relative differences between 2.4% and 2.8%.

For the Cluster-Robust Cox and Copula models, the combined standard errors ranged from 0.0193 to 0.0196, with relative differences between 7.6% and 8.9%. Similarly, Kendall's τ standard errors ranged from 0.0156 to 0.0159, compared with the benchmark value of 0.0157. Overall, the differences remained small, although slightly larger than in the three- and five-subset analyses, indicating some loss of precision with increased splitting.

3.10.5 Computational Cost of the Eight-Subset Analysis

The computational results, Table 3.40, show that the eight-subset analysis produced the largest reduction in computation time. Each subset required only about 6 to 7 seconds of elapsed time, with memory usage of approximately 0.1 MB. This confirms that increasing the number of subsets improves computational efficiency, although it may also increase variability in the subset-specific model estimates.

3.11 Data Description After Combined Q1 and Q2 Splitting

In the combined splitting scenario, the largest quartile (Q1) was divided into three approximately equal subclusters, while the second-largest quartile (Q2) was divided into

two approximately equal subclusters. The remaining quartiles, Q3 and Q4, were left unchanged.

The original dataset contains 10,513 observations distributed across 181 herds. After applying the combined Q1 and Q2 splitting strategy, the total number of clusters increased to 318. This increase reflects the division of 46 Q1 herds into three subclusters each and 45 Q2 herds into two subclusters each.

Table 3.41 summarises the distribution of observations across quartiles after the combined splitting procedure.

The results show that the combined splitting procedure modifies only the two largest quartiles. In Q1, the number of clusters increased from 46 to 138, while in Q2 the number of clusters increased from 45 to 90. The cluster structures of Q3 and Q4 remained unchanged.

The balance check showed that, for both Q1 and Q2, the difference between the smallest and largest subcluster within each original herd was at most one observation. This confirms that the subclusters were created in an approximately equal manner.

After the combined splitting, the median cluster size was 32 observations, with an interquartile range from 28 to 37 observations. The mean cluster size was 33.1. This shows that the combined splitting strategy substantially reduced the dominance of large clusters compared with the original clustering structure.

Importantly, the total number of observations remained unchanged at 10,513, with 9,939 events and 574 censored observations, corresponding to an overall censoring proportion of 5.46%. Therefore, the combined Q1 and Q2 splitting strategy changes only the clustering structure, while preserving the original survival outcomes and covariate distribution.

3.11.1 Results of Combined Q1 and Q2 Splitting

After splitting the largest quartile (Q1) into three subclusters and the second-largest quartile (Q2) into two subclusters, the models were re-estimated using the updated cluster structure. The regression coefficient for *Heifer* remained unchanged at $\hat{\beta} = -0.0603$, with a hazard ratio of 0.9414. This indicates that the marginal effect of *Heifer* is stable despite the modification of the cluster structure.

The standard Cox standard error remained 0.0205, while the cluster-robust standard error decreased to 0.0196. In the Sunclarco model, the estimated dependence parameter was $\hat{\theta} = 0.4429$ with standard error 0.0439. The corresponding Kendall's tau was $\hat{\tau} = 0.1813$ with standard error 0.0147.

The combined Q1 and Q2 splitting produced the same marginal regression estimate as the full data analysis, while slightly reducing the estimated dependence parameter and its standard error. This suggests that the marginal covariate effect is robust, whereas the

dependence estimates are more sensitive to the definition and size of clusters.

The results demonstrate that the regression coefficient remains identical across all scenarios, indicating robustness of the marginal effect. In contrast, the dependence parameter θ decreases as the cluster structure is progressively refined. The reduction is more pronounced when both Q1 and Q2 are split, compared to splitting Q1 alone. A similar pattern is observed for Kendall's τ and their standard errors, suggesting that larger clusters contribute to higher estimated dependence. These findings highlight that while marginal effects are stable, dependence estimates are sensitive to the underlying cluster structure.

3.11.2 Computational Cost of the Combined Q1 and Q2 Splitting Analysis

The computational results show that the combined Q1 and Q2 splitting strategy substantially increased the computational burden compared with both the original full-data analysis and the Q1-only splitting analysis. Table 3.44.

The elapsed computation time increased to approximately 503 seconds, with memory usage of 4.9 MB. This increase occurs because the copula likelihood must account for a larger number of clusters and a more complex dependence structure. These findings indicate that although splitting large clusters may help stabilize the dependence structure, increasing the number of clusters considerably increases the computational complexity of the copula estimation procedure.

3.12 Three-Subset Analysis After Q1 and Q2 Splitting

3.12.1 Data Description for the Three-Subset Analysis

After splitting Q1 into three subclusters and Q2 into two subclusters, the modified dataset was randomly divided into three approximately equal subsets at the cluster level. The splitting procedure was performed using the updated cluster identifiers to ensure that observations from the same modified cluster remained within the same subset. This preserved the within-cluster dependence structure required for the copula analysis.

Table 3.45 summarizes the data structure of the three subsets. Each subset contained approximately one-third of the total observations and modified clusters. The distribution of cows and clusters across subsets remained relatively balanced, allowing a fair comparison between the subset analyses and the Q1+Q2 full-data benchmark.

The number of events and censored observations was also well distributed across the subsets. Approximately 5.45% of the observations were censored overall, which was con-

sistent with the censoring proportion observed in the original dataset. This indicates that the random splitting procedure preserved the general survival and censoring characteristics of the data.

3.12.2 Subset-Specific Model Results

After splitting Q1 into three subclusters and Q2 into two subclusters, the updated dataset was randomly divided into three subsets at the cluster level. Each subset contained 106 clusters. Table 3.46 presents the subset-specific model results.

The subset-specific estimates remained relatively consistent after splitting Q1 into three subclusters and Q2 into two subclusters. The regression coefficients ranged from -0.0716 to -0.0450 , with hazard ratios between 0.9309 and 0.9560 , while standard errors varied from 0.0298 to 0.0420 .

Similarly, Kendall's τ estimates ranged from 0.1659 to 0.1995 , with standard errors between 0.0242 and 0.0343 , indicating that the within-cluster dependence structure remained present across all subsets after the combined Q1 and Q2 splitting strategy.

3.12.3 Combined Estimates from the Three Subsets

Table 3.47 presents the combined estimates. The combined estimates from the three-subset Q1+Q2 analysis remained highly consistent across all weighting methods. The regression estimates ranged from -0.0585 to -0.0611 , with hazard ratios between 0.9407 and 0.9432 , while standard errors varied only slightly from 0.0189 to 0.0206 .

Similarly, the combined Kendall's τ estimates ranged from 0.1800 to 0.1814 , with standard errors between 0.0148 and 0.0149 . The results indicate that the Q1+Q2 splitting strategy preserved both the marginal covariate effect and the within-cluster dependence structure very well.

3.12.4 Comparison with the Q1+Q2 Full-Data Analysis

The three-subset analysis after Q1 and Q2 splitting produced estimates that remained close to the Q1+Q2 full-data benchmark. The absolute differences for the regression estimates ranged from 0.0008 to 0.0016 , with relative differences between 1.3% and 2.9% . Table 3.48

For Kendall's τ , the absolute differences ranged from 0.0001 to 0.0013 , while the relative differences remained below 1% across all weighting methods. Information matrix and inverse variance weighting produced the closest agreement for Kendall's τ , whereas the regression estimates remained highly stable under all weighting approaches.

3.12.5 Standard Error Comparison for the Three-Subset Analysis

The combined standard errors from the three-subset Q1+Q2 analysis, Table 3.49, the Standard Cox model, the combined standard errors were identical to the benchmark value of 0.0205 across all weighting methods, resulting in zero absolute and relative differences.

For the Cluster-Robust Cox and Copula models, the combined standard errors ranged from 0.0189 to 0.0199, compared with the benchmark value of 0.0196, with relative differences between 1.5% and 3.6%. Similarly, Kendall's τ standard errors ranged from 0.0148 to 0.0149, compared with the benchmark value of 0.0147.

3.12.6 Computational Cost of the Three-Subset Analysis

The computational results show that the three-subset analysis reduced the elapsed computation time compared with the Q1+Q2 full-data analysis. The full Q1+Q2 analysis required approximately 503 seconds, whereas the three subsets together required about 157 seconds. Thus, random sample splitting substantially reduced computation time while maintaining similar combined estimates. Table 3.50

3.13 Five-Subset Analysis After Q1 and Q2 Splitting

3.13.1 Data Description for the Five-Subset Analysis

After applying the combined Q1 and Q2 splitting strategy, the updated dataset was randomly divided into five subsets at the modified-cluster level. This ensured that observations belonging to the same modified cluster remained together within the same subset. The five subsets were reasonably balanced in terms of observations and clusters Table 3.51.

3.13.2 Subset-Specific Model Results

Table 3.52 The subset-specific estimates remained relatively consistent across all five subsets after the Q1+Q2 splitting strategy. The regression coefficients ranged from -0.0885 to -0.0120 , with hazard ratios between 0.9153 and 0.9881, while standard errors varied from 0.0370 to 0.0584.

Similarly, Kendall's τ estimates ranged from 0.1494 to 0.2003, with standard errors between 0.0274 and 0.0378, indicating that the within-cluster dependence structure remained present across all subsets after the Q1+Q2 splitting procedure.

3.13.3 Combined Estimates from the Five Subsets

The five-subset Q1+Q2 analysis remained close across all weighting methods. The regression estimates ranged from -0.0582 to -0.0614 , with hazard ratios between 0.9405 and 0.9435, while standard errors varied from 0.0186 to 0.0205. Moreover, the combined Kendall's τ estimates ranged from 0.1769 to 0.1786, with standard errors between 0.0144 and 0.0149. Overall, the results indicate that the five-subset Q1+Q2 splitting strategy preserved both the marginal covariate effect and the within-cluster dependence structure reasonably well, although the dependence estimates were slightly lower than the benchmark full-data values. Table 3.53.

3.13.4 Comparison with the Q1+Q2 Full-Data Analysis

The five-subset Q1+Q2 analysis produced combined estimates that remained close to the corresponding full-data benchmark. The absolute differences for the regression estimates ranged from 0.0001 to 0.0021, with relative differences between 0.2% and 3.5%.

For Kendall's τ , the absolute differences ranged from 0.0027 to 0.0044, while the relative differences varied between 1.5% and 2.4%. Among the weighting approaches, cluster-number weighting produced the closest agreement with the benchmark analysis for both the regression coefficient and Kendall's τ . Table 3.54.

3.13.5 Standard Error Comparison for the Five-Subset Analysis

For the Standard Cox model, the combined standard errors were identical to the benchmark value of 0.0205 across all weighting methods, resulting in zero absolute and relative differences.

For the Cluster-Robust Cox and Copula models, the combined standard errors ranged from 0.0186 to 0.0195, compared with the benchmark value of 0.0196, with relative differences between 0.5% and 5.1%. Similarly, Kendall's τ standard errors ranged from 0.0144 to 0.0149, compared with the benchmark value of 0.0147. Overall, the differences remained small, indicating that the five-subset Q1+Q2 splitting strategy preserved most of the statistical precision of the benchmark analysis. Table 3.55.

3.13.6 Computational Cost of the Five-Subset Analysis

The five-subset analysis further reduced the computational burden compared with the three-subset analysis and the Q1+Q2 full-data analysis. Each subset required approximately 16 to 19 seconds of elapsed time, while memory usage remained low. This suggests that increasing the number of subsets improved computational efficiency while maintaining reasonably stable combined estimates. Table 3.56.

3.14 Eight-Subset Analysis After Q1 and Q2 Splitting

3.14.1 Data Description for the Eight-Subset Analysis

After applying the combined Q1 and Q2 splitting strategy, the modified dataset was randomly divided into eight subsets at the cluster level. Each subset contained approximately 39 to 40 modified clusters. The subsets were reasonably balanced with respect to the number of observations and clusters Table 3.57.

The distribution of observations and modified clusters remained balanced across the eight subsets. The censoring structure and overall characteristics of the data were same after the random splitting procedure.

3.14.2 Subset-Specific Model Results

The subset-specific estimates showed greater variability than in the three- and five-subset analyses after the Q1+Q2 splitting procedure. The regression coefficients ranged from -0.0961 to -0.0124 , with hazard ratios between 0.9084 and 0.9877, while standard errors varied from 0.0499 to 0.0639.

The Kendall's τ estimates ranged from 0.1524 to 0.2113, with standard errors between 0.0388 and 0.0502. Despite the increased variability, the regression estimates remained negative and Kendall's τ remained positive across all subsets, indicating preservation of both the marginal covariate effect and the within-cluster dependence structure after eight-subset Q1+Q2 splitting. Table 3.58.

3.14.3 Combined Estimates from the Eight Subsets

The combined estimates from the eight-subset Q1+Q2 analysis remained reasonably close across all weighting methods. The regression estimates ranged from -0.0560 to -0.0591 , with hazard ratios between 0.9426 and 0.9455, while standard errors varied from 0.0181 to 0.0205. The combined Kendall's τ estimates ranged from 0.1761 to 0.1776, with standard errors between 0.0140 and 0.0145. Although the regression estimates remained relatively stable, the dependence estimates were slightly smaller than the benchmark full-data values, indicating some loss of stability with increased splitting. Table 3.59.

3.14.4 Comparison with the Q1+Q2 Full-Data Analysis

Table 3.60 Overall, the eight-subset Q1+Q2 analysis preserved the direction of the regression effect and the within-cluster dependence structure, although the deviations from the

full-data benchmark were slightly larger than in the smaller splitting scenarios. The absolute differences for the regression estimates ranged from 0.0015 to 0.0043, with relative differences between 2.4% and 7.1%.

For Kendall's τ , the absolute differences ranged from 0.0037 to 0.0052, while the relative differences varied between 2.0% and 2.9%. Among the weighting methods, cluster-number weighting produced the closest agreement with the benchmark analysis for both the regression coefficient and Kendall's τ .

3.14.5 Standard Error Comparison for the Eight-Subset Analysis

The combined standard errors from the eight-subset Q1+Q2 analysis remained close to the corresponding full-data benchmark. For the Standard Cox model, the combined standard errors were identical to the benchmark value of 0.0205 across all weighting methods, resulting in zero absolute and relative differences.

For the Cluster-Robust Cox and Copula models, the combined standard errors ranged from 0.0181 to 0.0192, compared with the benchmark value of 0.0196, with relative differences between 2.0% and 7.1%. Similarly, Kendall's τ standard errors ranged from 0.0140 to 0.0145, compared with the benchmark value of 0.0147. Overall, the standard errors remained close to the benchmark values, although variability increased slightly compared with the three- and five-subset analyses. Table 3.61.

3.14.6 Computational Cost of the Eight-Subset Analysis

The eight-subset analysis produced the lowest computational burden among the Q1+Q2 splitting scenarios. Each subset required approximately 8 to 9 seconds of elapsed computation time with very low memory usage. This confirms that increasing the number of subsets substantially improves computational efficiency, although it may also slightly increase variability in the subset-specific estimates Table 3.62.

3.15 Overall Summary of Full Data and Sample-Splitting Results

3.15.1 Heifer Estimate Comparison

Table 3.63 summarizes the estimated Heifer effect across the original full data, random sample-splitting analyses, Q1-split analyses, and combined Q1+Q2-split analyses.

The estimated Heifer effect remained negative across all analyses, indicating a consistent direction of association. The three- and five-subset results were generally very close to the corresponding full-data benchmarks, while the eight-subset analyses showed slightly larger deviation, especially after Q1 and Q1+Q2 splitting.

3.15.2 Standard Error (Heifer) Comparison

The standard errors were broadly comparable across the full-data and sample-splitting analyses Table 3.64. In general, the combined standard errors became slightly smaller as the number of subsets increased, although this should be interpreted carefully because smaller subsets can show greater subset-level variability.

3.15.3 Kendall's τ Comparison

Table 3.65 summarizes the overall comparison of the combined Kendall's τ estimates across the original full-data analysis, Q1 splitting, and Q1+Q2 splitting scenarios. Overall, the estimated dependence remained relatively stable across all analyses, indicating that the within-cluster association structure was generally preserved after sample splitting.

For the original random splitting analysis, the Kendall's τ estimate decreased slightly from 0.1828 in the full-data analysis to 0.1826, 0.1807, and 0.1777 for the three-, five-, and eight-subset analyses, respectively. A similar pattern was observed for the Q1 splitting strategy, where the dependence estimates remained close to the benchmark full-data value but gradually decreased as the number of subsets increased.

For the Q1+Q2 splitting strategy, the full-data benchmark Kendall's τ estimate was 0.1813. The three-subset analysis produced a nearly identical estimate of 0.1814, whereas the five- and eight-subset analyses produced slightly smaller estimates of 0.1786 and 0.1776. These results suggest that smaller numbers of subsets preserved the dependence structure more effectively, while larger numbers of subsets introduced slightly greater variability in the estimated within-cluster association.

3.15.4 Standard Error τ Comparison

Table 3.66 presents the corresponding standard errors for Kendall's τ across the different analyses. The standard errors remained broadly comparable across the full-data and sample-splitting approaches, indicating that the statistical precision of the dependence estimates was largely maintained after splitting.

For the original full-data and Q1 full-data analyses, the benchmark standard error was approximately 0.0210. The three-subset analyses produced nearly identical values, while the five- and eight-subset analyses yielded slightly smaller standard errors around 0.0159. Similarly, for the Q1+Q2 analyses, the benchmark standard error was 0.0147, and the subset analyses produced values ranging from 0.0145 to 0.0149.

These findings indicate that the sample-splitting procedures preserved both the magnitude and precision of the dependence estimates reasonably well. However, as the number of subsets increased, the subset-specific variability became larger, particularly in the eight-subset analyses, suggesting a small loss of stability in the estimation of within-cluster dependence.

3.15.5 Computational Cost Comparison

The computational cost comparison shows a clear reduction in copula fitting time after random sample splitting. The Q1+Q2 full-data analysis required the largest computation time because it had the highest number of modified clusters Table 3.67. In contrast, the eight-subset analyses had the lowest total copula time, showing that sample splitting can substantially improve computational efficiency while keeping the combined estimates reasonably close to the full-data benchmarks.

Chapter 4

Discussion



Acknowledgement

I would like to acknowledge the use of digital writing support tools during the preparation of this thesis. Grammarly was used to check grammar, spelling, and general language clarity.

ChatGPT was used only as a language-support tool in the Methodology and Results chapters. Its use was limited to clarifying sentence structure, improving wording, and finding more appropriate expressions for interpreting the results. For the R code included in the appendix, ChatGPT was used only to reduce the volume of the displayed code and to remove repeated code where present. The statistical analysis, coding, interpretation, and final academic decisions remain my own responsibility.

Appendix for Chapter 3

.1 PART 1 TABLES

Table 3.1: Summary of the `insem` dataset used in the analysis

Characteristic	Value	
Number of observations	10,513	
Number of variables	5	
Number of herds (clusters)	181	
Number of events	9,939	
Number of censored observations	574	
Censoring proportion	5.46%	
Mean survival time (days)	111.6	
Median survival time (days)	91	
Minimum survival time (days)	0.5	
Maximum survival time (days)	329	
Number of Heifer = 0	6,313	
Number of Heifer = 1	4,200	
Cluster structure	Unbalanced (varying herd sizes)	

Table 3.2: Results from the full data analysis.

Model	Estimate	Standard Error	Hazard Ratio
Marginal Cox	−0.0603	0.0205	0.9414
Cluster-Robust Cox	−0.0603	0.0210	0.9414
Copula Model (β_{Heifer})	−0.0603	0.0210	0.9414

Table 3.3: Computational summary for the full data copula analysis

Analysis	Observations	Herds	User Time	System Time	Elapsed Time
Full data	10513	181	220.95	5.86	260.12

Measure	Value
Approximate Memory Used (MB)	0.1

Table 3.4: Three-subset data splitting summary with censoring percentages

Subset	Herds	Observations	Events / Censored	% Censored (Subset)	% Censored (Full Data)
1	61	3664	3442 / 222	6.06%	2.11%
2	60	3212	3051 / 161	5.01%	1.53%
3	60	3637	3446 / 191	5.25%	1.82%

Table 3.5: Subset-specific model results for the three-subset analysis.

Subset	Model	Estimate	Standard Error	Hazard Ratio
1	Standard Cox	−0.0192	0.0348	0.9809
1	Cluster-Robust Cox	−0.0192	0.0374	0.9809
1	Copula Model (β_{Heifer})	−0.0192	0.0374	0.9809
1	Kendall's tau	0.2048	0.0452	—
2	Standard Cox	−0.0787	0.0370	0.9244
2	Cluster-Robust Cox	−0.0787	0.0333	0.9244
2	Copula Model (β_{Heifer})	−0.0787	0.0333	0.9244
2	Kendall's tau	0.1748	0.0316	—
3	Standard Cox	−0.0889	0.0350	0.9149
3	Cluster-Robust Cox	−0.0889	0.0339	0.9149
3	Copula Model (β_{Heifer})	−0.0889	0.0339	0.9149
3	Kendall's tau	0.1613	0.0301	—

Table 3.6: Combined results from the three-subset analysis using different weighting methods.

Model	Weighting Method	Combined Estimate	Standard Error	Hazard Ratio
Standard Cox	Information matrix	−0.0615	0.0205	0.9403
Standard Cox	Inverse variance	−0.0615	0.0205	0.9403
Standard Cox	Cluster-number	−0.0620	0.0205	0.9398
Standard Cox	Cow-number	−0.0615	0.0205	0.9404
Cluster-Robust Cox	Information matrix	−0.0652	0.0200	0.9369
Cluster-Robust Cox	Inverse variance	−0.0652	0.0200	0.9369
Cluster-Robust Cox	Cluster-number	−0.0620	0.0202	0.9398
Cluster-Robust Cox	Cow-number	−0.0615	0.0203	0.9404
Copula Model (β_{Heifer})	Information matrix	−0.0652	0.0200	0.9369
Copula Model (β_{Heifer})	Inverse variance	−0.0652	0.0200	0.9369
Copula Model (β_{Heifer})	Cluster-number	−0.0620	0.0202	0.9398
Copula Model (β_{Heifer})	Cow-number	−0.0615	0.0203	0.9404
Kendall's tau	Information matrix	0.1747	0.0196	—
Kendall's tau	Inverse variance	0.1747	0.0196	—
Kendall's tau	Cluster-number	0.1804	0.0210	—
Kendall's tau	Cow-number	0.1806	0.0212	—

Table 3.7: Comparison of the three-subset combined estimates with the full-data benchmark.

Model	Weighting Method	Combined Est.	Full Data Est.	Absolute Diff.	Relative Diff.
Standard Cox	Information matrix	-0.0615	-0.0603	0.0012	0.0193
Standard Cox	Inverse variance	-0.0615	-0.0603	0.0012	0.0193
Standard Cox	Cluster-number	-0.0620	-0.0603	0.0017	0.0280
Standard Cox	Cow-number	-0.0615	-0.0603	0.0012	0.0193
Cluster-Robust Cox	Information matrix	-0.0652	-0.0603	0.0048	0.0799
Cluster-Robust Cox	Inverse variance	-0.0652	-0.0603	0.0048	0.0799
Cluster-Robust Cox	Cluster-number	-0.0620	-0.0603	0.0017	0.0280
Cluster-Robust Cox	Cow-number	-0.0615	-0.0603	0.0012	0.0193
Copula Model (β_{Heifer})	Information matrix	-0.0652	-0.0603	0.0048	0.0799
Copula Model (β_{Heifer})	Inverse variance	-0.0652	-0.0603	0.0048	0.0799
Copula Model (β_{Heifer})	Cluster-number	-0.0620	-0.0603	0.0017	0.0280
Copula Model (β_{Heifer})	Cow-number	-0.0615	-0.0603	0.0012	0.0193
Kendall's tau	Information matrix	0.1747	0.1828	0.0081	0.0443
Kendall's tau	Inverse variance	0.1747	0.1828	0.0081	0.0443
Kendall's tau	Cluster-number	0.1804	0.1828	0.0024	0.0131
Kendall's tau	Cow-number	0.1806	0.1828	0.0022	0.0120

Table 3.8: Comparison of the three-subset combined standard errors with the full-data benchmark.

Model	Weighting Method	Combined SE	Full Data SE	Absolute Diff.	Relative Diff.
Standard Cox	Information matrix	0.0205	0.0205	0.0000	0.0002
Standard Cox	Inverse variance	0.0205	0.0205	0.0000	0.0002
Standard Cox	Cluster-number	0.0205	0.0205	0.0000	0.0015
Standard Cox	Cow-number	0.0205	0.0205	0.0000	0.0002
Cluster-Robust Cox	Information matrix	0.0200	0.0210	0.0009	0.0440
Cluster-Robust Cox	Inverse variance	0.0200	0.0210	0.0009	0.0440
Cluster-Robust Cox	Cluster-number	0.0202	0.0210	0.0008	0.0381
Cluster-Robust Cox	Cow-number	0.0203	0.0210	0.0007	0.0333
Copula Model (β_{Heifer})	Information matrix	0.0200	0.0210	0.0009	0.0440
Copula Model (β_{Heifer})	Inverse variance	0.0200	0.0210	0.0009	0.0440
Copula Model (β_{Heifer})	Cluster-number	0.0202	0.0210	0.0008	0.0381
Copula Model (β_{Heifer})	Cow-number	0.0203	0.0210	0.0007	0.0333
Kendall's tau	Information matrix	0.0196	0.0210	0.0014	0.0667
Kendall's tau	Inverse variance	0.0196	0.0210	0.0014	0.0667
Kendall's tau	Cluster-number	0.0210	0.0210	0.0000	0.0012
Kendall's tau	Cow-number	0.0212	0.0210	0.0002	0.0095

Table 3.9: Computational summary for the three-subset copula analysis

Subset	Observations	Herds	User Time	System Time	Elapsed Time	Memory (MB)
1	3664	61	22.83	0.39	26.71	0.3
2	3212	60	25.37	0.78	35.00	0.3
3	3637	60	22.67	0.36	25.64	0.3

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Table 3.10: Five-subset data splitting summary with censoring percentages

Subset	Herds	Observations	Events / Censored	% Censored (Subset)	% Censored (Full Data)
1	37	2239	2090 / 149	6.66%	1.42%
2	36	2165	2084 / 81	3.74%	0.77%
3	36	1928	1798 / 130	6.74%	1.24%
4	36	2064	1993 / 71	3.44%	0.68%
5	36	2117	1974 / 143	6.76%	1.36%

Table 3.11: Subset-specific model results for the five-subset analysis.

Subset	Model	Estimate	Standard Error	Hazard Ratio
1	Standard Cox	−0.0678	0.0449	0.9345
1	Cluster-Robust Cox	−0.0678	0.0443	0.9345
1	Copula Model (β_{Heifer})	−0.0678	0.0443	0.9345
1	Kendall's tau	0.2248	0.0682	—
2	Standard Cox	0.0303	0.0443	1.0307
2	Cluster-Robust Cox	0.0303	0.0492	1.0307
2	Copula Model (β_{Heifer})	0.0303	0.0492	1.0307
2	Kendall's tau	0.1438	0.0374	—
3	Standard Cox	−0.0839	0.0484	0.9196
3	Cluster-Robust Cox	−0.0839	0.0353	0.9196
3	Copula Model (β_{Heifer})	−0.0839	0.0353	0.9196
3	Kendall's tau	0.2189	0.0445	—
4	Standard Cox	−0.1114	0.0460	0.8946
4	Cluster-Robust Cox	−0.1114	0.0487	0.8946
4	Copula Model (β_{Heifer})	−0.1114	0.0487	0.8946
4	Kendall's tau	0.1210	0.0181	—
5	Standard Cox	−0.0947	0.0461	0.9097
5	Cluster-Robust Cox	−0.0947	0.0460	0.9097
5	Copula Model (β_{Heifer})	−0.0947	0.0460	0.9097
5	Kendall's tau	0.1783	0.0437	—

Table 3.12: Combined results from the five-subset analysis.

Model	Weighting Method	Combined Estimate	Standard Error	Hazard Ratio
Standard Cox	Information matrix	−0.0636	0.0205	0.9384
Standard Cox	Inverse variance	−0.0636	0.0205	0.9384
Standard Cox	Cluster-number	−0.0655	0.0206	0.9366
Standard Cox	Cow-number	−0.0645	0.0205	0.9375
Cluster-Robust Cox	Information matrix	−0.0692	0.0195	0.9332
Cluster-Robust Cox	Inverse variance	−0.0692	0.0195	0.9332
Cluster-Robust Cox	Cluster-number	−0.0655	0.0201	0.9366
Cluster-Robust Cox	Cow-number	−0.0645	0.0203	0.9375
Copula Model (β_{Heifer})	Information matrix	−0.0692	0.0195	0.9332
Copula Model (β_{Heifer})	Inverse variance	−0.0692	0.0195	0.9332
Copula Model (β_{Heifer})	Cluster-number	−0.0655	0.0201	0.9366
Copula Model (β_{Heifer})	Cow-number	−0.0645	0.0203	0.9375
Kendall's tau	Information matrix	0.1446	0.0141	—
Kendall's tau	Inverse variance	0.1446	0.0141	—
Kendall's tau	Cluster-number	0.1776	0.0204	—
Kendall's tau	Cow-number	0.1773	0.0207	—

Table 3.13: Comparison of the five-subset combined estimates with the full-data benchmark.

Model	Weighting Method	Combined Est.	Full Data Est.	Absolute Diff.	Relative Diff.
Standard Cox	Information matrix	-0.0636	-0.0603	0.0033	0.0547
Standard Cox	Inverse variance	-0.0636	-0.0603	0.0033	0.0547
Standard Cox	Cluster-number	-0.0655	-0.0603	0.0052	0.0862
Standard Cox	Cow-number	-0.0645	-0.0603	0.0042	0.0696
Cluster-Robust Cox	Information matrix	-0.0692	-0.0603	0.0088	0.1459
Cluster-Robust Cox	Inverse variance	-0.0692	-0.0603	0.0088	0.1459
Cluster-Robust Cox	Cluster-number	-0.0655	-0.0603	0.0052	0.0862
Cluster-Robust Cox	Cow-number	-0.0645	-0.0603	0.0042	0.0696
Copula Model (β_{Heifer})	Information matrix	-0.0692	-0.0603	0.0088	0.1459
Copula Model (β_{Heifer})	Inverse variance	-0.0692	-0.0603	0.0088	0.1459
Copula Model (β_{Heifer})	Cluster-number	-0.0655	-0.0603	0.0052	0.0862
Copula Model (β_{Heifer})	Cow-number	-0.0645	-0.0603	0.0042	0.0696
Kendall's tau	Information matrix	0.1446	0.1828	0.0382	0.2090
Kendall's tau	Inverse variance	0.1446	0.1828	0.0382	0.2090
Kendall's tau	Cluster-number	0.1776	0.1828	0.0052	0.0284
Kendall's tau	Cow-number	0.1773	0.1828	0.0055	0.0301

Table 3.14: Comparison of the five-subset combined standard errors with the full-data benchmark.

Model	Weighting Method	Combined SE	Full Data SE	Absolute Diff.	Relative Diff.
Standard Cox	Information matrix	0.02053	0.02050	0.00002	0.0012
Standard Cox	Inverse variance	0.02053	0.02050	0.00002	0.0012
Standard Cox	Cluster-number	0.02056	0.02050	0.00006	0.0028
Standard Cox	Cow-number	0.02053	0.02050	0.00003	0.0015
Cluster-Robust Cox	Information matrix	0.01954	0.02096	0.00142	0.0677
Cluster-Robust Cox	Inverse variance	0.01954	0.02096	0.00142	0.0677
Cluster-Robust Cox	Cluster-number	0.02012	0.02096	0.00084	0.0401
Cluster-Robust Cox	Cow-number	0.02027	0.02096	0.00069	0.0329
Copula Model (β_{Heifer})	Information matrix	0.01954	0.02096	0.00142	0.0678
Copula Model (β_{Heifer})	Inverse variance	0.01954	0.02096	0.00142	0.0678
Copula Model (β_{Heifer})	Cluster-number	0.02012	0.02096	0.00084	0.0401
Copula Model (β_{Heifer})	Cow-number	0.02027	0.02096	0.00069	0.0329
Kendall's tau	Information matrix	0.01414	0.02103	0.00689	0.3276
Kendall's tau	Inverse variance	0.01414	0.02103	0.00689	0.3276
Kendall's tau	Cluster-number	0.02041	0.02103	0.00062	0.0295
Kendall's tau	Cow-number	0.02066	0.02103	0.00037	0.0176

Table 3.15: Computational summary for the five-subset copula analysis

Subset	Observations	Herds	User Time	System Time	Elapsed Time	Memory (MB)
1	2239	37	7.59	0.18	9.11	0.2
2	2165	36	7.00	0.11	10.68	0.2
3	1928	36	8.38	0.17	11.11	0.1
4	2064	36	7.92	0.15	8.22	0.2
5	2117	36	10.08	0.21	13.44	0.2

Table 3.16: Eight-subset data splitting summary with censoring percentages

Subset	Herds	Observations	Events / Censored	% Censored (Subset)	% Censored (Full Data)
1	23	1497	1412 / 85	5.68%	0.81%
2	23	1261	1182 / 79	6.26%	0.75%
3	22	1308	1247 / 61	4.66%	0.58%
4	23	1277	1203 / 74	5.79%	0.70%
5	22	1124	1051 / 73	6.49%	0.69%
6	23	1213	1175 / 38	3.13%	0.36%
7	22	1487	1443 / 44	2.96%	0.42%
8	23	1346	1226 / 120	8.92%	1.14%

Table 3.17: Subset-specific model results for the eight-subset analysis.

Subset	Model	Estimate	Standard Error	Hazard Ratio
1	Standard Cox	−0.1144	0.0546	0.8919
1	Cluster-Robust Cox	−0.1144	0.0429	0.8919
1	Copula Model (β_{Heifer})	−0.1144	0.0429	0.8919
1	Kendall's tau	0.2268	0.0918	—
2	Standard Cox	0.0254	0.0594	1.0257
2	Cluster-Robust Cox	0.0254	0.0704	1.0257
2	Copula Model (β_{Heifer})	0.0254	0.0704	1.0257
2	Kendall's tau	0.1840	0.0822	—
3	Standard Cox	0.0502	0.0573	1.0514
3	Cluster-Robust Cox	0.0502	0.0640	1.0514
3	Copula Model (β_{Heifer})	0.0502	0.0640	1.0514
3	Kendall's tau	0.1549	0.0560	—
4	Standard Cox	−0.0688	0.0589	0.9336
4	Cluster-Robust Cox	−0.0688	0.0371	0.9336
4	Copula Model (β_{Heifer})	−0.0688	0.0371	0.9336
4	Kendall's tau	0.1777	0.0463	—
5	Standard Cox	−0.1336	0.0635	0.8749
5	Cluster-Robust Cox	−0.1336	0.0681	0.8749
5	Copula Model (β_{Heifer})	−0.1336	0.0681	0.8749
5	Kendall's tau	0.2228	0.0606	—
6	Standard Cox	−0.0823	0.0599	0.9210
6	Cluster-Robust Cox	−0.0823	0.0532	0.9210
6	Copula Model (β_{Heifer})	−0.0823	0.0532	0.9210
6	Kendall's tau	0.1245	0.0236	—
7	Standard Cox	−0.0652	0.0539	0.9369
7	Cluster-Robust Cox	−0.0652	0.0449	0.9369
7	Copula Model (β_{Heifer})	−0.0652	0.0449	0.9369
7	Kendall's tau	0.1103	0.0194	—
8	Standard Cox	−0.0998	0.0585	0.9050
8	Cluster-Robust Cox	−0.0998	0.0606	0.9050
8	Copula Model (β_{Heifer})	−0.0998	0.0606	0.9050
8	Kendall's tau	0.2051	0.0576	—

Table 3.18: Combined results from the eight-subset analysis.

Model	Weighting Method	Combined Est.	Standard Error	Hazard Ratio
Standard Cox	Information matrix	-0.0603	0.0205	0.9414
Standard Cox	Inverse variance	-0.0603	0.0205	0.9414
Standard Cox	Cluster-number	-0.0613	0.0206	0.9406
Standard Cox	Cow-number	-0.0611	0.0205	0.9407
Cluster-Robust Cox	Information matrix	-0.0695	0.0181	0.9329
Cluster-Robust Cox	Inverse variance	-0.0695	0.0181	0.9329
Cluster-Robust Cox	Cluster-number	-0.0613	0.0199	0.9406
Cluster-Robust Cox	Cow-number	-0.0611	0.0196	0.9407
Copula Model (β_{Heifer})	Information matrix	-0.0695	0.0181	0.9329
Copula Model (β_{Heifer})	Inverse variance	-0.0695	0.0181	0.9329
Copula Model (β_{Heifer})	Cluster-number	-0.0613	0.0199	0.9406
Copula Model (β_{Heifer})	Cow-number	-0.0611	0.0196	0.9407
Kendall's tau	Information matrix	0.1358	0.0128	—
Kendall's tau	Inverse variance	0.1358	0.0128	—
Kendall's tau	Cluster-number	0.1760	0.0212	—
Kendall's tau	Cow-number	0.1753	0.0214	—

Table 3.19: Comparison of the eight-subset combined estimates with the full-data benchmark.

Model	Weighting Method	Combined Est.	Full Data Est.	Absolute Diff.	Relative Diff.
Standard Cox	Information matrix	-0.0603	-0.0603	0.0000	0.0000
Standard Cox	Inverse variance	-0.0603	-0.0603	0.0000	0.0000
Standard Cox	Cluster-number	-0.0613	-0.0603	0.0009	0.0149
Standard Cox	Cow-number	-0.0611	-0.0603	0.0008	0.0133
Cluster-Robust Cox	Information matrix	-0.0695	-0.0603	0.0091	0.1509
Cluster-Robust Cox	Inverse variance	-0.0695	-0.0603	0.0091	0.1509
Cluster-Robust Cox	Cluster-number	-0.0613	-0.0603	0.0009	0.0149
Cluster-Robust Cox	Cow-number	-0.0611	-0.0603	0.0008	0.0133
Copula Model (β_{Heifer})	Information matrix	-0.0695	-0.0603	0.0091	0.1509
Copula Model (β_{Heifer})	Inverse variance	-0.0695	-0.0603	0.0091	0.1509
Copula Model (β_{Heifer})	Cluster-number	-0.0613	-0.0603	0.0009	0.0149
Copula Model (β_{Heifer})	Cow-number	-0.0611	-0.0603	0.0008	0.0133
Kendall's tau	Information matrix	0.1358	0.1828	0.0470	0.2572
Kendall's tau	Inverse variance	0.1358	0.1828	0.0470	0.2572
Kendall's tau	Cluster-number	0.1760	0.1828	0.0068	0.0372
Kendall's tau	Cow-number	0.1753	0.1828	0.0076	0.0416

Table 3.20: Comparison of the eight-subset combined standard errors with the full-data benchmark.

Model	Weighting Method	Combined SE	Full Data SE	Absolute Diff.	Relative Diff.
Standard Cox	Information matrix	0.02052	0.02050	0.00002	0.0010
Standard Cox	Inverse variance	0.02052	0.02050	0.00002	0.0010
Standard Cox	Cluster-number	0.02062	0.02050	0.00012	0.0059
Standard Cox	Cow-number	0.02053	0.02050	0.00002	0.0012
Cluster-Robust Cox	Information matrix	0.01809	0.02096	0.00287	0.1369
Cluster-Robust Cox	Inverse variance	0.01809	0.02096	0.00287	0.1369
Cluster-Robust Cox	Cluster-number	0.01989	0.02096	0.00107	0.0511
Cluster-Robust Cox	Cow-number	0.01958	0.02096	0.00138	0.0659
Copula Model (β_{Heifer})	Information matrix	0.01809	0.02096	0.00287	0.1369
Copula Model (β_{Heifer})	Inverse variance	0.01809	0.02096	0.00287	0.1369
Copula Model (β_{Heifer})	Cluster-number	0.01989	0.02096	0.00107	0.0511
Copula Model (β_{Heifer})	Cow-number	0.01958	0.02096	0.00138	0.0659
Kendall's tau	Information matrix	0.01282	0.02103	0.00821	0.3904
Kendall's tau	Inverse variance	0.01282	0.02103	0.00821	0.3904
Kendall's tau	Cluster-number	0.02117	0.02103	0.00014	0.0069
Kendall's tau	Cow-number	0.02143	0.02103	0.00040	0.0190

Table 3.21: Computational summary for the eight-subset copula analysis

Subset	Observations	Herds	User Time	System Time	Elapsed Time	Memory (MB)
1	1497	23	3.47	0.05	3.62	0.1
2	1261	23	3.27	0.02	3.37	0.1
3	1308	22	3.30	0.03	4.45	0.1
4	1277	23	3.64	0.08	5.69	0.1
5	1124	22	2.70	0.11	2.87	0.1
6	1213	23	3.97	0.13	8.23	0.1
7	1487	22	5.67	0.13	8.12	0.1
8	1346	23	4.83	0.16	6.11	0.1

.2 PART 2 TABLES

Table 3.22: Summary of data after Q1 splitting

Quartile	Number of Herds	Number of Clusters	Observations	Censoring (%)
Q1 (Largest)	46	138	4,582	5.91
Q2	45	45	2,943	4.35
Q3	45	45	2,040	6.03
Q4 (Smallest)	45	45	948	5.49

Table 3.23: Overview of random sample splitting after Q1 splitting

Scenario	Subset	Number of Observations	Number of Clusters	Total Clusters
3 Subsets	Subset 1	3515	91	273
	Subset 2	3533	91	
	Subset 3	3465	91	
5 Subsets	Subset 1	2216	55	273
	Subset 2	1997	54	
	Subset 3	2045	55	
	Subset 4	2088	54	
	Subset 5	2167	55	
8 Subsets	Subset 1	1457	35	273
	Subset 2	1244	34	
	Subset 3	1286	34	
	Subset 4	1275	34	
	Subset 5	1267	34	
	Subset 6	1336	34	
	Subset 7	1244	34	
	Subset 8	1404	34	

Table 3.24: Comparison of full data analysis and Q1 splitting results

Parameter	Full Data	Q1 Split	Difference
$\hat{\beta}$ (Heifer)	-0.0603	-0.0603	0.0000
SE (Cox)	0.0205	0.0205	0.0000
SE (Robust)	0.0210	0.0200	-0.0010
$\hat{\theta}$	0.4475	0.4433	-0.0042
SE($\hat{\theta}$)	0.0630	0.0470	-0.0160
Kendall's τ	0.1828	0.1814	-0.0014
SE(τ)	0.0210	0.0157	-0.0053

Table 3.25: Computational summary for the initial Q1 full analysis without random splitting

Analysis	Observations	Clusters	User Time	System Time	Elapsed Time	Memory (MB)
Q1-split full data	10513	273	310.32	12.36	371.72	4.9

Table 3.26: Subset-specific model results for the three-subset analysis after Q1 splitting

Subset	Model	Estimate	Standard Error	Hazard Ratio
1	Standard Cox	-0.0851	0.0353	0.9184
1	Cluster-Robust Cox	-0.0851	0.0367	0.9184
1	Copula Model (β_{Heifer})	-0.0851	0.0367	0.9184
1	Kendall's τ	0.1914	0.0242	—
2	Standard Cox	-0.0451	0.0356	0.9559
2	Cluster-Robust Cox	-0.0451	0.0336	0.9559
2	Copula Model (β_{Heifer})	-0.0451	0.0336	0.9559
2	Kendall's τ	0.1725	0.0275	—
3	Standard Cox	-0.0514	0.0357	0.9499
3	Cluster-Robust Cox	-0.0514	0.0339	0.9499
3	Copula Model (β_{Heifer})	-0.0514	0.0339	0.9499
3	Kendall's τ	0.1838	0.0290	—

Table 3.27: Combined results from the three-subset analysis after Q1 splitting

Model	Weighting Method	Combined Estimate	Standard Error	Hazard Ratio
Standard Cox	Information matrix	-0.0608	0.0209	0.9410
Standard Cox	Inverse variance	-0.0608	0.0209	0.9410
Standard Cox	Cluster-number	-0.0605	0.0210	0.9413
Standard Cox	Cow-number	-0.0606	0.0209	0.9412
Cluster-Robust Cox	Information matrix	-0.0609	0.0211	0.9409
Cluster-Robust Cox	Inverse variance	-0.0609	0.0211	0.9409
Cluster-Robust Cox	Cluster-number	-0.0605	0.0212	0.9413
Cluster-Robust Cox	Cow-number	-0.0606	0.0211	0.9412
Copula Model	Information matrix	-0.0609	0.0211	0.9409
Copula Model	Inverse variance	-0.0609	0.0211	0.9409
Copula Model	Cluster-number	-0.0605	0.0212	0.9413
Copula Model	Cow-number	-0.0606	0.0211	0.9412
Kendall's τ	Information matrix	0.1821	0.0214	—
Kendall's τ	Inverse variance	0.1821	0.0214	—
Kendall's τ	Cluster-number	0.1826	0.0210	—
Kendall's τ	Cow-number	0.1824	0.0212	—

Table 3.28: Comparison of the three-subset Q1-splitting combined estimates with the Q1-split full-data benchmark

Model	Weighting Method	Combined Est.	Q1 Full Data Est.	Abs. Diff.	Rel. Diff.
Standard Cox	Information matrix	-0.0608	-0.0603	0.0005	0.0083
Standard Cox	Inverse variance	-0.0608	-0.0603	0.0005	0.0083
Standard Cox	Cluster-number	-0.0605	-0.0603	0.0002	0.0033
Standard Cox	Cow-number	-0.0606	-0.0603	0.0003	0.0050
Cluster-Robust Cox	Information matrix	-0.0609	-0.0603	0.0006	0.0100
Cluster-Robust Cox	Inverse variance	-0.0609	-0.0603	0.0006	0.0100
Cluster-Robust Cox	Cluster-number	-0.0605	-0.0603	0.0002	0.0033
Cluster-Robust Cox	Cow-number	-0.0606	-0.0603	0.0003	0.0050
Copula Model	Information matrix	-0.0609	-0.0603	0.0006	0.0100
Copula Model	Inverse variance	-0.0609	-0.0603	0.0006	0.0100
Copula Model	Cluster-number	-0.0605	-0.0603	0.0002	0.0033
Copula Model	Cow-number	-0.0606	-0.0603	0.0003	0.0050
Kendall's τ	Information matrix	0.1821	0.1814	0.0007	0.0039
Kendall's τ	Inverse variance	0.1821	0.1814	0.0007	0.0039
Kendall's τ	Cluster-number	0.1826	0.1814	0.0012	0.0066
Kendall's τ	Cow-number	0.1824	0.1814	0.0010	0.0055

Table 3.29: Comparison of the three-subset Q1-splitting combined standard errors with the Q1-split full-data benchmark

Model	Weighting Method	Combined SE	Q1 Full Data SE	Abs. Diff.	Rel. Diff.
Standard Cox	Information matrix	0.0209	0.0205	0.0004	0.0195
Standard Cox	Inverse variance	0.0209	0.0205	0.0004	0.0195
Standard Cox	Cluster-number	0.0210	0.0205	0.0005	0.0244
Standard Cox	Cow-number	0.0209	0.0205	0.0004	0.0195
Cluster-Robust Cox	Information matrix	0.0211	0.0200	0.0011	0.0550
Cluster-Robust Cox	Inverse variance	0.0211	0.0200	0.0011	0.0550
Cluster-Robust Cox	Cluster-number	0.0212	0.0200	0.0012	0.0600
Cluster-Robust Cox	Cow-number	0.0211	0.0200	0.0011	0.0550
Copula Model	Information matrix	0.0211	0.0200	0.0011	0.0550
Copula Model	Inverse variance	0.0211	0.0200	0.0011	0.0550
Copula Model	Cluster-number	0.0212	0.0200	0.0012	0.0600
Copula Model	Cow-number	0.0211	0.0200	0.0011	0.0550
Kendall's τ	Information matrix	0.0214	0.0157	0.0057	0.3631
Kendall's τ	Inverse variance	0.0214	0.0157	0.0057	0.3631
Kendall's τ	Cluster-number	0.0210	0.0157	0.0053	0.3376
Kendall's τ	Cow-number	0.0212	0.0157	0.0055	0.3503

Table 3.30: Computational summary for the three-subset analysis after Q1 splitting

Subset	Observations	Clusters	Original Herds	User Time	System Time	Elapsed Time	Memory (MB)
1	3515	91	79	33.45	0.86	40.00	0.4
2	3533	91	79	35.45	0.85	42.28	0.3
3	3465	91	75	36.77	0.80	43.19	0.3

Table 3.31: Subset-specific model results for the five-subset analysis after Q1 splitting

Subset	Model	Estimate	Standard Error	Hazard Ratio
1	Standard Cox	-0.0888	0.0444	0.9150
1	Cluster-Robust Cox	-0.0888	0.0478	0.9150
1	Copula Model	-0.0888	0.0478	0.9150
1	Kendall's τ	0.1454	0.0288	—
2	Standard Cox	-0.0619	0.0471	0.9400
2	Cluster-Robust Cox	-0.0619	0.0413	0.9400
2	Copula Model	-0.0619	0.0413	0.9400
2	Kendall's τ	0.1668	0.0367	—
3	Standard Cox	-0.0401	0.0467	0.9607
3	Cluster-Robust Cox	-0.0401	0.0442	0.9607
3	Copula Model	-0.0401	0.0442	0.9607
3	Kendall's τ	0.2120	0.0367	—
4	Standard Cox	-0.0384	0.0462	0.9624
4	Cluster-Robust Cox	-0.0384	0.0463	0.9624
4	Copula Model	-0.0384	0.0463	0.9624
4	Kendall's τ	0.2155	0.0362	—
5	Standard Cox	-0.0734	0.0450	0.9292
5	Cluster-Robust Cox	-0.0734	0.0422	0.9292
5	Copula Model	-0.0734	0.0422	0.9292
5	Kendall's τ	0.1643	0.0387	—

Table 3.32: Combined results from the five-subset analysis after Q1 splitting

Model	Weighting Method	Combined Estimate	Standard Error	Hazard Ratio
Standard Cox	Information matrix	-0.0612	0.0205	0.9406
Standard Cox	Inverse variance	-0.0612	0.0205	0.9406
Standard Cox	Cluster-number	-0.0606	0.0205	0.9412
Standard Cox	Cow-number	-0.0610	0.0205	0.9408
Cluster-Robust Cox	Information matrix	-0.0604	0.0197	0.9414
Cluster-Robust Cox	Inverse variance	-0.0604	0.0197	0.9414
Cluster-Robust Cox	Cluster-number	-0.0606	0.0199	0.9412
Cluster-Robust Cox	Cow-number	-0.0610	0.0199	0.9408
Copula Model	Information matrix	-0.0604	0.0197	0.9414
Copula Model	Inverse variance	-0.0604	0.0197	0.9414
Copula Model	Cluster-number	-0.0606	0.0199	0.9412
Copula Model	Cow-number	-0.0610	0.0199	0.9408
Kendall's τ	Information matrix	0.1773	0.0156	—
Kendall's τ	Inverse variance	0.1773	0.0156	—
Kendall's τ	Cluster-number	0.1807	0.0159	—
Kendall's τ	Cow-number	0.1802	0.0159	—

Table 3.33: Comparison of the five-subset combined estimates with the Q1-split full-data benchmark

Model	Weighting Method	Combined Est.	Full Data Est.	Abs. Diff.	Rel. Diff.
Standard Cox	Information matrix	-0.0612	-0.0603	0.0009	0.0149
Standard Cox	Inverse variance	-0.0612	-0.0603	0.0009	0.0149
Standard Cox	Cluster-number	-0.0606	-0.0603	0.0003	0.0050
Standard Cox	Cow-number	-0.0610	-0.0603	0.0007	0.0116
Cluster-Robust Cox	Information matrix	-0.0604	-0.0603	0.0001	0.0017
Cluster-Robust Cox	Inverse variance	-0.0604	-0.0603	0.0001	0.0017
Cluster-Robust Cox	Cluster-number	-0.0606	-0.0603	0.0003	0.0050
Cluster-Robust Cox	Cow-number	-0.0610	-0.0603	0.0007	0.0116
Copula Model	Information matrix	-0.0604	-0.0603	0.0001	0.0017
Copula Model	Inverse variance	-0.0604	-0.0603	0.0001	0.0017
Copula Model	Cluster-number	-0.0606	-0.0603	0.0003	0.0050
Copula Model	Cow-number	-0.0610	-0.0603	0.0007	0.0116
Kendall's τ	Information matrix	0.1773	0.1814	0.0041	0.0226
Kendall's τ	Inverse variance	0.1773	0.1814	0.0041	0.0226
Kendall's τ	Cluster-number	0.1807	0.1814	0.0007	0.0039
Kendall's τ	Cow-number	0.1802	0.1814	0.0012	0.0066

Table 3.34: Comparison of the five-subset combined standard errors with the Q1-split full-data benchmark

Model	Weighting Method	Combined SE	Full Data SE	Abs. Diff.	Rel. Diff.
Standard Cox	Information matrix	0.0205	0.0210	0.0005	0.0238
Standard Cox	Inverse variance	0.0205	0.0210	0.0005	0.0238
Standard Cox	Cluster-number	0.0205	0.0210	0.0005	0.0238
Standard Cox	Cow-number	0.0205	0.0210	0.0005	0.0238
Cluster-Robust Cox	Information matrix	0.0197	0.0212	0.0015	0.0708
Cluster-Robust Cox	Inverse variance	0.0197	0.0212	0.0015	0.0708
Cluster-Robust Cox	Cluster-number	0.0199	0.0212	0.0013	0.0613
Cluster-Robust Cox	Cow-number	0.0199	0.0212	0.0013	0.0613
Copula Model	Information matrix	0.0197	0.0212	0.0015	0.0708
Copula Model	Inverse variance	0.0197	0.0212	0.0015	0.0708
Copula Model	Cluster-number	0.0199	0.0212	0.0013	0.0613
Copula Model	Cow-number	0.0199	0.0212	0.0013	0.0613
Kendall's τ	Information matrix	0.0156	0.0157	0.0001	0.0064
Kendall's τ	Inverse variance	0.0156	0.0157	0.0001	0.0064
Kendall's τ	Cluster-number	0.0159	0.0157	0.0002	0.0127
Kendall's τ	Cow-number	0.0159	0.0157	0.0002	0.0127

Table 3.35: Computational summary for the five-subset analysis after Q1 splitting

Subset	Observations	Clusters	Original Herds	User Time	System Time	Elapsed Time	Memory (MB)
1	2216	55	51	15.22	0.26	17.67	0.2
2	1997	54	47	12.69	0.26	15.00	0.1
3	2045	55	50	13.55	0.29	16.08	0.2
4	2088	54	51	13.01	0.22	15.33	0.2
5	2167	55	48	13.27	0.38	15.66	0.2

Table 3.36: Subset-specific model results for the eight-subset analysis after Q1 splitting

Subset	Model	Estimate	Standard Error	Hazard Ratio
1	Standard Cox	-0.1316	0.0552	0.8767
1	Cluster-Robust Cox	-0.1316	0.0591	0.8767
1	Copula Model	-0.1316	0.0591	0.8767
1	Kendall's τ	0.1712	0.0366	—
2	Standard Cox	-0.0050	0.0593	0.9950
2	Cluster-Robust Cox	-0.0050	0.0547	0.9950
2	Copula Model	-0.0050	0.0547	0.9950
2	Kendall's τ	0.1438	0.0453	—
3	Standard Cox	-0.0926	0.0581	0.9115
3	Cluster-Robust Cox	-0.0926	0.0508	0.9115
3	Copula Model	-0.0926	0.0508	0.9115
3	Kendall's τ	0.1394	0.0473	—
4	Standard Cox	-0.0691	0.0605	0.9332
4	Cluster-Robust Cox	-0.0691	0.0551	0.9332
4	Copula Model	-0.0691	0.0551	0.9332
4	Kendall's τ	0.2351	0.0470	—
5	Standard Cox	-0.0341	0.0585	0.9665
5	Cluster-Robust Cox	-0.0341	0.0528	0.9665
5	Copula Model	-0.0341	0.0528	0.9665
5	Kendall's τ	0.1613	0.0385	—
6	Standard Cox	-0.0067	0.0574	0.9933
6	Cluster-Robust Cox	-0.0067	0.0511	0.9933
6	Copula Model	-0.0067	0.0511	0.9933
6	Kendall's τ	0.2229	0.0457	—
7	Standard Cox	-0.0315	0.0599	0.9690
7	Cluster-Robust Cox	-0.0315	0.0676	0.9690
7	Copula Model	-0.0315	0.0676	0.9690
7	Kendall's τ	0.1882	0.0453	—
8	Standard Cox	-0.0839	0.0560	0.9195
8	Cluster-Robust Cox	-0.0839	0.0508	0.9195
8	Copula Model	-0.0839	0.0508	0.9195
8	Kendall's τ	0.1676	0.0527	—

Table 3.37: Combined results from the eight-subset analysis after Q1 splitting

Model	Weighting Method	Combined Estimate	Standard Error	Hazard Ratio
Standard Cox	Information matrix	-0.0583	0.0205	0.9434
Standard Cox	Inverse variance	-0.0583	0.0205	0.9434
Standard Cox	Cluster-number	-0.0571	0.0205	0.9445
Standard Cox	Cow-number	-0.0584	0.0205	0.9433
Cluster-Robust Cox	Information matrix	-0.0567	0.0193	0.9449
Cluster-Robust Cox	Inverse variance	-0.0567	0.0193	0.9449
Cluster-Robust Cox	Cluster-number	-0.0571	0.0196	0.9445
Cluster-Robust Cox	Cow-number	-0.0584	0.0196	0.9433
Copula Model	Information matrix	-0.0567	0.0193	0.9449
Copula Model	Inverse variance	-0.0567	0.0193	0.9449
Copula Model	Cluster-number	-0.0571	0.0196	0.9445
Copula Model	Cow-number	-0.0584	0.0196	0.9433
Kendall's τ	Information matrix	0.1767	0.0156	—
Kendall's τ	Inverse variance	0.1767	0.0156	—
Kendall's τ	Cluster-number	0.1777	0.0159	—
Kendall's τ	Cow-number	0.1778	0.0159	—

Table 3.38: Comparison of the eight-subset combined estimates with the Q1-split full-data benchmark

Model	Weighting Method	Combined Est.	Full Data Est.	Abs. Diff.	Rel. Diff.
Standard Cox	Information matrix	-0.0583	-0.0603	0.0020	0.0332
Standard Cox	Inverse variance	-0.0583	-0.0603	0.0020	0.0332
Standard Cox	Cluster-number	-0.0571	-0.0603	0.0032	0.0531
Standard Cox	Cow-number	-0.0584	-0.0603	0.0019	0.0315
Cluster-Robust Cox	Information matrix	-0.0567	-0.0603	0.0036	0.0597
Cluster-Robust Cox	Inverse variance	-0.0567	-0.0603	0.0036	0.0597
Cluster-Robust Cox	Cluster-number	-0.0571	-0.0603	0.0032	0.0531
Cluster-Robust Cox	Cow-number	-0.0584	-0.0603	0.0019	0.0315
Copula Model	Information matrix	-0.0567	-0.0603	0.0036	0.0597
Copula Model	Inverse variance	-0.0567	-0.0603	0.0036	0.0597
Copula Model	Cluster-number	-0.0571	-0.0603	0.0032	0.0531
Copula Model	Cow-number	-0.0584	-0.0603	0.0019	0.0315
Kendall's τ	Information matrix	0.1767	0.1814	0.0047	0.0259
Kendall's τ	Inverse variance	0.1767	0.1814	0.0047	0.0259
Kendall's τ	Cluster-number	0.1777	0.1814	0.0037	0.0204
Kendall's τ	Cow-number	0.1778	0.1814	0.0036	0.0198

Table 3.39: Comparison of the eight-subset combined standard errors with the Q1-split full-data benchmark

Model	Weighting Method	Combined SE	Full Data SE	Abs. Diff.	Rel. Diff.
Standard Cox	Information matrix	0.0205	0.0210	0.0005	0.0238
Standard Cox	Inverse variance	0.0205	0.0210	0.0005	0.0238
Standard Cox	Cluster-number	0.0205	0.0210	0.0005	0.0238
Standard Cox	Cow-number	0.0205	0.0210	0.0005	0.0238
Cluster-Robust Cox	Information matrix	0.0193	0.0212	0.0019	0.0896
Cluster-Robust Cox	Inverse variance	0.0193	0.0212	0.0019	0.0896
Cluster-Robust Cox	Cluster-number	0.0196	0.0212	0.0016	0.0755
Cluster-Robust Cox	Cow-number	0.0196	0.0212	0.0016	0.0755
Copula Model	Information matrix	0.0193	0.0212	0.0019	0.0896
Copula Model	Inverse variance	0.0193	0.0212	0.0019	0.0896
Copula Model	Cluster-number	0.0196	0.0212	0.0016	0.0755
Copula Model	Cow-number	0.0196	0.0212	0.0016	0.0755
Kendall's τ	Information matrix	0.0156	0.0157	0.0001	0.0064
Kendall's τ	Inverse variance	0.0156	0.0157	0.0001	0.0064
Kendall's τ	Cluster-number	0.0159	0.0157	0.0002	0.0127
Kendall's τ	Cow-number	0.0159	0.0157	0.0002	0.0127

Table 3.40: Computational summary for the eight-subset analysis after Q1 splitting

Subset	Observations	Clusters	Original Herds	User Time	System Time	Elapsed Time	Memory (MB)
1	1457	35	34	6.29	0.10	7.20	0.1
2	1244	34	32	4.98	0.14	6.13	0.1
3	1286	34	30	5.39	0.08	6.22	0.1
4	1275	34	31	5.81	0.08	6.47	0.1
5	1267	34	33	5.92	0.09	6.13	0.1
6	1336	34	33	5.81	0.08	5.90	0.1
7	1244	34	32	5.68	0.14	5.86	0.1
8	1404	34	32	6.83	0.01	6.08	0.1

Table 3.41: Summary of data after combined Q1 and Q2 splitting

Quartile	Number of Herds	Number of Clusters	Observations	Censoring (%)
Q1 (Largest)	46	138	4,582	5.91
Q2	45	90	2,943	4.35
Q3	45	45	2,040	6.03
Q4 (Smallest)	45	45	948	5.49

Table 3.42: Cluster size distribution after combined Q1 and Q2 splitting

Scenario	Min	Q1	Median	Mean	Q3	Max
Q1 + Q2 splitting	1	28	32	33.1	37	58

Table 3.43: Comparison of model estimates across different cluster splitting strategies

Parameter	Full Data	Q1 Split	Q1+Q2 Split
$\hat{\beta}$ (Heifer)	-0.0603	-0.0603	-0.0603
SE (Cox)	0.0205	0.0205	0.0205
SE (Robust)	0.0210	0.0200	0.0196
$\hat{\theta}$	0.4475	0.4433	0.4429
SE($\hat{\theta}$)	0.0630	0.0470	0.0439
Kendall's τ	0.1828	0.1814	0.1813
SE(τ)	0.0210	0.0157	0.0147

Table 3.44 summarizes the computational performance of the combined Q1 and Q2 splitting analysis.

Table 3.44: Computational summary for the combined Q1 and Q2 splitting analysis

Analysis	Observations	Clusters	User Time	System Time	Elapsed Time	Memory (MB)
Q1+Q2 full data	10513	318	468.55	14.85	502.97	4.9

Table 3.45: Data description for the three-subset analysis after Q1 and Q2 splitting

Subset	Number of Observations	Number of Modified Clusters	Original Herds	Events	Censored
1	3547	106	84	3358	189
2	3565	106	90	3368	197
3	3401	106	90	3214	187
Total	10513	318	181	9940	573

Table 3.46: Subset-specific model results for the three-subset analysis after Q1 and Q2 splitting

Subset	Model	Estimate	Standard Error	Hazard Ratio
1	Standard Cox	-0.0589	0.0354	0.9428
1	Cluster-Robust Cox	-0.0589	0.0298	0.9428
1	Copula Model	-0.0589	0.0298	0.9428
1	Kendall's τ	0.1995	0.0242	—
2	Standard Cox	-0.0716	0.0352	0.9309
2	Cluster-Robust Cox	-0.0716	0.0302	0.9309
2	Copula Model	-0.0716	0.0302	0.9309
2	Kendall's τ	0.1747	0.0259	—
3	Standard Cox	-0.0450	0.0360	0.9560
3	Cluster-Robust Cox	-0.0450	0.0420	0.9560
3	Copula Model	-0.0450	0.0420	0.9560
3	Kendall's τ	0.1659	0.0343	—

Table 3.47: Combined results from the three-subset analysis after Q1 and Q2 splitting

Model	Weighting Method	Combined Estimate	Standard Error	Hazard Ratio
Standard Cox	Information matrix	-0.0587	0.0205	0.9430
Standard Cox	Inverse variance	-0.0587	0.0205	0.9430
Standard Cox	Cluster-number	-0.0585	0.0205	0.9432
Standard Cox	Cow-number	-0.0587	0.0205	0.9430
Cluster-Robust Cox	Information matrix	-0.0611	0.0189	0.9407
Cluster-Robust Cox	Inverse variance	-0.0611	0.0189	0.9407
Cluster-Robust Cox	Cluster-number	-0.0585	0.0199	0.9432
Cluster-Robust Cox	Cow-number	-0.0587	0.0198	0.9430
Copula Model	Information matrix	-0.0611	0.0189	0.9407
Copula Model	Inverse variance	-0.0611	0.0189	0.9407
Copula Model	Cluster-number	-0.0585	0.0199	0.9432
Copula Model	Cow-number	-0.0587	0.0198	0.9430
Kendall's τ	Information matrix	0.1814	0.0148	—
Kendall's τ	Inverse variance	0.1814	0.0148	—
Kendall's τ	Cluster-number	0.1800	0.0149	—
Kendall's τ	Cow-number	0.1802	0.0149	—

Table 3.48: Comparison of the three-subset combined estimates with the Q1+Q2 full-data benchmark

Model	Weighting Method	Combined Est.	Full Data Est.	Abs. Diff.	Rel. Diff.
Standard Cox	Information matrix	-0.0587	-0.0603	0.0016	0.0265
Standard Cox	Inverse variance	-0.0587	-0.0603	0.0016	0.0265
Standard Cox	Cluster-number	-0.0585	-0.0603	0.0018	0.0299
Standard Cox	Cow-number	-0.0587	-0.0603	0.0016	0.0265
Cluster-Robust Cox	Information matrix	-0.0611	-0.0603	0.0008	0.0133
Cluster-Robust Cox	Inverse variance	-0.0611	-0.0603	0.0008	0.0133
Cluster-Robust Cox	Cluster-number	-0.0585	-0.0603	0.0018	0.0299
Cluster-Robust Cox	Cow-number	-0.0587	-0.0603	0.0016	0.0265
Copula Model	Information matrix	-0.0611	-0.0603	0.0008	0.0133
Copula Model	Inverse variance	-0.0611	-0.0603	0.0008	0.0133
Copula Model	Cluster-number	-0.0585	-0.0603	0.0018	0.0299
Copula Model	Cow-number	-0.0587	-0.0603	0.0016	0.0265
Kendall's τ	Information matrix	0.1814	0.1813	0.0001	0.0006
Kendall's τ	Inverse variance	0.1814	0.1813	0.0001	0.0006
Kendall's τ	Cluster-number	0.1800	0.1813	0.0013	0.0072
Kendall's τ	Cow-number	0.1802	0.1813	0.0011	0.0061

Table 3.49: Comparison of the three-subset combined standard errors with the Q1+Q2 full-data benchmark

Model	Weighting Method	Combined SE	Full Data SE	Abs. Diff.	Rel. Diff.
Standard Cox	Information matrix	0.0205	0.0205	0.0000	0.0000
Standard Cox	Inverse variance	0.0205	0.0205	0.0000	0.0000
Standard Cox	Cluster-number	0.0205	0.0205	0.0000	0.0000
Standard Cox	Cow-number	0.0205	0.0205	0.0000	0.0000
Cluster-Robust Cox	Information matrix	0.0189	0.0196	0.0007	0.0357
Cluster-Robust Cox	Inverse variance	0.0189	0.0196	0.0007	0.0357
Cluster-Robust Cox	Cluster-number	0.0199	0.0196	0.0003	0.0153
Cluster-Robust Cox	Cow-number	0.0198	0.0196	0.0002	0.0102
Copula Model	Information matrix	0.0189	0.0196	0.0007	0.0357
Copula Model	Inverse variance	0.0189	0.0196	0.0007	0.0357
Copula Model	Cluster-number	0.0199	0.0196	0.0003	0.0153
Copula Model	Cow-number	0.0198	0.0196	0.0002	0.0102
Kendall's τ	Information matrix	0.0148	0.0147	0.0001	0.0068
Kendall's τ	Inverse variance	0.0148	0.0147	0.0001	0.0068
Kendall's τ	Cluster-number	0.0149	0.0147	0.0002	0.0136
Kendall's τ	Cow-number	0.0149	0.0147	0.0002	0.0136

Table 3.50: Computational summary for the three-subset analysis after Q1 and Q2 splitting

Subset	Observations	Clusters	Original Herds	User Time	System Time	Elapsed Time	Memory (MB)
1	3547	106	84	53.77	0.86	55.80	0.3
2	3565	106	90	55.86	0.64	58.36	0.3
3	3401	106	90	42.28	0.66	43.21	0.2

Table 3.51: Data description for the five-subset analysis after Q1 and Q2 splitting

Subset	Observations	Modified Clusters	Original Herds	Cluster Share
1	2168	64	58	20.13%
2	2056	63	56	19.81%
3	2179	64	58	20.13%
4	2025	63	59	19.81%
5	2085	64	62	20.13%
Total	10513	318	181	100%

Subset-Specific Model Results

Table 3.52: Subset-specific model results for the five-subset analysis after Q1 and Q2 splitting

Subset	Model	Estimate	Standard Error	Hazard Ratio
1	Standard Cox	-0.0554	0.0452	0.9461
1	Cluster-Robust Cox	-0.0554	0.0402	0.9461
1	Copula Model	-0.0554	0.0402	0.9461
1	Kendall's τ	0.1895	0.0274	—
2	Standard Cox	-0.0767	0.0463	0.9262
2	Cluster-Robust Cox	-0.0767	0.0379	0.9262
2	Copula Model	-0.0767	0.0379	0.9262
2	Kendall's τ	0.1866	0.0352	—
3	Standard Cox	-0.0120	0.0451	0.9881
3	Cluster-Robust Cox	-0.0120	0.0370	0.9881
3	Copula Model	-0.0120	0.0370	0.9881
3	Kendall's τ	0.2003	0.0378	—
4	Standard Cox	-0.0771	0.0474	0.9258
4	Cluster-Robust Cox	-0.0771	0.0584	0.9258
4	Copula Model	-0.0771	0.0584	0.9258
4	Kendall's τ	0.1655	0.0355	—
5	Standard Cox	-0.0885	0.0456	0.9153
5	Cluster-Robust Cox	-0.0885	0.0421	0.9153
5	Copula Model	-0.0885	0.0421	0.9153
5	Kendall's τ	0.1494	0.0294	—

Table 3.53: Combined results from the five-subset analysis after Q1 and Q2 splitting

Model	Weighting Method	Combined Estimate	Standard Error	Hazard Ratio
Standard Cox	Information matrix	-0.0614	0.0205	0.9404
Standard Cox	Inverse variance	-0.0614	0.0205	0.9404
Standard Cox	Cluster-number	-0.0618	0.0205	0.9401
Standard Cox	Cow-number	-0.0613	0.0205	0.9405
Cluster-Robust Cox	Information matrix	-0.0582	0.0186	0.9435
Cluster-Robust Cox	Inverse variance	-0.0582	0.0186	0.9435
Cluster-Robust Cox	Cluster-number	-0.0618	0.0196	0.9401
Cluster-Robust Cox	Cow-number	-0.0613	0.0195	0.9405
Copula Model	Information matrix	-0.0582	0.0186	0.9435
Copula Model	Inverse variance	-0.0582	0.0186	0.9435
Copula Model	Cluster-number	-0.0618	0.0196	0.9401
Copula Model	Cow-number	-0.0613	0.0195	0.9405
Kendall's τ	Information matrix	0.1769	0.0144	—
Kendall's τ	Inverse variance	0.1769	0.0144	—
Kendall's τ	Cluster-number	0.1783	0.0149	—
Kendall's τ	Cow-number	0.1786	0.0149	—

Table 3.54: Comparison of the five-subset combined estimates with the Q1+Q2 full-data benchmark

Model	Weighting Method	Combined Est.	Full Data Est.	Abs. Diff.	Rel. Diff.
Standard Cox	Information matrix	-0.0614	-0.0603	0.0011	0.0182
Standard Cox	Inverse variance	-0.0614	-0.0603	0.0011	0.0182
Standard Cox	Cluster-number	-0.0618	-0.0603	0.0015	0.0249
Standard Cox	Cow-number	-0.0613	-0.0603	0.0010	0.0166
Cluster-Robust Cox	Information matrix	-0.0582	-0.0603	0.0021	0.0348
Cluster-Robust Cox	Inverse variance	-0.0582	-0.0603	0.0021	0.0348
Cluster-Robust Cox	Cluster-number	-0.0618	-0.0603	0.0015	0.0249
Cluster-Robust Cox	Cow-number	-0.0613	-0.0603	0.0010	0.0166
Copula Model	Information matrix	-0.0582	-0.0603	0.0021	0.0348
Copula Model	Inverse variance	-0.0582	-0.0603	0.0021	0.0348
Copula Model	Cluster-number	-0.0618	-0.0603	0.0015	0.0249
Copula Model	Cow-number	-0.0613	-0.0603	0.0010	0.0166
Kendall's τ	Information matrix	0.1769	0.1813	0.0044	0.0243
Kendall's τ	Inverse variance	0.1769	0.1813	0.0044	0.0243
Kendall's τ	Cluster-number	0.1783	0.1813	0.0030	0.0165
Kendall's τ	Cow-number	0.1786	0.1813	0.0027	0.0149

Table 3.55: Comparison of the five-subset combined standard errors with the Q1+Q2 full-data benchmark

Model	Weighting Method	Combined SE	Full Data SE	Abs. Diff.	Rel. Diff.
Standard Cox	Information matrix	0.0205	0.0205	0.0000	0.0000
Standard Cox	Inverse variance	0.0205	0.0205	0.0000	0.0000
Standard Cox	Cluster-number	0.0205	0.0205	0.0000	0.0000
Standard Cox	Cow-number	0.0205	0.0205	0.0000	0.0000
Cluster-Robust Cox	Information matrix	0.0186	0.0196	0.0010	0.0510
Cluster-Robust Cox	Inverse variance	0.0186	0.0196	0.0010	0.0510
Cluster-Robust Cox	Cluster-number	0.0196	0.0196	0.0000	0.0000
Cluster-Robust Cox	Cow-number	0.0195	0.0196	0.0001	0.0051
Copula Model	Information matrix	0.0186	0.0196	0.0010	0.0510
Copula Model	Inverse variance	0.0186	0.0196	0.0010	0.0510
Copula Model	Cluster-number	0.0196	0.0196	0.0000	0.0000
Copula Model	Cow-number	0.0195	0.0196	0.0001	0.0051
Kendall's τ	Information matrix	0.0144	0.0147	0.0003	0.0204
Kendall's τ	Inverse variance	0.0144	0.0147	0.0003	0.0204
Kendall's τ	Cluster-number	0.0149	0.0147	0.0002	0.0136
Kendall's τ	Cow-number	0.0149	0.0147	0.0002	0.0136

Table 3.56: Computational summary for the five-subset analysis after Q1 and Q2 splitting

Subset	Observations	Clusters	Original Herds	User Time	System Time	Elapsed Time	Memory (MB)
1	2168	64	58	17.54	0.34	17.90	0.1
2	2056	63	56	16.81	0.25	17.06	0.1
3	2179	64	58	18.38	0.19	18.66	0.3
4	2025	63	59	15.50	0.29	15.81	0.2
5	2085	64	62	16.25	0.27	16.56	0.2

Table 3.57: Data description for the eight-subset analysis after Q1 and Q2 splitting

Subset	Observations	Modified Clusters	Original Herds	Cluster Share
1	1324	40	35	12.58%
2	1268	40	36	12.58%
3	1297	40	37	12.58%
4	1256	40	36	12.58%
5	1304	40	37	12.58%
6	1336	40	38	12.58%
7	1350	39	36	12.26%
8	1378	39	38	12.26%
Total	10513	318	181	100%

Table 3.58: Subset-specific model results for the eight-subset analysis after Q1 and Q2 splitting

Subset	Model	Estimate	Standard Error	Hazard Ratio
1	Standard Cox	-0.0961	0.0584	0.9084
1	Cluster-Robust Cox	-0.0961	0.0499	0.9084
1	Copula Model	-0.0961	0.0499	0.9084
1	Kendall's τ	0.1834	0.0395	—
2	Standard Cox	-0.0124	0.0562	0.9877
2	Cluster-Robust Cox	-0.0124	0.0605	0.9877
2	Copula Model	-0.0124	0.0605	0.9877
2	Kendall's τ	0.1695	0.0441	—
3	Standard Cox	-0.0488	0.0571	0.9524
3	Cluster-Robust Cox	-0.0488	0.0533	0.9524
3	Copula Model	-0.0488	0.0533	0.9524
3	Kendall's τ	0.1752	0.0470	—
4	Standard Cox	-0.0882	0.0580	0.9156
4	Cluster-Robust Cox	-0.0882	0.0551	0.9156
4	Copula Model	-0.0882	0.0551	0.9156
4	Kendall's τ	0.1524	0.0388	—
5	Standard Cox	-0.0411	0.0573	0.9597
5	Cluster-Robust Cox	-0.0411	0.0508	0.9597
5	Copula Model	-0.0411	0.0508	0.9597
5	Kendall's τ	0.2113	0.0456	—
6	Standard Cox	-0.0675	0.0591	0.9347
6	Cluster-Robust Cox	-0.0675	0.0517	0.9347
6	Copula Model	-0.0675	0.0517	0.9347
6	Kendall's τ	0.1940	0.0463	—
7	Standard Cox	-0.0217	0.0587	0.9785
7	Cluster-Robust Cox	-0.0217	0.0639	0.9785
7	Copula Model	-0.0217	0.0639	0.9785
7	Kendall's τ	0.1608	0.0502	—
8	Standard Cox	-0.0794	0.0569	0.9237
8	Cluster-Robust Cox	-0.0794	0.0521	0.9237
8	Copula Model	-0.0794	0.0521	0.9237
8	Kendall's τ	0.1729	0.0474	—

Table 3.59: Combined results from the eight-subset analysis after Q1 and Q2 splitting

Model	Weighting Method	Combined Estimate	Standard Error	Hazard Ratio
Standard Cox	Information matrix	-0.0591	0.0205	0.9426
Standard Cox	Inverse variance	-0.0591	0.0205	0.9426
Standard Cox	Cluster-number	-0.0584	0.0205	0.9433
Standard Cox	Cow-number	-0.0588	0.0205	0.9429
Cluster-Robust Cox	Information matrix	-0.0560	0.0182	0.9455
Cluster-Robust Cox	Inverse variance	-0.0560	0.0182	0.9455
Cluster-Robust Cox	Cluster-number	-0.0584	0.0192	0.9433
Cluster-Robust Cox	Cow-number	-0.0588	0.0191	0.9429
Copula Model	Information matrix	-0.0560	0.0182	0.9455
Copula Model	Inverse variance	-0.0560	0.0182	0.9455
Copula Model	Cluster-number	-0.0584	0.0192	0.9433
Copula Model	Cow-number	-0.0588	0.0191	0.9429
Kendall's τ	Information matrix	0.1761	0.0140	—
Kendall's τ	Inverse variance	0.1761	0.0140	—
Kendall's τ	Cluster-number	0.1774	0.0145	—
Kendall's τ	Cow-number	0.1776	0.0145	—

Table 3.60: Comparison of the eight-subset combined estimates with the Q1+Q2 full-data benchmark

Model	Weighting Method	Combined Est.	Full Data Est.	Abs. Diff.	Rel. Diff.
Standard Cox	Information matrix	-0.0591	-0.0603	0.0012	0.0199
Standard Cox	Inverse variance	-0.0591	-0.0603	0.0012	0.0199
Standard Cox	Cluster-number	-0.0584	-0.0603	0.0019	0.0315
Standard Cox	Cow-number	-0.0588	-0.0603	0.0015	0.0249
Cluster-Robust Cox	Information matrix	-0.0560	-0.0603	0.0043	0.0713
Cluster-Robust Cox	Inverse variance	-0.0560	-0.0603	0.0043	0.0713
Cluster-Robust Cox	Cluster-number	-0.0584	-0.0603	0.0019	0.0315
Cluster-Robust Cox	Cow-number	-0.0588	-0.0603	0.0015	0.0249
Copula Model	Information matrix	-0.0560	-0.0603	0.0043	0.0713
Copula Model	Inverse variance	-0.0560	-0.0603	0.0043	0.0713
Copula Model	Cluster-number	-0.0584	-0.0603	0.0019	0.0315
Copula Model	Cow-number	-0.0588	-0.0603	0.0015	0.0249
Kendall's τ	Information matrix	0.1761	0.1813	0.0052	0.0287
Kendall's τ	Inverse variance	0.1761	0.1813	0.0052	0.0287
Kendall's τ	Cluster-number	0.1774	0.1813	0.0039	0.0215
Kendall's τ	Cow-number	0.1776	0.1813	0.0037	0.0204

Table 3.61: Comparison of the eight-subset combined standard errors with the Q1+Q2 full-data benchmark

Model	Weighting Method	Combined SE	Full Data SE	Abs. Diff.	Rel. Diff.
Standard Cox	Information matrix	0.0205	0.0205	0.0000	0.0000
Standard Cox	Inverse variance	0.0205	0.0205	0.0000	0.0000
Standard Cox	Cluster-number	0.0205	0.0205	0.0000	0.0000
Standard Cox	Cow-number	0.0205	0.0205	0.0000	0.0000
Cluster-Robust Cox	Information matrix	0.0182	0.0196	0.0014	0.0714
Cluster-Robust Cox	Inverse variance	0.0182	0.0196	0.0014	0.0714
Cluster-Robust Cox	Cluster-number	0.0192	0.0196	0.0004	0.0204
Cluster-Robust Cox	Cow-number	0.0191	0.0196	0.0005	0.0255
Copula Model	Information matrix	0.0182	0.0196	0.0014	0.0714
Copula Model	Inverse variance	0.0182	0.0196	0.0014	0.0714
Copula Model	Cluster-number	0.0192	0.0196	0.0004	0.0204
Copula Model	Cow-number	0.0191	0.0196	0.0005	0.0255
Kendall's τ	Information matrix	0.0140	0.0147	0.0007	0.0476
Kendall's τ	Inverse variance	0.0140	0.0147	0.0007	0.0476
Kendall's τ	Cluster-number	0.0145	0.0147	0.0002	0.0136
Kendall's τ	Cow-number	0.0145	0.0147	0.0002	0.0136

Table 3.62: Computational summary for the eight-subset analysis after Q1 and Q2 splitting

Subset	Observations	Clusters	Original Herds	User Time	System Time	Elapsed Time	Memory (MB)
1	1324	40	35	8.24	0.10	8.53	0.1
2	1268	40	36	7.58	0.08	7.82	0.1
3	1297	40	37	7.91	0.06	8.14	0.1
4	1256	40	36	7.63	0.09	7.91	0.1
5	1304	40	37	7.95	0.11	8.24	0.1
6	1336	40	38	8.33	0.10	8.62	0.1
7	1350	39	36	8.45	0.12	8.76	0.1
8	1378	39	38	8.60	0.09	8.89	0.1

Table 3.63: Overall comparison of combined estimates across full-data and sample-splitting analyses

Analysis	Number of Subsets	Combined Estimate	Hazard Ratio
Original full data	—	-0.0603	0.9414
Original random splitting	3	-0.0603	0.9414
Original random splitting	5	-0.0603	0.9414
Original random splitting	8	-0.0603	0.9414
Q1 full data	—	-0.0603	0.9414
Q1 random splitting	3	-0.0609	0.9409
Q1 random splitting	5	-0.0604	0.9414
Q1 random splitting	8	-0.0567	0.9449
Q1+Q2 full data	—	-0.0603	0.9414
Q1+Q2 random splitting	3	-0.0611	0.9407
Q1+Q2 random splitting	5	-0.0582	0.9435
Q1+Q2 random splitting	8	-0.0560	0.9455

Table 3.64: Overall comparison of standard errors across full-data and sample-splitting analyses

Analysis	Number of Subsets	Standard Error
Original full data	—	0.0210
Original random splitting	3	0.0210
Original random splitting	5	0.0197
Original random splitting	8	0.0193
Q1 full data	—	0.0212
Q1 random splitting	3	0.0211
Q1 random splitting	5	0.0197
Q1 random splitting	8	0.0193
Q1+Q2 full data	—	0.0196
Q1+Q2 random splitting	3	0.0189
Q1+Q2 random splitting	5	0.0186
Q1+Q2 random splitting	8	0.0182

Table 3.65: Overall comparison of Kendall's τ estimates across full data and sample-splitting analyses

Analysis	Number of Subsets	Kendall's τ
Original full data	—	0.1828
Original random splitting	3	0.1826
Original random splitting	5	0.1807
Original random splitting	8	0.1777
Q1 full data	—	0.1828
Q1 random splitting	3	0.1826
Q1 random splitting	5	0.1807
Q1 random splitting	8	0.1778
Q1+Q2 full data	—	0.1813
Q1+Q2 random splitting	3	0.1814
Q1+Q2 random splitting	5	0.1786
Q1+Q2 random splitting	8	0.1776

Table 3.66: Overall comparison of Kendall's τ standard errors across full data and sample-splitting analyses

Analysis	Number of Subsets	Standard Error
Original full data	—	0.0210
Original random splitting	3	0.0210
Original random splitting	5	0.0159
Original random splitting	8	0.0159
Q1 full data	—	0.0210
Q1 random splitting	3	0.0210
Q1 random splitting	5	0.0159
Q1 random splitting	8	0.0159
Q1+Q2 full data	—	0.0147
Q1+Q2 random splitting	3	0.0149
Q1+Q2 random splitting	5	0.0149
Q1+Q2 random splitting	8	0.0145



Table 3.67: Overall comparison of computational cost across full-data and sample-splitting analyses

Analysis	Number of Subsets	Total Clusters	User Time (s)	System Time (s)	Elapsed Time (s)	Memory Used (MB)
Original full data	–	181	220.95	5.86	260.12	0.1
Original random splitting	3	181	70.87	1.53	87.35	0.3
Original random splitting	5	181	40.97	0.82	52.56	0.1–0.2
Original random splitting	8	181	30.85	0.71	42.46	0.1
Q1 full data	–	273	310.32	12.36	371.72	4.9
Q1 random splitting	3	273	105.67	2.51	125.47	0.3–0.4
Q1 random splitting	5	273	67.74	1.41	79.74	0.1–0.2
Q1 random splitting	8	273	46.71	0.72	49.99	0.1
Q1+Q2 full data	–	318	468.55	14.85	502.97	4.9
Q1+Q2 random splitting	3	318	151.91	2.16	157.37	0.2–0.3
Q1+Q2 random splitting	5	318	84.48	1.34	85.99	0.1–0.3
Q1+Q2 random splitting	8	318	64.69	0.75	66.91	0.1

Listing 1: R code for full data analysis

```
# 2. LOAD PACKAGES
library(survival)
library(dplyr)
library(ggplot2)
library(Suncclarco)

# 3. LOAD DATA
data(insem)

# 4. DATA DESCRIPTION

# Structure of the dataset
str(insem)

# First six observations
head(insem)

# Variable names
names(insem)

# Summary statistics
summary(insem)

# Number of observations
nrow(insem)

# Number of variables
ncol(insem)

# Number of unique herds (clusters)
length(unique(insem$Herd))

# Event vs censoring
table(insem$Status)

# Censoring proportion
mean(insem$Status == 0)

# Herd size distribution
table(insem$Herd)
summary(table(insem$Herd))

# Distribution of Heifer
table(insem$Heifer)

# 5. CREATE A SIMPLE SUMMARY TABLE
data_summary <- data.frame(
  Measure = c(
```

```

      "Number of observations",
      "Number of variables",
      "Number of herds",
      "Number of events",
      "Number censored",
      "Censoring proportion"
    ),
    Value = c(
      nrow(insem),
      ncol(insem),
      length(unique(insem$Herd)),
      sum(insem$Status == 1),
      sum(insem$Status == 0),
      round(mean(insem$Status == 0), 4)
    )
  )
)

data_summary

# 6. SURVIVAL OBJECT

surv_obj <- Surv(time = insem$Time, event = insem$Status)

# first few entries
head(surv_obj)

# 7. MARGINAL COX MODEL

cox_model <- coxph(Surv(Time, Status) ~ Heifer, data = insem)

summary(cox_model)

# Extract regression coefficient
coef(cox_model)

# Hazard ratio
exp(coef(cox_model))

# Standard error
sqrt(diag(vcov(cox_model)))

# 8. CLUSTER-ROBUST COX MODEL

cox_cluster <- coxph(Surv(Time, Status) ~ Heifer + cluster(Herd), data = insem)

summary(cox_cluster)

# Extract regression coefficient
coef(cox_cluster)

# Hazard ratio
exp(coef(cox_cluster))

# 9. FULL DATA COPULA MODEL USING SUNCLARCO + COMPUTATIONAL COST

gc()
mem_before <- sum(gc()[, 2])

time_copula_full <- system.time({

  copula_fit <- SunclarcoModel(
    data = insem,
    time = "Time",
    status = "Status",
    clusters = "Herd",
    covariates = "Heifer",
    stage = 2,
    copula = "Clayton",
    marginal = "Cox"
  )
})

gc()
mem_after <- sum(gc()[, 2])

```

```

memory_used_full <- mem_after - mem_before

cat("\n=====\\n")
cat("FULL DATA COPULA COMPUTATIONAL COST\\n")
cat("=====\\n")
cat("Copula user time:", time_copula_full["user.self"], "seconds\\n")
cat("Copula system time:", time_copula_full["sys.self"], "seconds\\n")
cat("Copula elapsed time:", time_copula_full["elapsed"], "seconds\\n")
cat("Approximate memory used:", memory_used_full, "MB\\n")

# 10. VIEW COPULA MODEL OUTPUT

copula_fit
str(copula_fit)

# 10B. FULL DATA COMPUTATIONAL COST SUMMARY TABLE

full_computational_summary <- data.frame(
  Analysis = "Full data",
  Observations = nrow(insem),
  Herds = length(unique(insem$Herd)),

  Copula_User_Time = as.numeric(time_copula_full["user.self"]),
  Copula_System_Time = as.numeric(time_copula_full["sys.self"]),
  Copula_Elapsed_Time = as.numeric(time_copula_full["elapsed"]),

  Approx_Memory_Used_MB = memory_used_full
)

full_computational_summary

# 11. EXTRACT PARAMETER ESTIMATES

# Copula model parameters
cop_beta <- as.numeric(copula_fit$Parameters["beta_Heifer", "Estimates"])
cop_se <- as.numeric(copula_fit$Parameters["beta_Heifer", "StandardErrors"])

theta_est <- as.numeric(copula_fit$Parameters["theta", "Estimates"])
theta_se <- as.numeric(copula_fit$Parameters["theta", "StandardErrors"])

# Kendall's tau
tau_est <- as.numeric(copula_fit$Kendall_Tau["Estimates"])
tau_se <- as.numeric(copula_fit$Kendall_Tau["StandardErrors"])

cop_beta
cop_se
theta_est
theta_se
tau_est
tau_se

# 12. CREATE RESULTS TABLE

results_table <- data.frame(
  Model = c("Marginal Cox", "Cluster-Robust Cox", "Copula Model"),
  Beta_Heifer = c(
    as.numeric(coef(cox_model)[1]),
    as.numeric(coef(cox_cluster)[1]),
    cop_beta
  ),
  SE = c(
    as.numeric(sqrt(diag(vcov(cox_model)))[1]),
    as.numeric(sqrt(diag(vcov(cox_cluster)))[1]),
    cop_se
  )
)

results_table$Hazard_Ratio <- exp(results_table$Beta_Heifer)

results_table

# 13. PRINT KEY RESULTS

```

```

cat("\n===== FULL DATA ANALYSIS RESULTS =====\n")
cat("Number of observations :", nrow(insem), "\n")
cat("Number of herds      :", length(unique(insem$Herd)), "\n")
cat("Censoring proportion  :", round(mean(insem$Status == 0), 4), "\n\n")

cat("Marginal Cox model:\n")
cat("  Beta (Heifer)  =", round(as.numeric(coef(cox_model)[1]), 6), "\n")
cat("  SE              =", round(as.numeric(sqrt(diag(vcov(cox_model)))[1]), 6), "\n")
cat("  Hazard Ratio    =", round(as.numeric(exp(coef(cox_model)[1])), 6), "\n\n")

cat("Cluster-robust Cox model:\n")
cat("  Beta (Heifer)  =", round(as.numeric(coef(cox_cluster)[1]), 6), "\n")
cat("  Robust SE      =", round(as.numeric(sqrt(diag(vcov(cox_cluster)))[1]), 6), "\n")
cat("  Hazard Ratio    =", round(as.numeric(exp(coef(cox_cluster)[1])), 6), "\n\n")

cat("Copula model (Clayton, stage = 2, marginal = Cox):\n")
cat("  Beta (Heifer)  =", round(cop_beta, 6), "\n")
cat("  Beta SE        =", round(cop_se, 6), "\n")
cat("  Hazard Ratio    =", round(exp(cop_beta), 6), "\n")
cat("  Theta          =", round(theta_est, 6), "\n")
cat("  Theta SE       =", round(theta_se, 6), "\n")
cat("  Kendall tau     =", round(tau_est, 6), "\n")
cat("  Tau SE         =", round(tau_se, 6), "\n")
cat("===== \n")

# 14. SAVE RESULTS

write.csv(data_summary, "data_summary_full_analysis.csv", row.names = FALSE)
write.csv(results_table, "results_table_full_analysis.csv", row.names = FALSE)

```

Listing 2: Part 1 Three Subset Random Splitting

```

# 1. LOAD PACKAGES

library(survival)
library(Sunclarco)

# 2. LOAD DATA

data(insem)

# 3. FULL-DATA BENCHMARK VALUES

full_std_beta <- -0.06034839
full_std_se   <- 0.02050279

full_cl_beta  <- -0.06034839
full_cl_se    <- 0.02096202

full_cop_beta <- -0.06034839
full_cop_se   <- 0.02096202

full_tau      <- 0.1828338
full_tau_se   <- 0.02103043

# 4. HELPER FUNCTIONS

combine_scalar <- function(beta, var, weights) {
  beta_comb <- sum(weights * beta) / sum(weights)
  var_comb  <- sum((weights^2) * var) / (sum(weights)^2)
  se_comb   <- sqrt(var_comb)

  list(beta = beta_comb, var = var_comb, se = se_comb)
}

evaluation_metrics <- function(combined_est, combined_se, reference_est, reference_se) {
  data.frame(
    Combined_Estimate = combined_est,
    Reference_Estimate = reference_est,
    Absolute_Difference = abs(combined_est - reference_est),
    Relative_Difference = abs(combined_est - reference_est) / abs(reference_est),
    Combined_SE        = combined_se,
    Reference_SE        = reference_se,

```

```

    SE_Difference      = abs(combined_se - reference_se),
    Bias              = combined_est - reference_est
  )
}

round_df <- function(df, digits = 4) {
  df[sapply(df, is.numeric)] <- round(df[sapply(df, is.numeric)], digits)
  df
}

# 5. RANDOM SPLIT INTO 3 SUBSETS

set.seed(123)

herds <- sample(unique(insem$Herd))
herd_groups <- split(herds, cut(seq_along(herds), 3, labels = FALSE))

subset_list <- lapply(herd_groups, function(h) {
  insem[insem$Herd %in% h, ]
})

cat("\n===== \n")
cat("SUBSET SIZES: 3-SUBSET ANALYSIS \n")
cat("===== \n")

cat("\nObservations in each subset: \n")
print(sapply(subset_list, nrow))

cat("\nHerds in each subset: \n")
print(sapply(subset_list, function(x) length(unique(x$Herd))))

cat("\nEvents in each subset: \n")
print(sapply(subset_list, function(x) sum(x$Status == 1)))

cat("\nCensored observations in each subset: \n")
print(sapply(subset_list, function(x) sum(x$Status == 0)))

# 6. FIT MODELS IN EACH SUBSET + COMPUTATIONAL COST

results <- lapply(seq_along(subset_list), function(i) {

  dat <- subset_list[[i]]

  cat("\n===== \n")
  cat("FITTING MODELS FOR SUBSET", i, "\n")
  cat("===== \n")

  # Memory before fitting
  gc()
  mem_before <- sum(gc()[, 2])

  # Runtime for all three models in this subset
  time_total <- system.time({

    # Standard Cox model
    cox_standard <- coxph(Surv(Time, Status) ~ Heifer, data = dat)

    # Cluster-robust Cox model
    cox_cluster <- coxph(Surv(Time, Status) ~ Heifer + cluster(Herd), data = dat)

    # Runtime only for Sunclarco copula model
    time_copula <- system.time({
      copula_fit <- SunclarcoModel(
        data      = dat,
        time      = "Time",
        status     = "Status",
        clusters   = "Herd",
        covariates = "Heifer",
        stage      = 2,
        copula     = "Clayton",
        marginal   = "Cox"
      )
    })
  })

  # Memory after fitting
  gc()
  mem_after <- sum(gc()[, 2])

```

```

memory_used <- mem_after - mem_before

cat("\n--- Standard Cox Model: Subset", i, "---\n")
print(summary(cox_standard))

cat("\n--- Cluster-Robust Cox Model: Subset", i, "---\n")
print(summary(cox_cluster))

cat("\n--- Sunclarco Copula Model: Subset", i, "---\n")
print(copula_fit)

cat("\n--- Computational Cost: Subset", i, "---\n")
cat("Total elapsed time:", time_total["elapsed"], "seconds\n")
cat("Copula elapsed time:", time_copula["elapsed"], "seconds\n")
cat("Approximate memory used:", memory_used, "MB\n")

# Extract standard Cox results
std_beta <- as.numeric(coef(cox_standard)[1])
std_var <- as.numeric(vcov(cox_standard)[1, 1])
std_se <- sqrt(std_var)
std_hr <- exp(std_beta)

# Extract cluster-robust Cox results
cl_beta <- as.numeric(coef(cox_cluster)[1])
cl_var_robust <- as.numeric(vcov(cox_cluster)[1, 1])
cl_var_naive <- as.numeric(cox_cluster$naive.var[1, 1])
cl_se_robust <- sqrt(cl_var_robust)
cl_se_naive <- sqrt(cl_var_naive)
cl_hr <- exp(cl_beta)

# Extract copula results
cop_beta <- as.numeric(copula_fit$Parameters["beta_Heifer", "Estimates"])
cop_se <- as.numeric(copula_fit$Parameters["beta_Heifer", "StandardErrors"])
cop_var <- cop_se^2
cop_hr <- exp(cop_beta)

theta_est <- as.numeric(copula_fit$Parameters["theta", "Estimates"])
theta_se <- as.numeric(copula_fit$Parameters["theta", "StandardErrors"])

tau_est <- as.numeric(copula_fit$Kendall_Tau["Estimates"])
tau_se <- as.numeric(copula_fit$Kendall_Tau["StandardErrors"])
tau_var <- tau_se^2

list(
  subset_id = i,
  n_obs = nrow(dat),
  n_clusters = length(unique(dat$Herd)),

  total_user_time = as.numeric(time_total["user.self"]),
  total_system_time = as.numeric(time_total["sys.self"]),
  total_elapsed_time = as.numeric(time_total["elapsed"]),

  copula_user_time = as.numeric(time_copula["user.self"]),
  copula_system_time = as.numeric(time_copula["sys.self"]),
  copula_elapsed_time = as.numeric(time_copula["elapsed"]),

  memory_used_MB = memory_used,

  std_beta = std_beta,
  std_var = std_var,
  std_se = std_se,
  std_hr = std_hr,

  cl_beta = cl_beta,
  cl_var_robust = cl_var_robust,
  cl_var_naive = cl_var_naive,
  cl_se_robust = cl_se_robust,
  cl_se_naive = cl_se_naive,
  cl_hr = cl_hr,

  cop_beta = cop_beta,
  cop_var = cop_var,
  cop_se = cop_se,
  cop_hr = cop_hr,
  theta_est = theta_est,
  theta_se = theta_se,

  tau_est = tau_est,
  tau_var = tau_var,
  tau_se = tau_se

```

```

)
})

# 7A. COMPUTATIONAL COST SUMMARY TABLE

computational_summary <- data.frame(
  Subset = sapply(results, function(x) x$subset_id),
  Observations = sapply(results, function(x) x$n_obs),
  Herds = sapply(results, function(x) x$n_clusters),

  Total_User_Time = sapply(results, function(x) x$total_user_time),
  Total_System_Time = sapply(results, function(x) x$total_system_time),
  Total_Elapsed_Time = sapply(results, function(x) x$total_elapsed_time),

  Copula_User_Time = sapply(results, function(x) x$copula_user_time),
  Copula_System_Time = sapply(results, function(x) x$copula_system_time),
  Copula_Elapsed_Time = sapply(results, function(x) x$copula_elapsed_time),

  Approx_Memory_Used_MB = sapply(results, function(x) x$memory_used_MB)
)

cat("\n===== \n")
cat("COMPUTATIONAL COST SUMMARY \n")
cat("===== \n")
print(round_df(computational_summary, 4))

# 7. SUBSET SUMMARY TABLE

subset_summary <- data.frame(
  Subset = sapply(results, function(x) x$subset_id),
  Observations = sapply(results, function(x) x$n_obs),
  Herds = sapply(results, function(x) x$n_clusters),

  Standard_Cox_Beta = sapply(results, function(x) x$std_beta),
  Standard_Cox_SE = sapply(results, function(x) x$std_se),
  Standard_Cox_HR = sapply(results, function(x) x$std_hr),

  Cluster_Cox_Beta = sapply(results, function(x) x$cl_beta),
  Cluster_Cox_Robust_SE = sapply(results, function(x) x$cl_se_robust),
  Cluster_Cox_Naive_SE = sapply(results, function(x) x$cl_se_naive),
  Cluster_Cox_HR = sapply(results, function(x) x$cl_hr),

  Copula_Beta = sapply(results, function(x) x$cop_beta),
  Copula_SE = sapply(results, function(x) x$cop_se),
  Copula_HR = sapply(results, function(x) x$cop_hr),

  Theta = sapply(results, function(x) x$theta_est),
  Theta_SE = sapply(results, function(x) x$theta_se),

  Tau = sapply(results, function(x) x$tau_est),
  Tau_SE = sapply(results, function(x) x$tau_se)
)

cat("\n===== \n")
cat("SUBSET-SPECIFIC RESULTS \n")
cat("===== \n")
print(round_df(subset_summary, 4))

# 8. EXTRACT VECTORS FOR COMBINATION

n_clusters <- sapply(results, function(x) x$n_clusters)
n_cows <- sapply(results, function(x) x$n_obs)

std_beta <- sapply(results, function(x) x$std_beta)
std_var <- sapply(results, function(x) x$std_var)

cl_beta <- sapply(results, function(x) x$cl_beta)
cl_var_robust <- sapply(results, function(x) x$cl_var_robust)

cop_beta <- sapply(results, function(x) x$cop_beta)
cop_var <- sapply(results, function(x) x$cop_var)

tau_beta <- sapply(results, function(x) x$tau_est)
tau_var <- sapply(results, function(x) x$tau_var)

# 9. COMBINED ESTIMATES

```

```

# Standard Cox
std_comb_info <- combine_scalar(std_beta, std_var, 1 / std_var)
std_comb_iv   <- combine_scalar(std_beta, std_var, 1 / std_var)
std_comb_cln  <- combine_scalar(std_beta, std_var, n_clusters)
std_comb_cows <- combine_scalar(std_beta, std_var, n_cows)

# Cluster-Robust Cox
cl_comb_info <- combine_scalar(cl_beta, cl_var_robust, 1 / cl_var_robust)
cl_comb_iv   <- combine_scalar(cl_beta, cl_var_robust, 1 / cl_var_robust)
cl_comb_cln  <- combine_scalar(cl_beta, cl_var_robust, n_clusters)
cl_comb_cows <- combine_scalar(cl_beta, cl_var_robust, n_cows)

# Copula beta
cop_comb_info <- combine_scalar(cop_beta, cop_var, 1 / cop_var)
cop_comb_iv   <- combine_scalar(cop_beta, cop_var, 1 / cop_var)
cop_comb_cln  <- combine_scalar(cop_beta, cop_var, n_clusters)
cop_comb_cows <- combine_scalar(cop_beta, cop_var, n_cows)

# Kendall's tau
tau_comb_info <- combine_scalar(tau_beta, tau_var, 1 / tau_var)
tau_comb_iv   <- combine_scalar(tau_beta, tau_var, 1 / tau_var)
tau_comb_cln  <- combine_scalar(tau_beta, tau_var, n_clusters)
tau_comb_cows <- combine_scalar(tau_beta, tau_var, n_cows)

# 10. FINAL COMBINED TABLES

standard_combined_table <- data.frame(
  Method = c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
  Beta   = c(std_comb_info$beta, std_comb_iv$beta, std_comb_cln$beta, std_comb_cows$beta),
  SE     = c(std_comb_info$se,   std_comb_iv$se,   std_comb_cln$se,   std_comb_cows$se),
  HR     = exp(c(std_comb_info$beta, std_comb_iv$beta, std_comb_cln$beta, std_comb_cows$beta))
)

cluster_combined_table <- data.frame(
  Method = c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
  Beta   = c(cl_comb_info$beta, cl_comb_iv$beta, cl_comb_cln$beta, cl_comb_cows$beta),
  SE     = c(cl_comb_info$se,   cl_comb_iv$se,   cl_comb_cln$se,   cl_comb_cows$se),
  HR     = exp(c(cl_comb_info$beta, cl_comb_iv$beta, cl_comb_cln$beta, cl_comb_cows$beta))
)

copula_combined_table <- data.frame(
  Method = c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
  Beta   = c(cop_comb_info$beta, cop_comb_iv$beta, cop_comb_cln$beta, cop_comb_cows$beta),
  SE     = c(cop_comb_info$se,   cop_comb_iv$se,   cop_comb_cln$se,   cop_comb_cows$se),
  HR     = exp(c(cop_comb_info$beta, cop_comb_iv$beta, cop_comb_cln$beta, cop_comb_cows$beta))
)

tau_combined_table <- data.frame(
  Method = c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
  Tau    = c(tau_comb_info$beta, tau_comb_iv$beta, tau_comb_cln$beta, tau_comb_cows$beta),
  SE     = c(tau_comb_info$se,   tau_comb_iv$se,   tau_comb_cln$se,   tau_comb_cows$se)
)

cat("\n===== \n")
cat("COMBINED STANDARD COX RESULTS\n")
cat("===== \n")
print(round_df(standard_combined_table, 4))

cat("\n===== \n")
cat("COMBINED CLUSTER-ROBUST COX RESULTS\n")
cat("===== \n")
print(round_df(cluster_combined_table, 4))

cat("\n===== \n")
cat("COMBINED COPULA RESULTS\n")
cat("===== \n")
print(round_df(copula_combined_table, 4))

cat("\n===== \n")
cat("COMBINED KENDALL'S TAU RESULTS\n")
cat("===== \n")
print(round_df(tau_combined_table, 4))

# 11. EVALUATION AGAINST FULL DATA

eval_standard <- do.call(rbind, lapply(1:nrow(standard_combined_table), function(i) {
  cbind(Method = standard_combined_table$Method[i],

```



```

        evaluation_metrics(standard_combined_table$Beta[i],
                           standard_combined_table$SE[i],
                           full_std_beta,
                           full_std_se))
    )))

eval_cluster <- do.call(rbind, lapply(1:nrow(cluster_combined_table), function(i) {
  cbind(Method = cluster_combined_table$Method[i],
        evaluation_metrics(cluster_combined_table$Beta[i],
                           cluster_combined_table$SE[i],
                           full_cl_beta,
                           full_cl_se))
})))

eval_copula <- do.call(rbind, lapply(1:nrow(copula_combined_table), function(i) {
  cbind(Method = copula_combined_table$Method[i],
        evaluation_metrics(copula_combined_table$Beta[i],
                           copula_combined_table$SE[i],
                           full_cop_beta,
                           full_cop_se))
})))

eval_tau <- do.call(rbind, lapply(1:nrow(tau_combined_table), function(i) {
  cbind(Method = tau_combined_table$Method[i],
        evaluation_metrics(tau_combined_table$Tau[i],
                           tau_combined_table$SE[i],
                           full_tau,
                           full_tau_se))
})))

cat("\n===== \n")
cat("EVALUATION: STANDARD COX\n")
cat("===== \n")
print(round_df(as.data.frame(eval_standard), 4))

cat("\n===== \n")
cat("EVALUATION: CLUSTER-ROBUST COX\n")
cat("===== \n")
print(round_df(as.data.frame(eval_cluster), 4))

cat("\n===== \n")
cat("EVALUATION: COPULA MODEL\n")
cat("===== \n")
print(round_df(as.data.frame(eval_copula), 4))

cat("\n===== \n")
cat("EVALUATION: KENDALL'S TAU\n")
cat("===== \n")
print(round_df(as.data.frame(eval_tau), 4))

```

Listing 3: R code for five-subset sample splitting analysis

```

# 1. LOAD PACKAGES
library(survival)
library(Suncclarco)

# 2. LOAD DATA
data(insem)

# 3. FULL-DATA BENCHMARK VALUES
full_std_beta <- -0.06034839
full_std_se <- 0.02050279

full_cl_beta <- -0.06034839
full_cl_se <- 0.02096202

full_cop_beta <- -0.06034839
full_cop_se <- 0.02096202

full_tau <- 0.1828338
full_tau_se <- 0.02103043

# 4. HELPER FUNCTIONS
combine_scalar <- function(beta, var, weights) {
  beta_comb <- sum(weights * beta) / sum(weights)
  var_comb <- sum((weights^2) * var) / (sum(weights)^2)
  se_comb <- sqrt(var_comb)
  list(beta = beta_comb, var = var_comb, se = se_comb)
}

```

```

evaluation_metrics <- function(combined_est, combined_se, reference_est, reference_se) {
  data.frame(
    Combined_Estimate = combined_est,
    Reference_Estimate = reference_est,
    Absolute_Difference = abs(combined_est - reference_est),
    Relative_Difference = abs(combined_est - reference_est) / abs(reference_est),
    Combined_SE = combined_se,
    Reference_SE = reference_se,
    SE_Difference = abs(combined_se - reference_se),
    Bias = combined_est - reference_est
  )
}

round_df <- function(df, digits = 4) {
  df[sapply(df, is.numeric)] <- round(df[sapply(df, is.numeric)], digits)
  df
}

# 5. RANDOM SPLIT INTO 5 SUBSETS
set.seed(123)

herds <- sample(unique(insem$Herd))
herd_groups <- split(herds, cut(seq_along(herds), 5, labels = FALSE))

subset_list <- lapply(herd_groups, function(h) {
  insem[insem$Herd %in% h, ]
})

cat("\n===== \n")
cat("SUBSET SIZES: 5-SUBSET ANALYSIS \n")
cat("===== \n")

cat("\nObservations in each subset: \n")
print(sapply(subset_list, nrow))

cat("\nHerds in each subset: \n")
print(sapply(subset_list, function(x) length(unique(x$Herd))))

cat("\nEvents in each subset: \n")
print(sapply(subset_list, function(x) sum(x$Status == 1)))

cat("\nCensored observations in each subset: \n")
print(sapply(subset_list, function(x) sum(x$Status == 0)))

# 6. FIT MODELS IN EACH SUBSET + COMPUTATIONAL COST
results <- lapply(seq_along(subset_list), function(i) {
  dat <- subset_list[[i]]

  cat("\n===== \n")
  cat("FITTING MODELS FOR SUBSET", i, "\n")
  cat("===== \n")

  gc()
  mem_before <- sum(gc()[, 2])

  time_total <- system.time({
    cox_standard <- coxph(Surv(Time, Status) ~ Heifer, data = dat)
    cox_cluster <- coxph(Surv(Time, Status) ~ Heifer + cluster(Herd), data = dat)

    time_copula <- system.time({
      copula_fit <- SunclarcoModel(
        data = dat,
        time = "Time",
        status = "Status",
        clusters = "Herd",
        covariates = "Heifer",
        stage = 2,
        copula = "Clayton",
        marginal = "Cox"
      )
    })
  })

  gc()
  mem_after <- sum(gc()[, 2])
  memory_used <- mem_after - mem_before

  cat("\n--- Standard Cox Model: Subset", i, "--- \n")
  print(summary(cox_standard))

```

```

cat("\n--- Cluster-Robust Cox Model: Subset", i, "---\n")
print(summary(cox_cluster))

cat("\n--- Sunclarco Copula Model: Subset", i, "---\n")
print(copula_fit)

cat("\n--- Computational Cost: Subset", i, "---\n")
cat("Total elapsed time:", time_total["elapsed"], "seconds\n")
cat("Copula elapsed time:", time_copula["elapsed"], "seconds\n")
cat("Approximate memory used:", memory_used, "MB\n")

std_beta <- as.numeric(coef(cox_standard)[1])
std_var <- as.numeric(vcov(cox_standard)[1, 1])
std_se <- sqrt(std_var)
std_hr <- exp(std_beta)

cl_beta <- as.numeric(coef(cox_cluster)[1])
cl_var_robust <- as.numeric(vcov(cox_cluster)[1, 1])
cl_var_naive <- as.numeric(cox_cluster$naive.var[1, 1])
cl_se_robust <- sqrt(cl_var_robust)
cl_se_naive <- sqrt(cl_var_naive)
cl_hr <- exp(cl_beta)

cop_beta <- as.numeric(copula_fit$Parameters["beta_Heifer", "Estimates"])
cop_se <- as.numeric(copula_fit$Parameters["beta_Heifer", "StandardErrors"])
cop_var <- cop_se^2
cop_hr <- exp(cop_beta)

theta_est <- as.numeric(copula_fit$Parameters["theta", "Estimates"])
theta_se <- as.numeric(copula_fit$Parameters["theta", "StandardErrors"])

tau_est <- as.numeric(copula_fit$Kendall_Tau["Estimates"])
tau_se <- as.numeric(copula_fit$Kendall_Tau["StandardErrors"])
tau_var <- tau_se^2

list(
  subset_id = i,
  n_obs = nrow(dat),
  n_clusters = length(unique(dat$Herd)),

  total_user_time = as.numeric(time_total["user.self"]),
  total_system_time = as.numeric(time_total["sys.self"]),
  total_elapsed_time = as.numeric(time_total["elapsed"]),

  copula_user_time = as.numeric(time_copula["user.self"]),
  copula_system_time = as.numeric(time_copula["sys.self"]),
  copula_elapsed_time = as.numeric(time_copula["elapsed"]),

  memory_used_MB = memory_used,

  std_beta = std_beta,
  std_var = std_var,
  std_se = std_se,
  std_hr = std_hr,

  cl_beta = cl_beta,
  cl_var_robust = cl_var_robust,
  cl_var_naive = cl_var_naive,
  cl_se_robust = cl_se_robust,
  cl_se_naive = cl_se_naive,
  cl_hr = cl_hr,

  cop_beta = cop_beta,
  cop_var = cop_var,
  cop_se = cop_se,
  cop_hr = cop_hr,
  theta_est = theta_est,
  theta_se = theta_se,

  tau_est = tau_est,
  tau_var = tau_var,
  tau_se = tau_se
)
})

# 7A. COMPUTATIONAL COST SUMMARY TABLE
computational_summary5 <- data.frame(
  Subset = sapply(results, function(x) x$subset_id),
  Observations = sapply(results, function(x) x$n_obs),

```

```

Herds = sapply(results, function(x) x$n_clusters),

Total_User_Time = sapply(results, function(x) x$total_user_time),
Total_System_Time = sapply(results, function(x) x$total_system_time),
Total_Elapsed_Time = sapply(results, function(x) x$total_elapsed_time),

Copula_User_Time = sapply(results, function(x) x$copula_user_time),
Copula_System_Time = sapply(results, function(x) x$copula_system_time),
Copula_Elapsed_Time = sapply(results, function(x) x$copula_elapsed_time),

Approx_Memory_Used_MB = sapply(results, function(x) x$memory_used_MB)
)

cat("\n===== \n")
cat("COMPUTATIONAL COST SUMMARY: 5-SUBSET ANALYSIS\n")
cat("===== \n")
print(round_df(computational_summary5, 4))

# 7. SUBSET SUMMARY TABLE
subset_summary <- data.frame(
  Subset = sapply(results, function(x) x$subset_id),
  Observations = sapply(results, function(x) x$n_obs),
  Herds = sapply(results, function(x) x$n_clusters),

  Standard_Cox_Beta = sapply(results, function(x) x$std_beta),
  Standard_Cox_SE = sapply(results, function(x) x$std_se),
  Standard_Cox_HR = sapply(results, function(x) x$std_hr),

  Cluster_Cox_Beta = sapply(results, function(x) x$cl_beta),
  Cluster_Cox_Robust_SE = sapply(results, function(x) x$cl_se_robust),
  Cluster_Cox_Naive_SE = sapply(results, function(x) x$cl_se_naive),
  Cluster_Cox_HR = sapply(results, function(x) x$cl_hr),

  Copula_Beta = sapply(results, function(x) x$cop_beta),
  Copula_SE = sapply(results, function(x) x$cop_se),
  Copula_HR = sapply(results, function(x) x$cop_hr),

  Theta = sapply(results, function(x) x$theta_est),
  Theta_SE = sapply(results, function(x) x$theta_se),

  Tau = sapply(results, function(x) x$tau_est),
  Tau_SE = sapply(results, function(x) x$tau_se)
)

cat("\n===== \n")
cat("SUBSET-SPECIFIC RESULTS\n")
cat("===== \n")
print(round_df(subset_summary, 4))

# 8. EXTRACT VECTORS FOR COMBINATION
n_clusters <- sapply(results, function(x) x$n_clusters)
n_cows <- sapply(results, function(x) x$n_obs)

std_beta <- sapply(results, function(x) x$std_beta)
std_var <- sapply(results, function(x) x$std_var)

cl_beta <- sapply(results, function(x) x$cl_beta)
cl_var_robust <- sapply(results, function(x) x$cl_var_robust)

cop_beta <- sapply(results, function(x) x$cop_beta)
cop_var <- sapply(results, function(x) x$cop_var)

tau_beta <- sapply(results, function(x) x$tau_est)
tau_var <- sapply(results, function(x) x$tau_var)

# 9. COMBINED ESTIMATES
std_comb_info <- combine_scalar(std_beta, std_var, 1 / std_var)
std_comb_iv <- combine_scalar(std_beta, std_var, 1 / std_var)
std_comb_cln <- combine_scalar(std_beta, std_var, n_clusters)
std_comb_cows <- combine_scalar(std_beta, std_var, n_cows)

cl_comb_info <- combine_scalar(cl_beta, cl_var_robust, 1 / cl_var_robust)
cl_comb_iv <- combine_scalar(cl_beta, cl_var_robust, 1 / cl_var_robust)
cl_comb_cln <- combine_scalar(cl_beta, cl_var_robust, n_clusters)
cl_comb_cows <- combine_scalar(cl_beta, cl_var_robust, n_cows)

cop_comb_info <- combine_scalar(cop_beta, cop_var, 1 / cop_var)
cop_comb_iv <- combine_scalar(cop_beta, cop_var, 1 / cop_var)
cop_comb_cln <- combine_scalar(cop_beta, cop_var, n_clusters)
cop_comb_cows <- combine_scalar(cop_beta, cop_var, n_cows)

```

```

tau_comb_info <- combine_scalar(tau_beta, tau_var, 1 / tau_var)
tau_comb_iv <- combine_scalar(tau_beta, tau_var, 1 / tau_var)
tau_comb_cln <- combine_scalar(tau_beta, tau_var, n_clusters)
tau_comb_cows <- combine_scalar(tau_beta, tau_var, n_cows)

# 10. FINAL COMBINED TABLES
standard_combined_table <- data.frame(
  Method = c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
  Beta = c(std_comb_info$beta, std_comb_iv$beta, std_comb_cln$beta, std_comb_cows$beta),
  SE = c(std_comb_info$se, std_comb_iv$se, std_comb_cln$se, std_comb_cows$se),
  HR = exp(c(std_comb_info$beta, std_comb_iv$beta, std_comb_cln$beta, std_comb_cows$beta))
)

cluster_combined_table <- data.frame(
  Method = c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
  Beta = c(cl_comb_info$beta, cl_comb_iv$beta, cl_comb_cln$beta, cl_comb_cows$beta),
  SE = c(cl_comb_info$se, cl_comb_iv$se, cl_comb_cln$se, cl_comb_cows$se),
  HR = exp(c(cl_comb_info$beta, cl_comb_iv$beta, cl_comb_cln$beta, cl_comb_cows$beta))
)

copula_combined_table <- data.frame(
  Method = c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
  Beta = c(cop_comb_info$beta, cop_comb_iv$beta, cop_comb_cln$beta, cop_comb_cows$beta),
  SE = c(cop_comb_info$se, cop_comb_iv$se, cop_comb_cln$se, cop_comb_cows$se),
  HR = exp(c(cop_comb_info$beta, cop_comb_iv$beta, cop_comb_cln$beta, cop_comb_cows$beta))
)

tau_combined_table <- data.frame(
  Method = c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
  Tau = c(tau_comb_info$beta, tau_comb_iv$beta, tau_comb_cln$beta, tau_comb_cows$beta),
  SE = c(tau_comb_info$se, tau_comb_iv$se, tau_comb_cln$se, tau_comb_cows$se)
)

cat("\n===== \n")
cat("COMBINED STANDARD COX RESULTS \n")
cat("===== \n")
print(round_df(standard_combined_table, 4))

cat("\n===== \n")
cat("COMBINED CLUSTER-ROBUST COX RESULTS \n")
cat("===== \n")
print(round_df(cluster_combined_table, 4))

cat("\n===== \n")
cat("COMBINED COPULA RESULTS \n")
cat("===== \n")
print(round_df(copula_combined_table, 4))

cat("\n===== \n")
cat("COMBINED KENDALL'S TAU RESULTS \n")
cat("===== \n")
print(round_df(tau_combined_table, 4))

# 11. EVALUATION AGAINST FULL DATA
eval_standard <- do.call(rbind, lapply(1:nrow(standard_combined_table), function(i) {
  cbind(Method = standard_combined_table$Method[i],
        evaluation_metrics(standard_combined_table$Beta[i],
                          standard_combined_table$SE[i],
                          full_std_beta,
                          full_std_se))
}))

eval_cluster <- do.call(rbind, lapply(1:nrow(cluster_combined_table), function(i) {
  cbind(Method = cluster_combined_table$Method[i],
        evaluation_metrics(cluster_combined_table$Beta[i],
                          cluster_combined_table$SE[i],
                          full_cl_beta,
                          full_cl_se))
}))

eval_copula <- do.call(rbind, lapply(1:nrow(copula_combined_table), function(i) {
  cbind(Method = copula_combined_table$Method[i],
        evaluation_metrics(copula_combined_table$Beta[i],
                          copula_combined_table$SE[i],
                          full_cop_beta,
                          full_cop_se))
}))

eval_tau <- do.call(rbind, lapply(1:nrow(tau_combined_table), function(i) {

```

```

      cbind(Method = tau_combined_table$Method[i],
            evaluation_metrics(tau_combined_table$Tau[i],
                              tau_combined_table$SE[i],
                              full_tau,
                              full_tau_se))
    )))

cat("\n=====\n")
cat("EVALUATION: STANDARD COX\n")
cat("=====\n")
print(round_df(as.data.frame(eval_standard), 4))

cat("\n=====\n")
cat("EVALUATION: CLUSTER-ROBUST COX\n")
cat("=====\n")
print(round_df(as.data.frame(eval_cluster), 4))

cat("\n=====\n")
cat("EVALUATION: COPULA MODEL\n")
cat("=====\n")
print(round_df(as.data.frame(eval_copula), 4))

cat("\n=====\n")
cat("EVALUATION: KENDALL'S TAU\n")
cat("=====\n")
print(round_df(as.data.frame(eval_tau), 4))

```

Listing 4: R code for eight-subset sample splitting analysis

```

# 1. LOAD PACKAGES

library(survival)
library(Suncclarco)

# 2. LOAD DATA

data(insem)

# 3. FULL-DATA BENCHMARK VALUES

full_std_beta <- -0.06034839
full_std_se   <- 0.02050279

full_cl_beta  <- -0.06034839
full_cl_se    <- 0.02096202

full_cop_beta <- -0.06034839
full_cop_se   <- 0.02096202

full_tau      <- 0.1828338
full_tau_se   <- 0.02103043

# 4. HELPER FUNCTIONS

combine_scalar <- function(beta, var, weights) {
  beta_comb <- sum(weights * beta) / sum(weights)
  var_comb  <- sum((weights^2) * var) / (sum(weights)^2)
  se_comb   <- sqrt(var_comb)

  list(beta = beta_comb, var = var_comb, se = se_comb)
}

evaluation_metrics <- function(combined_est, combined_se, reference_est, reference_se) {
  data.frame(
    Combined_Estimate = combined_est,
    Reference_Estimate = reference_est,
    Absolute_Difference = abs(combined_est - reference_est),
    Relative_Difference = abs(combined_est - reference_est) / abs(reference_est),
    Combined_SE        = combined_se,
    Reference_SE       = reference_se,
    SE_Difference       = abs(combined_se - reference_se),
    Bias               = combined_est - reference_est
  )
}

round_df <- function(df, digits = 4) {

```

```

df[sapply(df, is.numeric)] <- round(df[sapply(df, is.numeric)], digits)
df
}

# 5. RANDOM SPLIT INTO 8 SUBSETS

set.seed(123)

herds <- sample(unique(insem$Herd))
herd_groups <- split(herds, cut(seq_along(herds), 8, labels = FALSE))

subset_list <- lapply(herd_groups, function(h) {
  insem[insem$Herd %in% h, ]
})

cat("\n===== \n")
cat("SUBSET SIZES: 8-SUBSET ANALYSIS \n")
cat("===== \n")

cat("\nObservations in each subset: \n")
print(sapply(subset_list, nrow))

cat("\nHerds in each subset: \n")
print(sapply(subset_list, function(x) length(unique(x$Herd))))

cat("\nEvents in each subset: \n")
print(sapply(subset_list, function(x) sum(x$Status == 1)))

cat("\nCensored observations in each subset: \n")
print(sapply(subset_list, function(x) sum(x$Status == 0)))

# 6. FIT MODELS IN EACH SUBSET + COMPUTATIONAL COST

results <- lapply(seq_along(subset_list), function(i) {

  dat <- subset_list[[i]]

  cat("\n===== \n")
  cat("FITTING MODELS FOR SUBSET", i, "\n")
  cat("===== \n")

  gc()
  mem_before <- sum(gc()[, 2])

  time_total <- system.time({

    cox_standard <- coxph(Surv(Time, Status) ~ Heifer, data = dat)

    cox_cluster <- coxph(Surv(Time, Status) ~ Heifer + cluster(Herd), data = dat)

    time_copula <- system.time({
      copula_fit <- SunclarcoModel(
        data = dat,
        time = "Time",
        status = "Status",
        clusters = "Herd",
        covariates = "Heifer",
        stage = 2,
        copula = "Clayton",
        marginal = "Cox"
      )
    })
  })

  gc()
  mem_after <- sum(gc()[, 2])
  memory_used <- mem_after - mem_before

  cat("\n--- Standard Cox Model: Subset", i, "--- \n")
  print(summary(cox_standard))

  cat("\n--- Cluster-Robust Cox Model: Subset", i, "--- \n")
  print(summary(cox_cluster))

  cat("\n--- Sunclarco Copula Model: Subset", i, "--- \n")
  print(copula_fit)

  cat("\n--- Computational Cost: Subset", i, "--- \n")

```

```

cat("Total elapsed time:", time_total["elapsed"], "seconds\n")
cat("Copula elapsed time:", time_copula["elapsed"], "seconds\n")
cat("Approximate memory used:", memory_used, "MB\n")

std_beta <- as.numeric(coef(cox_standard)[1])
std_var <- as.numeric(vcov(cox_standard)[1, 1])
std_se <- sqrt(std_var)
std_hr <- exp(std_beta)

cl_beta <- as.numeric(coef(cox_cluster)[1])
cl_var_robust <- as.numeric(vcov(cox_cluster)[1, 1])
cl_var_naive <- as.numeric(cox_cluster$naive.var[1, 1])
cl_se_robust <- sqrt(cl_var_robust)
cl_se_naive <- sqrt(cl_var_naive)
cl_hr <- exp(cl_beta)

cop_beta <- as.numeric(copula_fit$Parameters["beta_Heifer", "Estimates"])
cop_se <- as.numeric(copula_fit$Parameters["beta_Heifer", "StandardErrors"])
cop_var <- cop_se^2
cop_hr <- exp(cop_beta)

theta_est <- as.numeric(copula_fit$Parameters["theta", "Estimates"])
theta_se <- as.numeric(copula_fit$Parameters["theta", "StandardErrors"])

tau_est <- as.numeric(copula_fit$Kendall_Tau["Estimates"])
tau_se <- as.numeric(copula_fit$Kendall_Tau["StandardErrors"])
tau_var <- tau_se^2

list(
  subset_id = i,
  n_obs = nrow(dat),
  n_clusters = length(unique(dat$Herd)),

  total_user_time = as.numeric(time_total["user.self"]),
  total_system_time = as.numeric(time_total["sys.self"]),
  total_elapsed_time = as.numeric(time_total["elapsed"]),

  copula_user_time = as.numeric(time_copula["user.self"]),
  copula_system_time = as.numeric(time_copula["sys.self"]),
  copula_elapsed_time = as.numeric(time_copula["elapsed"]),

  memory_used_MB = memory_used,

  std_beta = std_beta,
  std_var = std_var,
  std_se = std_se,
  std_hr = std_hr,

  cl_beta = cl_beta,
  cl_var_robust = cl_var_robust,
  cl_var_naive = cl_var_naive,
  cl_se_robust = cl_se_robust,
  cl_se_naive = cl_se_naive,
  cl_hr = cl_hr,

  cop_beta = cop_beta,
  cop_var = cop_var,
  cop_se = cop_se,
  cop_hr = cop_hr,
  theta_est = theta_est,
  theta_se = theta_se,

  tau_est = tau_est,
  tau_var = tau_var,
  tau_se = tau_se
)
})

# 7A. COMPUTATIONAL COST SUMMARY TABLE

computational_summary8 <- data.frame(
  Subset = sapply(results, function(x) x$subset_id),
  Observations = sapply(results, function(x) x$n_obs),
  Herds = sapply(results, function(x) x$n_clusters),

  Total_User_Time = sapply(results, function(x) x$total_user_time),
  Total_System_Time = sapply(results, function(x) x$total_system_time),
  Total_Elapsed_Time = sapply(results, function(x) x$total_elapsed_time),

```



```

Copula_User_Time = sapply(results, function(x) x$copula_user_time),
Copula_System_Time = sapply(results, function(x) x$copula_system_time),
Copula_Elapsed_Time = sapply(results, function(x) x$copula_elapsed_time),

Approx_Memory_Used_MB = sapply(results, function(x) x$memory_used_MB)
)

cat("\n===== \n")
cat("COMPUTATIONAL COST SUMMARY: 8-SUBSET ANALYSIS\n")
cat("===== \n")
print(round_df(computational_summary8, 4))

# 7. SUBSET SUMMARY TABLE

subset_summary <- data.frame(
  Subset = sapply(results, function(x) x$subset_id),
  Observations = sapply(results, function(x) x$n_obs),
  Herds = sapply(results, function(x) x$n_clusters),

  Standard_Cox_Beta = sapply(results, function(x) x$std_beta),
  Standard_Cox_SE = sapply(results, function(x) x$std_se),
  Standard_Cox_HR = sapply(results, function(x) x$std_hr),

  Cluster_Cox_Beta = sapply(results, function(x) x$cl_beta),
  Cluster_Cox_Robust_SE = sapply(results, function(x) x$cl_se_robust),
  Cluster_Cox_Naive_SE = sapply(results, function(x) x$cl_se_naive),
  Cluster_Cox_HR = sapply(results, function(x) x$cl_hr),

  Copula_Beta = sapply(results, function(x) x$cop_beta),
  Copula_SE = sapply(results, function(x) x$cop_se),
  Copula_HR = sapply(results, function(x) x$cop_hr),

  Theta = sapply(results, function(x) x$theta_est),
  Theta_SE = sapply(results, function(x) x$theta_se),

  Tau = sapply(results, function(x) x$tau_est),
  Tau_SE = sapply(results, function(x) x$tau_se)
)

cat("\n===== \n")
cat("SUBSET-SPECIFIC RESULTS\n")
cat("===== \n")
print(round_df(subset_summary, 4))

# 8. EXTRACT VECTORS FOR COMBINATION

n_clusters <- sapply(results, function(x) x$n_clusters)
n_cows <- sapply(results, function(x) x$n_obs)

std_beta <- sapply(results, function(x) x$std_beta)
std_var <- sapply(results, function(x) x$std_var)

cl_beta <- sapply(results, function(x) x$cl_beta)
cl_var_robust <- sapply(results, function(x) x$cl_var_robust)

cop_beta <- sapply(results, function(x) x$cop_beta)
cop_var <- sapply(results, function(x) x$cop_var)

tau_beta <- sapply(results, function(x) x$tau_est)
tau_var <- sapply(results, function(x) x$tau_var)

# 9. COMBINED ESTIMATES

# Standard Cox
std_comb_info <- combine_scalar(std_beta, std_var, 1 / std_var)
std_comb_iv <- combine_scalar(std_beta, std_var, 1 / std_var)
std_comb_cln <- combine_scalar(std_beta, std_var, n_clusters)
std_comb_cows <- combine_scalar(std_beta, std_var, n_cows)

# Cluster-Robust Cox
cl_comb_info <- combine_scalar(cl_beta, cl_var_robust, 1 / cl_var_robust)
cl_comb_iv <- combine_scalar(cl_beta, cl_var_robust, 1 / cl_var_robust)
cl_comb_cln <- combine_scalar(cl_beta, cl_var_robust, n_clusters)
cl_comb_cows <- combine_scalar(cl_beta, cl_var_robust, n_cows)

# Copula beta
cop_comb_info <- combine_scalar(cop_beta, cop_var, 1 / cop_var)

```

```

cop_comb_iv <- combine_scalar(cop_beta, cop_var, 1 / cop_var)
cop_comb_cln <- combine_scalar(cop_beta, cop_var, n_clusters)
cop_comb_cows <- combine_scalar(cop_beta, cop_var, n_cows)

# Kendall's tau
tau_comb_info <- combine_scalar(tau_beta, tau_var, 1 / tau_var)
tau_comb_iv <- combine_scalar(tau_beta, tau_var, 1 / tau_var)
tau_comb_cln <- combine_scalar(tau_beta, tau_var, n_clusters)
tau_comb_cows <- combine_scalar(tau_beta, tau_var, n_cows)

# 10. FINAL COMBINED TABLES

standard_combined_table <- data.frame(
  Method = c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
  Beta = c(std_comb_info$beta, std_comb_iv$beta, std_comb_cln$beta, std_comb_cows$beta),
  SE = c(std_comb_info$se, std_comb_iv$se, std_comb_cln$se, std_comb_cows$se),
  HR = exp(c(std_comb_info$beta, std_comb_iv$beta, std_comb_cln$beta, std_comb_cows$beta))
)

cluster_combined_table <- data.frame(
  Method = c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
  Beta = c(cl_comb_info$beta, cl_comb_iv$beta, cl_comb_cln$beta, cl_comb_cows$beta),
  SE = c(cl_comb_info$se, cl_comb_iv$se, cl_comb_cln$se, cl_comb_cows$se),
  HR = exp(c(cl_comb_info$beta, cl_comb_iv$beta, cl_comb_cln$beta, cl_comb_cows$beta))
)

copula_combined_table <- data.frame(
  Method = c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
  Beta = c(cop_comb_info$beta, cop_comb_iv$beta, cop_comb_cln$beta, cop_comb_cows$beta),
  SE = c(cop_comb_info$se, cop_comb_iv$se, cop_comb_cln$se, cop_comb_cows$se),
  HR = exp(c(cop_comb_info$beta, cop_comb_iv$beta, cop_comb_cln$beta, cop_comb_cows$beta))
)

tau_combined_table <- data.frame(
  Method = c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
  Tau = c(tau_comb_info$beta, tau_comb_iv$beta, tau_comb_cln$beta, tau_comb_cows$beta),
  SE = c(tau_comb_info$se, tau_comb_iv$se, tau_comb_cln$se, tau_comb_cows$se)
)

cat("\n===== \n")
cat("COMBINED STANDARD COX RESULTS\n")
cat("===== \n")
print(round_df(standard_combined_table, 4))

cat("\n===== \n")
cat("COMBINED CLUSTER-ROBUST COX RESULTS\n")
cat("===== \n")
print(round_df(cluster_combined_table, 4))

cat("\n===== \n")
cat("COMBINED COPULA RESULTS\n")
cat("===== \n")
print(round_df(copula_combined_table, 4))

cat("\n===== \n")
cat("COMBINED KENDALL'S TAU RESULTS\n")
cat("===== \n")
print(round_df(tau_combined_table, 4))

# 11. EVALUATION AGAINST FULL DATA

eval_standard <- do.call(rbind, lapply(1:nrow(standard_combined_table), function(i) {
  cbind(Method = standard_combined_table$Method[i],
    evaluation_metrics(standard_combined_table$Beta[i],
      standard_combined_table$SE[i],
      full_std_beta,
      full_std_se))
}))

eval_cluster <- do.call(rbind, lapply(1:nrow(cluster_combined_table), function(i) {
  cbind(Method = cluster_combined_table$Method[i],
    evaluation_metrics(cluster_combined_table$Beta[i],
      cluster_combined_table$SE[i],
      full_cl_beta,
      full_cl_se))
}))

eval_copula <- do.call(rbind, lapply(1:nrow(copula_combined_table), function(i) {

```

```

      cbind(Method = copula_combined_table$Method[i],
            evaluation_metrics(copula_combined_table$Beta[i],
                              copula_combined_table$SE[i],
                              full_cop_beta,
                              full_cop_se))
    )))

eval_tau <- do.call(rbind, lapply(1:nrow(tau_combined_table), function(i) {
  cbind(Method = tau_combined_table$Method[i],
        evaluation_metrics(tau_combined_table$Tau[i],
                          tau_combined_table$SE[i],
                          full_tau,
                          full_tau_se))
})))

cat("\n===== \n")
cat("EVALUATION: STANDARD COX\n")
cat("===== \n")
print(round_df(as.data.frame(eval_standard), 4))

cat("\n===== \n")
cat("EVALUATION: CLUSTER-ROBUST COX\n")
cat("===== \n")
print(round_df(as.data.frame(eval_cluster), 4))

cat("\n===== \n")
cat("EVALUATION: COPULA MODEL\n")
cat("===== \n")
print(round_df(as.data.frame(eval_copula), 4))

cat("\n===== \n")
cat("EVALUATION: KENDALL'S TAU\n")
cat("===== \n")
print(round_df(as.data.frame(eval_tau), 4))

```

Listing 5: R code for Q1-split analysis

```

# 1. LOAD PACKAGES

library(Sunclarco)
library(survival)
library(dplyr)

# 2. LOAD DATA

data(insem)

# 3. FULL-DATA BENCHMARK VALUES

full_std_beta <- -0.06034839
full_std_se <- 0.02050279

full_cl_beta <- -0.06034839
full_cl_se <- 0.02096202

full_cop_beta <- -0.06034839
full_cop_se <- 0.02096202

full_tau <- 0.1828338
full_tau_se <- 0.02103043

# 4. HELPER FUNCTIONS

combine_scalar <- function(beta, var, weights) {
  beta_comb <- sum(weights * beta) / sum(weights)
  var_comb <- sum((weights^2) * var) / (sum(weights)^2)
  se_comb <- sqrt(var_comb)

  list(beta = beta_comb, var = var_comb, se = se_comb)
}

evaluation_metrics <- function(combined_est, combined_se, reference_est, reference_se) {
  data.frame(
    Combined_Estimate = combined_est,
    Reference_Estimate = reference_est,
    Absolute_Difference = abs(combined_est - reference_est),

```

```

    Relative_Difference = abs(combined_est - reference_est) / abs(reference_est),
    Combined_SE         = combined_se,
    Reference_SE        = reference_se,
    SE_Difference        = abs(combined_se - reference_se),
    Bias                 = combined_est - reference_est
  )
}

round_df <- function(df, digits = 4) {
  df[sapply(df, is.numeric)] <- round(df[sapply(df, is.numeric)], digits)
  df
}

# 5. CREATE HERD SIZE QUANTILES
# Q1 = largest herds, Q4 = smallest herds

herd_quartiles <- insem %>%
  count(Herd, name = "n_cows") %>%
  arrange(desc(n_cows)) %>%
  mutate(herd_quartile = ntile(desc(n_cows), 4))

insem_q <- insem %>%
  left_join(herd_quartiles, by = "Herd")

# 6. Q1 SPLITTING: LARGEST HERDS INTO 3 SUBCLUSTERS

data_q1_split <- insem_q %>%
  group_by(Herd) %>%
  mutate(
    subcluster = if_else(
      herd_quartile == 1,
      ntile(row_number(), 3),
      1L
    )
  ) %>%
  ungroup() %>%
  group_by(Herd, subcluster) %>%
  mutate(cluster_id = cur_group_id()) %>%
  ungroup()

data_q1_split <- as.data.frame(data_q1_split)

# 7. DESCRIPTION AFTER Q1 SPLITTING

q1_cluster_summary <- data.frame(
  Description = c(
    "Original number of herds",
    "Number of Q1 herds split",
    "Final number of clusters after Q1 splitting",
    "Number of observations",
    "Number of events",
    "Number censored"
  ),
  Value = c(
    n_distinct(insem$Herd),
    n_distinct(data_q1_split$Herd[data_q1_split$herd_quartile == 1]),
    n_distinct(data_q1_split$cluster_id),
    nrow(data_q1_split),
    sum(data_q1_split$Status == 1),
    sum(data_q1_split$Status == 0)
  )
)

q1_by_quartile <- data_q1_split %>%
  group_by(herd_quartile) %>%
  summarise(
    original_herds = n_distinct(Herd),
    final_clusters = n_distinct(cluster_id),
    observations = n(),
    events = sum(Status == 1),
    censored = sum(Status == 0),
    censoring_percent = round(mean(Status == 0) * 100, 2),
    .groups = "drop"
  )

q1_cluster_size_summary <- data_q1_split %>%
  count(cluster_id, name = "cluster_size") %>%

```

```

summarise(
  min_cluster_size = min(cluster_size),
  q1_cluster_size = quantile(cluster_size, 0.25),
  median_cluster_size = median(cluster_size),
  mean_cluster_size = round(mean(cluster_size), 2),
  q3_cluster_size = quantile(cluster_size, 0.75),
  max_cluster_size = max(cluster_size)
)

q1_cluster_summary
q1_by_quartile
q1_cluster_size_summary

# 8. Q1-SPLIT FULL-DATA ANALYSIS + COMPUTATIONAL COST

gc()
mem_before_q1_full <- sum(gc()[, 2])

time_q1_full <- system.time({

  cox_standard_q1 <- coxph(
    Surv(Time, Status) ~ Heifer,
    data = data_q1_split
  )

  cox_robust_q1 <- coxph(
    Surv(Time, Status) ~ Heifer + cluster(cluster_id),
    data = data_q1_split
  )

  time_q1_copula_full <- system.time({
    sun_q1 <- SunclarcoModel(
      data      = data_q1_split,
      time      = "Time",
      status    = "Status",
      clusters   = "cluster_id",
      covariates = "Heifer",
      stage     = 2,
      copula     = "Clayton",
      marginal   = "Cox",
      verbose    = FALSE
    )
  })
})

gc()
mem_after_q1_full <- sum(gc()[, 2])
memory_q1_full <- mem_after_q1_full - mem_before_q1_full

q1_full_results <- data.frame(
  Model = c("Standard Cox", "Cluster-Robust Cox", "Q1-Split Copula"),
  Beta = c(
    as.numeric(coef(cox_standard_q1)[1]),
    as.numeric(coef(cox_robust_q1)[1]),
    as.numeric(sun_q1$Parameters["beta_Heifer", "Estimates"])
  ),
  SE = c(
    sqrt(as.numeric(vcov(cox_standard_q1)[1, 1])),
    sqrt(as.numeric(vcov(cox_robust_q1)[1, 1])),
    as.numeric(sun_q1$Parameters["beta_Heifer", "StandardErrors"])
  )
)

q1_full_results$HR <- exp(q1_full_results$Beta)

q1_full_tau <- as.numeric(sun_q1$Kendall_Tau["Estimates"])
q1_full_tau_se <- as.numeric(sun_q1$Kendall_Tau["StandardErrors"])

q1_full_cost <- data.frame(
  Analysis = "Q1-split full data",
  Observations = nrow(data_q1_split),
  Clusters = length(unique(data_q1_split$cluster_id)),
  Total_User_Time = as.numeric(time_q1_full["user.self"]),
  Total_System_Time = as.numeric(time_q1_full["sys.self"]),
  Total_Elapsed_Time = as.numeric(time_q1_full["elapsed"]),
  Copula_User_Time = as.numeric(time_q1_copula_full["user.self"]),
  Copula_System_Time = as.numeric(time_q1_copula_full["sys.self"]),
  Copula_Elapsed_Time = as.numeric(time_q1_copula_full["elapsed"]),
  Approx_Memory_Used_MB = memory_q1_full
)

```

```

)

round_df(q1_full_results, 4)
round_df(q1_full_cost, 4)

# 9. FUNCTION FOR Q1-SPLIT SUBSET ANALYSIS

run_q1_subset_analysis <- function(data_input, n_subsets, seed = 123) {

  set.seed(seed)

  clusters <- sample(unique(data_input$cluster_id))
  cluster_groups <- split(
    clusters,
    cut(seq_along(clusters), n_subsets, labels = FALSE)
  )

  subset_list <- lapply(cluster_groups, function(c1) {
    data_input[data_input$cluster_id %in% c1, ]
  })

  cat("\n=====\\n")
  cat(n_subsets, "SUBSET ANALYSIS AFTER Q1 SPLITTING\\n")
  cat("=====\\n")

  results <- lapply(seq_along(subset_list), function(i) {

    dat <- subset_list[[i]]

    cat("\n=====\\n")
    cat("FITTING MODELS FOR SUBSET", i, "\\n")
    cat("=====\\n")

    gc()
    mem_before <- sum(gc()[, 2])

    time_total <- system.time({

      cox_standard <- coxph(
        Surv(Time, Status) ~ Heifer,
        data = dat
      )

      cox_cluster <- coxph(
        Surv(Time, Status) ~ Heifer + cluster(cluster_id),
        data = dat
      )

      time_copula <- system.time({
        copula_fit <- SunclarcoModel(
          data = dat,
          time = "Time",
          status = "Status",
          clusters = "cluster_id",
          covariates = "Heifer",
          stage = 2,
          copula = "Clayton",
          marginal = "Cox",
          verbose = FALSE
        )
      })
    })

    gc()
    mem_after <- sum(gc()[, 2])
    memory_used <- mem_after - mem_before

    # Standard Cox
    std_beta <- as.numeric(coef(cox_standard)[1])
    std_var <- as.numeric(vcov(cox_standard)[1, 1])
    std_info <- 1 / std_var
    std_se <- sqrt(std_var)
    std_hr <- exp(std_beta)

    # Cluster-robust Cox
    cl_beta <- as.numeric(coef(cox_cluster)[1])
    cl_var_robust <- as.numeric(vcov(cox_cluster)[1, 1])
    cl_var_naive <- as.numeric(cox_cluster$naive.var[1, 1])
  })
}

```

```

# Information matrix
cl_info <- 1 / cl_var_naive

cl_se_robust <- sqrt(cl_var_robust)
cl_se_naive <- sqrt(cl_var_naive)
cl_hr <- exp(cl_beta)

# Sunclarco copula
cop_beta <- as.numeric(copula_fit$Parameters["beta_Heifer", "Estimates"])
cop_se <- as.numeric(copula_fit$Parameters["beta_Heifer", "StandardErrors"])
cop_var <- cop_se^2

# Sunclarco
cop_info <- 1 / cop_var

cop_hr <- exp(cop_beta)

theta_est <- as.numeric(copula_fit$Parameters["theta", "Estimates"])
theta_se <- as.numeric(copula_fit$Parameters["theta", "StandardErrors"])

tau_est <- as.numeric(copula_fit$Kendall_Tau["Estimates"])
tau_se <- as.numeric(copula_fit$Kendall_Tau["StandardErrors"])
tau_var <- tau_se^2
tau_info <- 1 / tau_var

list(
  subset_id = i,
  n_obs = nrow(dat),
  n_clusters = length(unique(dat$cluster_id)),
  n_original_herds = length(unique(dat$Herd)),

  total_user_time = as.numeric(time_total["user.self"]),
  total_system_time = as.numeric(time_total["sys.self"]),
  total_elapsed_time = as.numeric(time_total["elapsed"]),

  copula_user_time = as.numeric(time_copula["user.self"]),
  copula_system_time = as.numeric(time_copula["sys.self"]),
  copula_elapsed_time = as.numeric(time_copula["elapsed"]),

  memory_used_MB = memory_used,

  std_beta = std_beta,
  std_var = std_var,
  std_info = std_info,
  std_se = std_se,
  std_hr = std_hr,

  cl_beta = cl_beta,
  cl_var_robust = cl_var_robust,
  cl_var_naive = cl_var_naive,
  cl_info = cl_info,
  cl_se_robust = cl_se_robust,
  cl_se_naive = cl_se_naive,
  cl_hr = cl_hr,

  cop_beta = cop_beta,
  cop_var = cop_var,
  cop_info = cop_info,
  cop_se = cop_se,
  cop_hr = cop_hr,

  theta_est = theta_est,
  theta_se = theta_se,

  tau_est = tau_est,
  tau_var = tau_var,
  tau_info = tau_info,
  tau_se = tau_se
)
})

subset_summary <- data.frame(
  Split = paste0(n_subsets, "-subset"),
  Subset = sapply(results, function(x) x$subset_id),
  Observations = sapply(results, function(x) x$n_obs),
  Clusters = sapply(results, function(x) x$n_clusters),
  Original_Herds = sapply(results, function(x) x$n_original_herds),

  Standard_Cox_Beta = sapply(results, function(x) x$std_beta),

```

```

Standard_Cox_SE = sapply(results, function(x) x$std_se),
Standard_Cox_HR = sapply(results, function(x) x$std_hr),

Cluster_Cox_Beta = sapply(results, function(x) x$cl_beta),
Cluster_Cox_Robust_SE = sapply(results, function(x) x$cl_se_robust),
Cluster_Cox_Naive_SE = sapply(results, function(x) x$cl_se_naive),
Cluster_Cox_HR = sapply(results, function(x) x$cl_hr),

Copula_Beta = sapply(results, function(x) x$cop_beta),
Copula_SE = sapply(results, function(x) x$cop_se),
Copula_HR = sapply(results, function(x) x$cop_hr),

Theta = sapply(results, function(x) x$theta_est),
Theta_SE = sapply(results, function(x) x$theta_se),

Tau = sapply(results, function(x) x$tau_est),
Tau_SE = sapply(results, function(x) x$tau_se)
)

computational_summary <- data.frame(
  Split = paste0(n_subsets, "-subset"),
  Subset = sapply(results, function(x) x$subset_id),
  Observations = sapply(results, function(x) x$n_obs),
  Clusters = sapply(results, function(x) x$n_clusters),
  Original_Herds = sapply(results, function(x) x$n_original_herds),

  Total_User_Time = sapply(results, function(x) x$total_user_time),
  Total_System_Time = sapply(results, function(x) x$total_system_time),
  Total_Elapsed_Time = sapply(results, function(x) x$total_elapsed_time),

  Copula_User_Time = sapply(results, function(x) x$copula_user_time),
  Copula_System_Time = sapply(results, function(x) x$copula_system_time),
  Copula_Elapsed_Time = sapply(results, function(x) x$copula_elapsed_time),

  Approx_Memory_Used_MB = sapply(results, function(x) x$memory_used_MB)
)

n_clusters <- sapply(results, function(x) x$n_clusters)
n_cows <- sapply(results, function(x) x$n_obs)

std_beta <- sapply(results, function(x) x$std_beta)
std_var <- sapply(results, function(x) x$std_var)
std_info <- sapply(results, function(x) x$std_info)

cl_beta <- sapply(results, function(x) x$cl_beta)
cl_var_robust <- sapply(results, function(x) x$cl_var_robust)
cl_info <- sapply(results, function(x) x$cl_info)

cop_beta <- sapply(results, function(x) x$cop_beta)
cop_var <- sapply(results, function(x) x$cop_var)
cop_info <- sapply(results, function(x) x$cop_info)

tau_beta <- sapply(results, function(x) x$tau_est)
tau_var <- sapply(results, function(x) x$tau_var)
tau_info <- sapply(results, function(x) x$tau_info)

# Standard Cox
std_comb_info <- combine_scalar(std_beta, std_var, std_info)
std_comb_iv <- combine_scalar(std_beta, std_var, 1 / std_var)
std_comb_cln <- combine_scalar(std_beta, std_var, n_clusters)
std_comb_cows <- combine_scalar(std_beta, std_var, n_cows)

# Cluster-Robust Cox
# Information matrix uses naive/model-based information
# Inverse variance uses robust sandwich variance
cl_comb_info <- combine_scalar(cl_beta, cl_var_robust, 1 / cl_info, cl_info)
cl_comb_iv <- combine_scalar(cl_beta, cl_var_robust, 1 / cl_var_robust)
cl_comb_cln <- combine_scalar(cl_beta, cl_var_robust, n_clusters)
cl_comb_cows <- combine_scalar(cl_beta, cl_var_robust, n_cows)

# Copula beta
# Suncilarco information-type weight uses jackknife inverse variance
cop_comb_info <- combine_scalar(cop_beta, cop_var, cop_info)
cop_comb_iv <- combine_scalar(cop_beta, cop_var, 1 / cop_var)
cop_comb_cln <- combine_scalar(cop_beta, cop_var, n_clusters)
cop_comb_cows <- combine_scalar(cop_beta, cop_var, n_cows)

# Kendall's tau
tau_comb_info <- combine_scalar(tau_beta, tau_var, tau_info)
tau_comb_iv <- combine_scalar(tau_beta, tau_var, 1 / tau_var)

```



```

tau_comb_cln <- combine_scalar(tau_beta, tau_var, n_clusters)
tau_comb_cows <- combine_scalar(tau_beta, tau_var, n_cows)

combined_table <- data.frame(
  Split = paste0(n_subsets, "-subset"),
  Model = rep(
    c("Standard Cox", "Cluster-Robust Cox", "Copula Model", "Kendall tau"),
    each = 4
  ),
  Method = rep(
    c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
    times = 4
  ),
  Estimate = c(
    std_comb_info$beta, std_comb_iv$beta, std_comb_cln$beta, std_comb_cows$beta,
    cl_comb_info$beta, cl_comb_iv$beta, cl_comb_cln$beta, cl_comb_cows$beta,
    cop_comb_info$beta, cop_comb_iv$beta, cop_comb_cln$beta, cop_comb_cows$beta,
    tau_comb_info$beta, tau_comb_iv$beta, tau_comb_cln$beta, tau_comb_cows$beta
  ),
  SE = c(
    std_comb_info$se, std_comb_iv$se, std_comb_cln$se, std_comb_cows$se,
    cl_comb_info$se, cl_comb_iv$se, cl_comb_cln$se, cl_comb_cows$se,
    cop_comb_info$se, cop_comb_iv$se, cop_comb_cln$se, cop_comb_cows$se,
    tau_comb_info$se, tau_comb_iv$se, tau_comb_cln$se, tau_comb_cows$se
  )
)

list(
  results = results,
  subset_summary = subset_summary,
  computational_summary = computational_summary,
  combined_table = combined_table
)
}

```

9. FUNCTION FOR Q1-SPLIT SUBSET ANALYSIS

```

run_q1_subset_analysis <- function(data_input, n_subsets, seed = 123) {

  set.seed(seed)

  clusters <- sample(unique(data_input$cluster_id))
  cluster_groups <- split(
    clusters,
    cut(seq_along(clusters), n_subsets, labels = FALSE)
  )

  subset_list <- lapply(cluster_groups, function(cl) {
    data_input[data_input$cluster_id %in% cl, ]
  })

  cat("\n===== \n")
  cat(n_subsets, "SUBSET ANALYSIS AFTER Q1 SPLITTING \n")
  cat("===== \n")

  cat("\nObservations in each subset: \n")
  print(sapply(subset_list, nrow))

  cat("\nClusters in each subset: \n")
  print(sapply(subset_list, function(x) length(unique(x$cluster_id))))

  results <- lapply(seq_along(subset_list), function(i) {

    dat <- subset_list[[i]]

    cat("\n===== \n")
    cat("FITTING MODELS FOR SUBSET", i, " \n")
    cat("===== \n")

    gc()
    mem_before <- sum(gc()[, 2])

    time_total <- system.time({

      cox_standard <- coxph(
        Surv(Time, Status) ~ Heifer,
        data = dat
      )
    })
  })
}

```

```

cox_cluster <- coxph(
  Surv(Time, Status) ~ Heifer + cluster(cluster_id),
  data = dat
)

time_copula <- system.time({
  copula_fit <- SunclarcoModel(
    data      = dat,
    time      = "Time",
    status    = "Status",
    clusters  = "cluster_id",
    covariates = "Heifer",
    stage     = 2,
    copula    = "Clayton",
    marginal  = "Cox",
    verbose   = FALSE
  )
})
})

gc()
mem_after <- sum(gc())[, 2])
memory_used <- mem_after - mem_before

std_beta <- as.numeric(coef(cox_standard)[1])
std_var  <- as.numeric(vcov(cox_standard)[1, 1])
std_se   <- sqrt(std_var)
std_hr   <- exp(std_beta)

cl_beta  <- as.numeric(coef(cox_cluster)[1])
cl_var_robust <- as.numeric(vcov(cox_cluster)[1, 1])
cl_var_naive <- as.numeric(cox_cluster$naive.var[1, 1])
cl_se_robust <- sqrt(cl_var_robust)
cl_se_naive <- sqrt(cl_var_naive)
cl_hr       <- exp(cl_beta)

cop_beta <- as.numeric(copula_fit$Parameters["beta_Heifer", "Estimates"])
cop_se   <- as.numeric(copula_fit$Parameters["beta_Heifer", "StandardErrors"])
cop_var  <- cop_se^2
cop_hr   <- exp(cop_beta)

theta_est <- as.numeric(copula_fit$Parameters["theta", "Estimates"])
theta_se  <- as.numeric(copula_fit$Parameters["theta", "StandardErrors"])

tau_est <- as.numeric(copula_fit$Kendall_Tau["Estimates"])
tau_se  <- as.numeric(copula_fit$Kendall_Tau["StandardErrors"])
tau_var <- tau_se^2

list(
  subset_id = i,
  n_obs     = nrow(dat),
  n_clusters = length(unique(dat$cluster_id)),
  n_original_herds = length(unique(dat$Herd)),

  total_user_time = as.numeric(time_total["user.self"]),
  total_system_time = as.numeric(time_total["sys.self"]),
  total_elapsed_time = as.numeric(time_total["elapsed"]),

  copula_user_time = as.numeric(time_copula["user.self"]),
  copula_system_time = as.numeric(time_copula["sys.self"]),
  copula_elapsed_time = as.numeric(time_copula["elapsed"]),

  memory_used_MB = memory_used,

  std_beta = std_beta,
  std_var = std_var,
  std_se = std_se,
  std_hr = std_hr,

  cl_beta = cl_beta,
  cl_var_robust = cl_var_robust,
  cl_var_naive = cl_var_naive,
  cl_se_robust = cl_se_robust,
  cl_se_naive = cl_se_naive,
  cl_hr = cl_hr,

  cop_beta = cop_beta,
  cop_var = cop_var,
  cop_se = cop_se,
  cop_hr = cop_hr,

```

```

    theta_est = theta_est,
    theta_se = theta_se,

    tau_est = tau_est,
    tau_var = tau_var,
    tau_se = tau_se
  )
})

subset_summary <- data.frame(
  Split = paste0(n_subsets, "-subset"),
  Subset = sapply(results, function(x) x$subset_id),
  Observations = sapply(results, function(x) x$n_obs),
  Clusters = sapply(results, function(x) x$n_clusters),
  Original_Herds = sapply(results, function(x) x$n_original_herds),

  Standard_Cox_Beta = sapply(results, function(x) x$std_beta),
  Standard_Cox_SE = sapply(results, function(x) x$std_se),
  Standard_Cox_HR = sapply(results, function(x) x$std_hr),

  Cluster_Cox_Beta = sapply(results, function(x) x$cl_beta),
  Cluster_Cox_Robust_SE = sapply(results, function(x) x$cl_se_robust),
  Cluster_Cox_Naive_SE = sapply(results, function(x) x$cl_se_naive),
  Cluster_Cox_HR = sapply(results, function(x) x$cl_hr),

  Copula_Beta = sapply(results, function(x) x$cop_beta),
  Copula_SE = sapply(results, function(x) x$cop_se),
  Copula_HR = sapply(results, function(x) x$cop_hr),

  Theta = sapply(results, function(x) x$theta_est),
  Theta_SE = sapply(results, function(x) x$theta_se),

  Tau = sapply(results, function(x) x$tau_est),
  Tau_SE = sapply(results, function(x) x$tau_se)
)

computational_summary <- data.frame(
  Split = paste0(n_subsets, "-subset"),
  Subset = sapply(results, function(x) x$subset_id),
  Observations = sapply(results, function(x) x$n_obs),
  Clusters = sapply(results, function(x) x$n_clusters),
  Original_Herds = sapply(results, function(x) x$n_original_herds),

  Total_User_Time = sapply(results, function(x) x$total_user_time),
  Total_System_Time = sapply(results, function(x) x$total_system_time),
  Total_Elapsed_Time = sapply(results, function(x) x$total_elapsed_time),

  Copula_User_Time = sapply(results, function(x) x$copula_user_time),
  Copula_System_Time = sapply(results, function(x) x$copula_system_time),
  Copula_Elapsed_Time = sapply(results, function(x) x$copula_elapsed_time),

  Approx_Memory_Used_MB = sapply(results, function(x) x$memory_used_MB)
)

n_clusters <- sapply(results, function(x) x$n_clusters)
n_cows <- sapply(results, function(x) x$n_obs)

std_beta <- sapply(results, function(x) x$std_beta)
std_var <- sapply(results, function(x) x$std_var)

cl_beta <- sapply(results, function(x) x$cl_beta)
cl_var_robust <- sapply(results, function(x) x$cl_var_robust)

cop_beta <- sapply(results, function(x) x$cop_beta)
cop_var <- sapply(results, function(x) x$cop_var)

tau_beta <- sapply(results, function(x) x$tau_est)
tau_var <- sapply(results, function(x) x$tau_var)

std_comb_iv <- combine_scalar(std_beta, std_var, 1 / std_var)
std_comb_cln <- combine_scalar(std_beta, std_var, n_clusters)
std_comb_cows <- combine_scalar(std_beta, std_var, n_cows)

cl_comb_iv <- combine_scalar(cl_beta, cl_var_robust, 1 / cl_var_robust)
cl_comb_cln <- combine_scalar(cl_beta, cl_var_robust, n_clusters)
cl_comb_cows <- combine_scalar(cl_beta, cl_var_robust, n_cows)

cop_comb_iv <- combine_scalar(cop_beta, cop_var, 1 / cop_var)
cop_comb_cln <- combine_scalar(cop_beta, cop_var, n_clusters)
cop_comb_cows <- combine_scalar(cop_beta, cop_var, n_cows)

```

```

tau_comb_iv <- combine_scalar(tau_beta, tau_var, 1 / tau_var)
tau_comb_cln <- combine_scalar(tau_beta, tau_var, n_clusters)
tau_comb_cows <- combine_scalar(tau_beta, tau_var, n_cows)

combined_table <- data.frame(
  Split = paste0(n_subsets, "-subset"),
  Model = rep(c("Standard Cox", "Cluster-Robust Cox", "Copula Model", "Kendall tau"), each = 3),
  Method = rep(c("Inverse variance", "Cluster-number", "Cow-number"), times = 4),
  Estimate = c(
    std_comb_iv$beta, std_comb_cln$beta, std_comb_cows$beta,
    cl_comb_iv$beta, cl_comb_cln$beta, cl_comb_cows$beta,
    cop_comb_iv$beta, cop_comb_cln$beta, cop_comb_cows$beta,
    tau_comb_iv$beta, tau_comb_cln$beta, tau_comb_cows$beta
  ),
  SE = c(
    std_comb_iv$sse, std_comb_cln$sse, std_comb_cows$sse,
    cl_comb_iv$sse, cl_comb_cln$sse, cl_comb_cows$sse,
    cop_comb_iv$sse, cop_comb_cln$sse, cop_comb_cows$sse,
    tau_comb_iv$sse, tau_comb_cln$sse, tau_comb_cows$sse
  )
)

list(
  results = results,
  subset_summary = subset_summary,
  computational_summary = computational_summary,
  combined_table = combined_table
)
}

# 10. RUN 3-, 5-, AND 8-SUBSET ANALYSES AFTER Q1 SPLITTING

analysis_3 <- run_q1_subset_analysis(data_q1_split, n_subsets = 3, seed = 123)
analysis_5 <- run_q1_subset_analysis(data_q1_split, n_subsets = 5, seed = 123)
analysis_8 <- run_q1_subset_analysis(data_q1_split, n_subsets = 8, seed = 123)

# 11. SUBSET-SPECIFIC RESULTS

all_subset_results <- bind_rows(
  analysis_3$subset_summary,
  analysis_5$subset_summary,
  analysis_8$subset_summary
)

round_df(all_subset_results, 4)

# 12. COMPUTATIONAL COST FOR ALL SUBSET ANALYSES

all_computational_cost <- bind_rows(
  analysis_3$computational_summary,
  analysis_5$computational_summary,
  analysis_8$computational_summary
)

round_df(all_computational_cost, 4)

# 13. COMBINED ESTIMATES FOR 3, 5, AND 8 SUBSETS

all_combined_results <- bind_rows(
  analysis_3$combined_table,
  analysis_5$combined_table,
  analysis_8$combined_table
)

round_df(all_combined_results, 4)

# 14. EVALUATION AGAINST Q1-SPLIT FULL-DATA ANALYSIS

q1_ref_std_beta <- as.numeric(coef(cox_standard_q1)[1])
q1_ref_std_se <- sqrt(as.numeric(vcov(cox_standard_q1)[1, 1]))

q1_ref_cl_beta <- as.numeric(coef(cox_robust_q1)[1])
q1_ref_cl_se <- sqrt(as.numeric(vcov(cox_robust_q1)[1, 1]))

```

```

q1_ref_cop_beta <- as.numeric(sun_q1$Parameters["beta_Heifer", "Estimates"])
q1_ref_cop_se   <- as.numeric(sun_q1$Parameters["beta_Heifer", "StandardErrors"])

q1_ref_tau      <- as.numeric(sun_q1$Kendall_Tau["Estimates"])
q1_ref_tau_se   <- as.numeric(sun_q1$Kendall_Tau["StandardErrors"])

eval_against_q1 <- all_combined_results %>%
  rowwise() %>%
  mutate(
    Reference_Estimate = case_when(
      Model == "Standard Cox" ~ q1_ref_std_beta,
      Model == "Cluster-Robust Cox" ~ q1_ref_cl_beta,
      Model == "Copula Model" ~ q1_ref_cop_beta,
      Model == "Kendall tau" ~ q1_ref_tau
    ),
    Reference_SE = case_when(
      Model == "Standard Cox" ~ q1_ref_std_se,
      Model == "Cluster-Robust Cox" ~ q1_ref_cl_se,
      Model == "Copula Model" ~ q1_ref_cop_se,
      Model == "Kendall tau" ~ q1_ref_tau_se
    ),
    Absolute_Difference = abs(Estimate - Reference_Estimate),
    Relative_Difference = abs(Estimate - Reference_Estimate) / abs(Reference_Estimate),
    SE_Difference = abs(SE - Reference_SE),
    Bias = Estimate - Reference_Estimate
  ) %>%
  ungroup()

round_df(eval_against_q1, 4)

# 15. EVALUATION AGAINST ORIGINAL FULL-DATA BENCHMARK

eval_against_original_full <- all_combined_results %>%
  rowwise() %>%
  mutate(
    Reference_Estimate = case_when(
      Model == "Standard Cox" ~ full_std_beta,
      Model == "Cluster-Robust Cox" ~ full_cl_beta,
      Model == "Copula Model" ~ full_cop_beta,
      Model == "Kendall tau" ~ full_tau
    ),
    Reference_SE = case_when(
      Model == "Standard Cox" ~ full_std_se,
      Model == "Cluster-Robust Cox" ~ full_cl_se,
      Model == "Copula Model" ~ full_cop_se,
      Model == "Kendall tau" ~ full_tau_se
    ),
    Absolute_Difference = abs(Estimate - Reference_Estimate),
    Relative_Difference = abs(Estimate - Reference_Estimate) / abs(Reference_Estimate),
    SE_Difference = abs(SE - Reference_SE),
    Bias = Estimate - Reference_Estimate
  ) %>%
  ungroup()

round_df(eval_against_original_full, 4)

# 16. COMPUTATIONAL COST COMPARISON

cost_by_split <- all_computational_cost %>%
  group_by(Split) %>%
  summarise(
    Total_Observations = sum(Observations),
    Total_Clusters = sum(Clusters),
    Total_Elapsed_Time = sum(Total_Elapsed_Time),
    Total_Copula_Elapsed_Time = sum(Copula_Elapsed_Time),
    Mean_Copula_Elapsed_Time = mean(Copula_Elapsed_Time),
    Max_Copula_Elapsed_Time = max(Copula_Elapsed_Time),
    Total_Memory_MB = sum(Approx_Memory_Used_MB),
    .groups = "drop"
  )

q1_full_cost_short <- q1_full_cost %>%
  transmute(
    Split = "Q1-split full data",
    Total_Observations = Observations,
    Total_Clusters = Clusters,
    Total_Elapsed_Time = Total_Elapsed_Time,
    Total_Copula_Elapsed_Time = Copula_Elapsed_Time,

```

```

    Mean_Copula_Elapsed_Time = Copula_Elapsed_Time,
    Max_Copula_Elapsed_Time = Copula_Elapsed_Time,
    Total_Memory_MB = Approx_Memory_Used_MB
  )

computational_comparison <- bind_rows(
  q1_full_cost_short,
  cost_by_split
)

round_df(computational_comparison, 4)

# 17. SAVE IMPORTANT TABLES

write.csv(q1_cluster_summary, "q1_cluster_summary.csv", row.names = FALSE)
write.csv(q1_by_quartile, "q1_by_quartile.csv", row.names = FALSE)
write.csv(q1_cluster_size_summary, "q1_cluster_size_summary.csv", row.names = FALSE)

write.csv(q1_full_results, "q1_full_results.csv", row.names = FALSE)
write.csv(q1_full_cost, "q1_full_computational_cost.csv", row.names = FALSE)

write.csv(all_subset_results, "q1_subset_specific_results.csv", row.names = FALSE)
write.csv(all_computational_cost, "q1_all_subset_computational_cost.csv", row.names = FALSE)
write.csv(all_combined_results, "q1_combined_results.csv", row.names = FALSE)

write.csv(eval_against_q1, "q1_eval_against_q1_full.csv", row.names = FALSE)
write.csv(eval_against_original_full, "q1_eval_against_original_full.csv", row.names = FALSE)
write.csv(computational_comparison, "q1_computational_comparison.csv", row.names = FALSE)

```

Listing 6: R code for Q1 and Q2 split analysis

```

# 1. LOAD PACKAGES

library(Sunclarco)
library(survival)
library(dplyr)

# 2. LOAD DATA

data(insem)

# 3. FULL-DATA BENCHMARK VALUES

full_std_beta <- -0.06034839
full_std_se <- 0.02050279

full_cl_beta <- -0.06034839
full_cl_se <- 0.02096202

full_cop_beta <- -0.06034839
full_cop_se <- 0.02096202

full_tau <- 0.1828338
full_tau_se <- 0.02103043

# 4. HELPER FUNCTIONS

combine_scalar <- function(beta, var, weights) {
  beta_comb <- sum(weights * beta) / sum(weights)
  var_comb <- sum((weights^2) * var) / (sum(weights)^2)
  se_comb <- sqrt(var_comb)

  list(beta = beta_comb, var = var_comb, se = se_comb)
}

evaluation_metrics <- function(combined_est, combined_se, reference_est, reference_se) {
  data.frame(
    Combined_Estimate = combined_est,
    Reference_Estimate = reference_est,
    Absolute_Difference = abs(combined_est - reference_est),
    Relative_Difference = abs(combined_est - reference_est) / abs(reference_est),
    Combined_SE = combined_se,
    Reference_SE = reference_se,
    SE_Difference = abs(combined_se - reference_se),
    Bias = combined_est - reference_est
  )
}

```

```

round_df <- function(df, digits = 4) {
  df[sapply(df, is.numeric)] <- round(df[sapply(df, is.numeric)], digits)
  df
}

# 5. CREATE HERD SIZE QUANTILES
# Q1 = largest herds, Q4 = smallest herds

herd_quartiles <- insem %>%
  count(Herd, name = "n_cows") %>%
  arrange(desc(n_cows)) %>%
  mutate(herd_quartile = ntile(desc(n_cows), 4))

insem_q <- insem %>%
  left_join(herd_quartiles, by = "Herd")

# 6. Q1 SPLITTING: LARGEST HERDS INTO 3 SUBCLUSTERS

data_q1_split <- insem_q %>%
  group_by(Herd) %>%
  mutate(
    subcluster = if_else(
      herd_quartile == 1,
      ntile(row_number(), 3),
      1L
    )
  ) %>%
  ungroup() %>%
  group_by(Herd, subcluster) %>%
  mutate(cluster_id = cur_group_id()) %>%
  ungroup()

data_q1_split <- as.data.frame(data_q1_split)

# 6. Q1 + Q2 SPLITTING
# Q1 split into 3 subclusters
# Q2 split into 2 subclusters

data_q1_q2_split <- insem_q %>%
  group_by(Herd) %>%
  mutate(
    subcluster = case_when(
      herd_quartile == 1 ~ ntile(row_number(), 3),
      herd_quartile == 2 ~ ntile(row_number(), 2),
      TRUE ~ 1L
    )
  ) %>%
  ungroup() %>%
  group_by(Herd, subcluster) %>%
  mutate(cluster_id = cur_group_id()) %>%
  ungroup()

data_q1_q2_split <- as.data.frame(data_q1_q2_split)

# 7. DESCRIPTION AFTER Q1 + Q2 SPLITTING

q1_q2_cluster_summary <- data.frame(
  Description = c(
    "Original number of herds",
    "Number of Q1 herds split",
    "Number of Q2 herds split",
    "Final number of clusters after Q1 + Q2 splitting",
    "Number of observations",
    "Number of events",
    "Number censored"
  ),
  Value = c(
    n_distinct(insem$Herd),
    n_distinct(data_q1_q2_split$Herd[data_q1_q2_split$herd_quartile == 1]),
    n_distinct(data_q1_q2_split$Herd[data_q1_q2_split$herd_quartile == 2]),
    n_distinct(data_q1_q2_split$cluster_id),
    nrow(data_q1_q2_split),
    sum(data_q1_q2_split$Status == 1),
    sum(data_q1_q2_split$Status == 0)
  )
)

q1_q2_by_quartile <- data_q1_q2_split %>%
  group_by(herd_quartile) %>%
  summarise(

```

```

    original_herds = n_distinct(Herd),
    final_clusters = n_distinct(cluster_id),
    observations = n(),
    events = sum(Status == 1),
    censored = sum(Status == 0),
    censoring_percent = round(mean(Status == 0) * 100, 2),
    .groups = "drop"
  )

q1_q2_cluster_size_summary <- data_q1_q2_split %>%
  count(cluster_id, name = "cluster_size") %>%
  summarise(
    min_cluster_size = min(cluster_size),
    q1_cluster_size = quantile(cluster_size, 0.25),
    median_cluster_size = median(cluster_size),
    mean_cluster_size = round(mean(cluster_size), 2),
    q3_cluster_size = quantile(cluster_size, 0.75),
    max_cluster_size = max(cluster_size)
  )

q1_q2_cluster_summary
q1_q2_by_quartile
q1_q2_cluster_size_summary

# 8. Q1 + Q2 SPLIT FULL-DATA ANALYSIS + COMPUTATIONAL COST

gc()
mem_before_q1_q2_full <- sum(gc())[, 2])

time_q1_q2_full <- system.time({

  cox_standard_q1_q2 <- coxph(
    Surv(Time, Status) ~ Heifer,
    data = data_q1_q2_split
  )

  cox_robust_q1_q2 <- coxph(
    Surv(Time, Status) ~ Heifer + cluster(cluster_id),
    data = data_q1_q2_split
  )

  time_q1_q2_copula_full <- system.time({
    sun_q1_q2 <- SunclarcoModel(
      data = data_q1_q2_split,
      time = "Time",
      status = "Status",
      clusters = "cluster_id",
      covariates = "Heifer",
      stage = 2,
      copula = "Clayton",
      marginal = "Cox",
      verbose = FALSE
    )
  })
})

gc()
mem_after_q1_q2_full <- sum(gc())[, 2])
memory_q1_q2_full <- mem_after_q1_q2_full - mem_before_q1_q2_full

q1_q2_full_results <- data.frame(
  Model = c("Standard Cox", "Cluster-Robust Cox", "Q1+Q2 Split Copula"),
  Beta = c(
    as.numeric(coef(cox_standard_q1_q2)[1]),
    as.numeric(coef(cox_robust_q1_q2)[1]),
    as.numeric(sun_q1_q2$Parameters["beta_Heifer", "Estimates"])
  ),
  SE = c(
    sqrt(as.numeric(vcov(cox_standard_q1_q2)[1, 1])),
    sqrt(as.numeric(vcov(cox_robust_q1_q2)[1, 1])),
    as.numeric(sun_q1_q2$Parameters["beta_Heifer", "StandardErrors"])
  )
)

q1_q2_full_results$HR <- exp(q1_q2_full_results$Beta)

q1_q2_full_tau <- as.numeric(sun_q1_q2$Kendall_Tau["Estimates"])
q1_q2_full_tau_se <- as.numeric(sun_q1_q2$Kendall_Tau["StandardErrors"])

q1_q2_full_cost <- data.frame(

```



```

Analysis = "Q1+Q2 split full data",
Observations = nrow(data_q1_q2_split),
Clusters = length(unique(data_q1_q2_split$cluster_id)),
Total_User_Time = as.numeric(time_q1_q2_full["user.self"]),
Total_System_Time = as.numeric(time_q1_q2_full["sys.self"]),
Total_Elapsed_Time = as.numeric(time_q1_q2_full["elapsed"]),
Copula_User_Time = as.numeric(time_q1_q2_copula_full["user.self"]),
Copula_System_Time = as.numeric(time_q1_q2_copula_full["sys.self"]),
Copula_Elapsed_Time = as.numeric(time_q1_q2_copula_full["elapsed"]),
Approx_Memory_Used_MB = memory_q1_q2_full
)

round_df(q1_q2_full_results, 4)
round_df(q1_q2_full_cost, 4)

# 9. FUNCTION FOR Q1 + Q2 SPLIT SUBSET ANALYSIS

run_q1_q2_subset_analysis <- function(data_input, n_subsets, seed = 123) {

  set.seed(seed)

  clusters <- sample(unique(data_input$cluster_id))
  cluster_groups <- split(
    clusters,
    cut(seq_along(clusters), n_subsets, labels = FALSE)
  )

  subset_list <- lapply(cluster_groups, function(cl) {
    data_input[data_input$cluster_id %in% cl, ]
  })

  cat("\n===== \n")
  cat(n_subsets, "SUBSET ANALYSIS AFTER Q1 + Q2 SPLITTING \n")
  cat("===== \n")

  cat("\nObservations in each subset: \n")
  print(sapply(subset_list, nrow))

  cat("\nClusters in each subset: \n")
  print(sapply(subset_list, function(x) length(unique(x$cluster_id))))

  results <- lapply(seq_along(subset_list), function(i) {

    dat <- subset_list[[i]]

    cat("\n===== \n")
    cat("FITTING MODELS FOR SUBSET", i, "\n")
    cat("===== \n")

    gc()
    mem_before <- sum(gc()[, 2])

    time_total <- system.time({

      cox_standard <- coxph(
        Surv(Time, Status) ~ Heifer,
        data = dat
      )

      cox_cluster <- coxph(
        Surv(Time, Status) ~ Heifer + cluster(cluster_id),
        data = dat
      )

      time_copula <- system.time({
        copula_fit <- SunclarcoModel(
          data = dat,
          time = "Time",
          status = "Status",
          clusters = "cluster_id",
          covariates = "Heifer",
          stage = 2,
          copula = "Clayton",
          marginal = "Cox",
          verbose = FALSE
        )
      })
    })

    gc()

```

```

mem_after <- sum(gc()[, 2])
memory_used <- mem_after - mem_before

std_beta <- as.numeric(coef(cox_standard)[1])
std_var <- as.numeric(vcov(cox_standard)[1, 1])
std_info <- 1 / std_var
std_se <- sqrt(std_var)
std_hr <- exp(std_beta)

cl_beta <- as.numeric(coef(cox_cluster)[1])
cl_var_robust <- as.numeric(vcov(cox_cluster)[1, 1])
cl_var_naive <- as.numeric(cox_cluster$naive.var[1, 1])
cl_info <- 1 / cl_var_naive
cl_se_robust <- sqrt(cl_var_robust)
cl_se_naive <- sqrt(cl_var_naive)
cl_hr <- exp(cl_beta)

cop_beta <- as.numeric(copula_fit$Parameters["beta_Heifer", "Estimates"])
cop_se <- as.numeric(copula_fit$Parameters["beta_Heifer", "StandardErrors"])
cop_var <- cop_se^2
cop_info <- 1 / cop_var
cop_hr <- exp(cop_beta)

theta_est <- as.numeric(copula_fit$Parameters["theta", "Estimates"])
theta_se <- as.numeric(copula_fit$Parameters["theta", "StandardErrors"])

tau_est <- as.numeric(copula_fit$Kendall_Tau["Estimates"])
tau_se <- as.numeric(copula_fit$Kendall_Tau["StandardErrors"])
tau_var <- tau_se^2
tau_info <- 1 / tau_var

list(
  subset_id = i,
  n_obs = nrow(dat),
  n_clusters = length(unique(dat$cluster_id)),
  n_original_herds = length(unique(dat$Herd)),

  total_user_time = as.numeric(time_total["user.self"]),
  total_system_time = as.numeric(time_total["sys.self"]),
  total_elapsed_time = as.numeric(time_total["elapsed"]),

  copula_user_time = as.numeric(time_copula["user.self"]),
  copula_system_time = as.numeric(time_copula["sys.self"]),
  copula_elapsed_time = as.numeric(time_copula["elapsed"]),

  memory_used_MB = memory_used,

  std_beta = std_beta,
  std_var = std_var,
  std_info = std_info,
  std_se = std_se,
  std_hr = std_hr,

  cl_beta = cl_beta,
  cl_var_robust = cl_var_robust,
  cl_var_naive = cl_var_naive,
  cl_info = cl_info,
  cl_se_robust = cl_se_robust,
  cl_se_naive = cl_se_naive,
  cl_hr = cl_hr,

  cop_beta = cop_beta,
  cop_var = cop_var,
  cop_info = cop_info,
  cop_se = cop_se,
  cop_hr = cop_hr,

  theta_est = theta_est,
  theta_se = theta_se,

  tau_est = tau_est,
  tau_var = tau_var,
  tau_info = tau_info,
  tau_se = tau_se
)
})

subset_summary <- data.frame(
  Split = paste0(n_subsets, "-subset"),
  Subset = sapply(results, function(x) x$subset_id),

```

```

Observations = sapply(results, function(x) x$n_obs),
Clusters = sapply(results, function(x) x$n_clusters),
Original_Herds = sapply(results, function(x) x$n_original_herds),

Standard_Cox_Beta = sapply(results, function(x) x$std_beta),
Standard_Cox_SE = sapply(results, function(x) x$std_se),
Standard_Cox_HR = sapply(results, function(x) x$std_hr),

Cluster_Cox_Beta = sapply(results, function(x) x$cl_beta),
Cluster_Cox_Robust_SE = sapply(results, function(x) x$cl_se_robust),
Cluster_Cox_Naive_SE = sapply(results, function(x) x$cl_se_naive),
Cluster_Cox_HR = sapply(results, function(x) x$cl_hr),

Copula_Beta = sapply(results, function(x) x$cop_beta),
Copula_SE = sapply(results, function(x) x$cop_se),
Copula_HR = sapply(results, function(x) x$cop_hr),

Theta = sapply(results, function(x) x$theta_est),
Theta_SE = sapply(results, function(x) x$theta_se),

Tau = sapply(results, function(x) x$tau_est),
Tau_SE = sapply(results, function(x) x$tau_se)
)

computational_summary <- data.frame(
  Split = paste0(n_subsets, "-subset"),
  Subset = sapply(results, function(x) x$subset_id),
  Observations = sapply(results, function(x) x$n_obs),
  Clusters = sapply(results, function(x) x$n_clusters),
  Original_Herds = sapply(results, function(x) x$n_original_herds),

  Total_User_Time = sapply(results, function(x) x$total_user_time),
  Total_System_Time = sapply(results, function(x) x$total_system_time),
  Total_Elapsed_Time = sapply(results, function(x) x$total_elapsed_time),

  Copula_User_Time = sapply(results, function(x) x$copula_user_time),
  Copula_System_Time = sapply(results, function(x) x$copula_system_time),
  Copula_Elapsed_Time = sapply(results, function(x) x$copula_elapsed_time),

  Approx_Memory_Used_MB = sapply(results, function(x) x$memory_used_MB)
)

n_clusters <- sapply(results, function(x) x$n_clusters)
n_cows <- sapply(results, function(x) x$n_obs)

std_beta <- sapply(results, function(x) x$std_beta)
std_var <- sapply(results, function(x) x$std_var)
std_info <- sapply(results, function(x) x$std_info)

cl_beta <- sapply(results, function(x) x$cl_beta)
cl_var_robust <- sapply(results, function(x) x$cl_var_robust)
cl_info <- sapply(results, function(x) x$cl_info)

cop_beta <- sapply(results, function(x) x$cop_beta)
cop_var <- sapply(results, function(x) x$cop_var)
cop_info <- sapply(results, function(x) x$cop_info)

tau_beta <- sapply(results, function(x) x$tau_est)
tau_var <- sapply(results, function(x) x$tau_var)
tau_info <- sapply(results, function(x) x$tau_info)

std_comb_info <- combine_scalar(std_beta, std_var, std_info)
std_comb_iv <- combine_scalar(std_beta, std_var, 1 / std_var)
std_comb_cln <- combine_scalar(std_beta, std_var, n_clusters)
std_comb_cows <- combine_scalar(std_beta, std_var, n_cows)

cl_comb_info <- combine_scalar(cl_beta, 1 / cl_info, cl_info)
cl_comb_iv <- combine_scalar(cl_beta, cl_var_robust, 1 / cl_var_robust)
cl_comb_cln <- combine_scalar(cl_beta, cl_var_robust, n_clusters)
cl_comb_cows <- combine_scalar(cl_beta, cl_var_robust, n_cows)

cop_comb_info <- combine_scalar(cop_beta, cop_var, cop_info)
cop_comb_iv <- combine_scalar(cop_beta, cop_var, 1 / cop_var)
cop_comb_cln <- combine_scalar(cop_beta, cop_var, n_clusters)
cop_comb_cows <- combine_scalar(cop_beta, cop_var, n_cows)

tau_comb_info <- combine_scalar(tau_beta, tau_var, tau_info)
tau_comb_iv <- combine_scalar(tau_beta, tau_var, 1 / tau_var)
tau_comb_cln <- combine_scalar(tau_beta, tau_var, n_clusters)
tau_comb_cows <- combine_scalar(tau_beta, tau_var, n_cows)

```

```

combined_table <- data.frame(
  Split = paste0(n_subsets, "-subset"),
  Model = rep(
    c("Standard Cox", "Cluster-Robust Cox", "Copula Model", "Kendall tau"),
    each = 4
  ),
  Method = rep(
    c("Information matrix", "Inverse variance", "Cluster-number", "Cow-number"),
    times = 4
  ),
  Estimate = c(
    std_comb_info$beta, std_comb_iv$beta, std_comb_cln$beta, std_comb_cows$beta,
    cl_comb_info$beta, cl_comb_iv$beta, cl_comb_cln$beta, cl_comb_cows$beta,
    cop_comb_info$beta, cop_comb_iv$beta, cop_comb_cln$beta, cop_comb_cows$beta,
    tau_comb_info$beta, tau_comb_iv$beta, tau_comb_cln$beta, tau_comb_cows$beta
  ),
  SE = c(
    std_comb_info$se, std_comb_iv$se, std_comb_cln$se, std_comb_cows$se,
    cl_comb_info$se, cl_comb_iv$se, cl_comb_cln$se, cl_comb_cows$se,
    cop_comb_info$se, cop_comb_iv$se, cop_comb_cln$se, cop_comb_cows$se,
    tau_comb_info$se, tau_comb_iv$se, tau_comb_cln$se, tau_comb_cows$se
  )
)

list(
  results = results,
  subset_summary = subset_summary,
  computational_summary = computational_summary,
  combined_table = combined_table
)
}

# 10. RUN 3-, 5-, AND 8-SUBSET ANALYSES AFTER Q1 + Q2 SPLITTING

analysis_q1_q2_3 <- run_q1_q2_subset_analysis(data_q1_q2_split, n_subsets = 3, seed = 123)
analysis_q1_q2_5 <- run_q1_q2_subset_analysis(data_q1_q2_split, n_subsets = 5, seed = 123)
analysis_q1_q2_8 <- run_q1_q2_subset_analysis(data_q1_q2_split, n_subsets = 8, seed = 123)

# 11. SUBSET-SPECIFIC RESULTS

all_q1_q2_subset_results <- bind_rows(
  analysis_q1_q2_3$subset_summary,
  analysis_q1_q2_5$subset_summary,
  analysis_q1_q2_8$subset_summary
)

round_df(all_q1_q2_subset_results, 4)

# 12. COMPUTATIONAL COST FOR ALL SUBSET ANALYSES

all_q1_q2_computational_cost <- bind_rows(
  analysis_q1_q2_3$computational_summary,
  analysis_q1_q2_5$computational_summary,
  analysis_q1_q2_8$computational_summary
)

round_df(all_q1_q2_computational_cost, 4)

# 13. COMBINED ESTIMATES FOR 3, 5, AND 8 SUBSETS

all_q1_q2_combined_results <- bind_rows(
  analysis_q1_q2_3$combined_table,
  analysis_q1_q2_5$combined_table,
  analysis_q1_q2_8$combined_table
)

round_df(all_q1_q2_combined_results, 4)

# 14. EVALUATION AGAINST Q1 + Q2 SPLIT FULL-DATA ANALYSIS

q1_q2_ref_std_beta <- as.numeric(coef(cox_standard_q1_q2)[1])
q1_q2_ref_std_se <- sqrt(as.numeric(vcov(cox_standard_q1_q2)[1, 1]))

q1_q2_ref_cl_beta <- as.numeric(coef(cox_robust_q1_q2)[1])
q1_q2_ref_cl_se <- sqrt(as.numeric(vcov(cox_robust_q1_q2)[1, 1]))

q1_q2_ref_cop_beta <- as.numeric(sun_q1_q2$Parameters["beta_Heifer", "Estimates"])
q1_q2_ref_cop_se <- as.numeric(sun_q1_q2$Parameters["beta_Heifer", "StandardErrors"])

```

```

q1_q2_ref_tau    <- as.numeric(sun_q1_q2$Kendall_Tau["Estimates"])
q1_q2_ref_tau_se <- as.numeric(sun_q1_q2$Kendall_Tau["StandardErrors"])

eval_q1_q2_against_q1_q2_full <- all_q1_q2_combined_results %>%
  rowwise() %>%
  mutate(
    Reference_Estimate = case_when(
      Model == "Standard Cox" ~ q1_q2_ref_std_beta,
      Model == "Cluster-Robust Cox" ~ q1_q2_ref_cl_beta,
      Model == "Copula Model" ~ q1_q2_ref_cop_beta,
      Model == "Kendall tau" ~ q1_q2_ref_tau
    ),
    Reference_SE = case_when(
      Model == "Standard Cox" ~ q1_q2_ref_std_se,
      Model == "Cluster-Robust Cox" ~ q1_q2_ref_cl_se,
      Model == "Copula Model" ~ q1_q2_ref_cop_se,
      Model == "Kendall tau" ~ q1_q2_ref_tau_se
    ),
    Absolute_Difference = abs(Estimate - Reference_Estimate),
    Relative_Difference = abs(Estimate - Reference_Estimate) / abs(Reference_Estimate),
    SE_Difference = abs(SE - Reference_SE),
    Bias = Estimate - Reference_Estimate
  ) %>%
  ungroup()

round_df(eval_q1_q2_against_q1_q2_full, 4)

# 15. EVALUATION AGAINST ORIGINAL FULL-DATA BENCHMARK

eval_q1_q2_against_original_full <- all_q1_q2_combined_results %>%
  rowwise() %>%
  mutate(
    Reference_Estimate = case_when(
      Model == "Standard Cox" ~ full_std_beta,
      Model == "Cluster-Robust Cox" ~ full_cl_beta,
      Model == "Copula Model" ~ full_cop_beta,
      Model == "Kendall tau" ~ full_tau
    ),
    Reference_SE = case_when(
      Model == "Standard Cox" ~ full_std_se,
      Model == "Cluster-Robust Cox" ~ full_cl_se,
      Model == "Copula Model" ~ full_cop_se,
      Model == "Kendall tau" ~ full_tau_se
    ),
    Absolute_Difference = abs(Estimate - Reference_Estimate),
    Relative_Difference = abs(Estimate - Reference_Estimate) / abs(Reference_Estimate),
    SE_Difference = abs(SE - Reference_SE),
    Bias = Estimate - Reference_Estimate
  ) %>%
  ungroup()

round_df(eval_q1_q2_against_original_full, 4)

# 16. COMPUTATIONAL COST COMPARISON

q1_q2_cost_by_split <- all_q1_q2_computational_cost %>%
  group_by(Split) %>%
  summarise(
    Total_Observations = sum(Observations),
    Total_Clusters = sum(Clusters),
    Total_Elapsed_Time = sum(Total_Elapsed_Time),
    Total_Copula_Elapsed_Time = sum(Copula_Elapsed_Time),
    Mean_Copula_Elapsed_Time = mean(Copula_Elapsed_Time),
    Max_Copula_Elapsed_Time = max(Copula_Elapsed_Time),
    Total_Memory_MB = sum(Approx_Memory_Used_MB),
    .groups = "drop"
  )

q1_q2_full_cost_short <- q1_q2_full_cost %>%
  transmute(
    Split = "Q1+Q2 full data",
    Total_Observations = Observations,
    Total_Clusters = Clusters,
    Total_Elapsed_Time = Total_Elapsed_Time,
    Total_Copula_Elapsed_Time = Copula_Elapsed_Time,
    Mean_Copula_Elapsed_Time = Copula_Elapsed_Time,
    Max_Copula_Elapsed_Time = Copula_Elapsed_Time,
    Total_Memory_MB = Approx_Memory_Used_MB
  )

```

```

q1_q2_computational_comparison <- bind_rows(
  q1_q2_full_cost_short,
  q1_q2_cost_by_split
)

round_df(q1_q2_computational_comparison, 4)

# 17. SAVE TABLES

write.csv(q1_q2_cluster_summary, "q1_q2_cluster_summary.csv", row.names = FALSE)
write.csv(q1_q2_by_quartile, "q1_q2_by_quartile.csv", row.names = FALSE)
write.csv(q1_q2_cluster_size_summary, "q1_q2_cluster_size_summary.csv", row.names = FALSE)

write.csv(q1_q2_full_results, "q1_q2_full_results.csv", row.names = FALSE)
write.csv(q1_q2_full_cost, "q1_q2_full_computational_cost.csv", row.names = FALSE)

write.csv(all_q1_q2_subset_results, "q1_q2_subset_specific_results.csv", row.names = FALSE)
write.csv(all_q1_q2_computational_cost, "q1_q2_all_subset_computational_cost.csv", row.names = FALSE)
write.csv(all_q1_q2_combined_results, "q1_q2_combined_results.csv", row.names = FALSE)

write.csv(eval_q1_q2_against_q1_q2_full, "q1_q2_eval_against_q1_q2_full.csv", row.names = FALSE)
write.csv(eval_q1_q2_against_original_full, "q1_q2_eval_against_original_full.csv", row.names = FALSE)
write.csv(q1_q2_computational_comparison, "q1_q2_computational_comparison.csv", row.names = FALSE)

```